The construction of four-dimensional quantum Yang–Mills theory

Under construction. Version of November 11, 2026: this manuscript is incomplete. Sections are added and revised as they are written and checked against the formal proof; see the status table at `https://github.com/scottnarmstrong/yang-mills-manuscript`.

Abstract

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Part I
The theorem and the architecture of its proof

1 Introduction

2 Setting

Lattice observables are local functions of parallel transports; their limiting fields are distinguished by all separated correlations.

The faithful unitary realization $G \subset U(n_G)$ is fixed, with G closed and its real Lie algebra $\mathfrak{g}$ simple and nonabelian; connectedness is not assumed. Throughout,

$$\langle X, Y\rangle_{\mathfrak{g}} = -n_G^{-1} \operatorname{Re}\operatorname{tr}(XY), \qquad d_{\mathfrak{g}} = \dim_{\mathbb{R}} \mathfrak{g}\,. \tag{2.1}$$

Here $\mathbb{N} = \{0, 1, \ldots\}$, coordinate indices run from 1 to 4, $|x|$ is Euclidean and $|x|_\infty = \max_a |x^a|$. The block-spin penalty is a fixed $\mathsf{b} > 0$. Time is the fourth coordinate: $t(x) = x^4$, $\vec{x} = (x^1, x^2, x^3)$, $\mathsf{m}(x) = (x^4)^2 - |\vec{x}|^2$ and $\overline{V}_+ = \{\mathbf{p} : |\vec{\mathbf{p}}| \le \mathbf{p}^4\}$. Fourier phases use $e^{i\mathbf{p}\cdot x}$ with the Euclidean pairing; Hilbert inner products are antilinear in the first variable. Spacetime integrals use Lebesgue measure. We include arity zero: empty products are 1, and empty sums and the maximum of an empty degree list are 0.

The ultraviolet, volume and renormalization indices are N, p, m, with $0 \le m \le N$; blocking decreases m. Odd ratios $\ell_m \ge 3$ give $D_m = \prod_{i<m} \ell_i$, $a_m = D_m^{-1}$ and $s_m = \log D_m$. The level-m torus has $L_p D_m$ sites per side. Tokens $(d, \mathbf{c})$ have insertions $O^{(d)}_{N,p}(\mathbf{c}, y)$; Wilson expectation is $\langle\cdot\rangle_{N,p}$. Fields (d, A) are written A when the degree is specified. Correlations $S(A_1(x_1), \ldots, A_n(x_n))$ and their derivatives concern separated points. Constants may depend on G, its realization, b and the fixed construction; further dependence and uniformity are stated.

The lattice theory. The matrix realization fixes the Wilson action and coupling normalization.

Definition 2.1 (Gauge group and Casimir)**.** A gauge group is an integer $n_G \ge 1$, a closed subgroup $G \subset U(n_G)$ and a real subspace $\mathfrak{g} \subset M_{n_G}(\mathbb{C})$ such that

$$X \in \mathfrak{g} \quad \Longleftrightarrow \quad \exp(tX) \in G \quad \text{for every } t \in \mathbb{R}\,. \tag{2.2}$$

Simplicity means that $[\mathfrak{g}, \mathfrak{g}] \ne 0$ and every real subspace $I_{\mathrm{id}} \subset \mathfrak{g}$ with $[\mathfrak{g}, I_{\mathrm{id}}] \subset I_{\mathrm{id}}$ is 0 or $\mathfrak{g}$. With the fixed inner product and an orthonormal basis $(e^{\mathfrak{g}}_\alpha)_{\alpha=1}^{d_{\mathfrak{g}}}$, define

$$C_{\mathfrak{g}} = -d_{\mathfrak{g}}^{-1} \sum_\alpha \operatorname{tr}_{\mathfrak{g}}(\operatorname{ad}(e^{\mathfrak{g}}_\alpha)^2), \qquad b_0 = \frac{11 C_{\mathfrak{g}}}{48\pi^2}, \qquad \kappa_G = \frac{11 C_{\mathfrak{g}}}{24\pi^2}\,. \tag{2.3}$$

Also put $\operatorname{pr}(u) = 1 - n_G^{-1} \operatorname{Re}\operatorname{tr} u$ and $\operatorname{ch}(u) = n_G^{-1/2} \|u - \mathrm{Id}_{n_G}\|_{\mathrm{F}}$, the chord to the identity, so $\operatorname{ch}(u)^2 = 2\operatorname{pr}(u)$.

Definition 2.2 (Configurations and tests)**.** For $n \ge 0$, let $\mathrm{Conf}_n = \{x \in (\mathbb{R}^4)^n : x_i \ne x_j \ (i \ne j)\}$ and $\mathrm{Gen}_n = \{x : x_i^a \ne x_j^a \ (i \ne j,\ a = 1, \ldots, 4)\}$. Write $\mathrm{sep}_\infty(x) = \min_{i \ne j} |x_i - x_j|_\infty$, with value $+\infty$ if $n \le 1$. A separated test is a globally smooth complex function with compact closed support contained in Conf_n. Write $\mathcal{S}_n = \mathcal{S}(\mathbb{R}^{4n}; \mathbb{C})$ and $\mathcal{S} = \mathcal{S}_1$. For a side $L > 0$, put

$$\mathrm{NoHit}_{L,n} = \{x : x_i - x_j \notin L\mathbb{Z}^4 \ (i \ne j)\}\,. \tag{2.4}$$

The group $H_4 \subset O(4)$ consists of the signed permutation matrices. Its reflection in the ath coordinate is $\mathsf{s}_a^{\mathrm{refl}}$.

Definition 2.3 (Wilson lattice theory)**.** For an integer $n_{\mathrm{tor}} \geq 1$ put $\Lambda_{n_{\mathrm{tor}}} = (\mathbb{Z}/n_{\mathrm{tor}}\mathbb{Z})^4$. A configuration assigns $U(y,a) \in G$ to each positive link (y,a); a reversed link has inverse value. Define

$$U_{y;ac} = U(y,a)U(y+e_a,c)U(y+e_c,a)^{-1}U(y,c)^{-1}, \tag{2.5}$$

$$E^{\mathrm{W}}_{n_{\mathrm{tor}}}(U) = \sum_{y\in\Lambda_{n_{\mathrm{tor}}}} \sum_{a<c} \mathrm{pr}(U_{y;ac}), \tag{2.6}$$

$$\mu_{n_{\mathrm{tor}},\beta}(dU) = Z^{-1}_{n_{\mathrm{tor}},\beta} e^{-\beta E^{\mathrm{W}}_{n_{\mathrm{tor}}}(U)} \prod_{y,a} dh(U(y,a))\,. \tag{2.7}$$

Here dh is probability Haar measure on G and $Z_{n_{\mathrm{tor}},\beta}$ is the integral of the numerator. A gauge transformation is $U^{\gamma_{\mathrm{g}}}(y,a) = \gamma_{\mathrm{g}}(y)U(y,a)\gamma_{\mathrm{g}}(y+e_a)^{-1}$. Lattice isometries act on links by transport, with inversion on reversed links. Let $r_{n_{\mathrm{tor}}}(j)$ be the representative of j in $(-n_{\mathrm{tor}}/2, n_{\mathrm{tor}}/2]$.

Definition 2.4 (Wilson families and regularity)**.** A Wilson family consists of strictly increasing positive integer sequences $(L_p), (D_N)$, real numbers $\beta_{N,p}$, probability Haar measure on G, a set $\mathcal{T}$ and functions $I_{N,p}(v,y;U) \in \mathbb{C}$ for $v \in \mathcal{T}$ and $y \in \Lambda_{L_pD_N}$. Here (D_N) is arbitrary; for the constructed family $D_N = \prod_{i<N} \ell_i$. Put $a_N = D_N^{-1}$, $\hat{y} = a_N(r_{L_pD_N}(y^a))_{a=1}^4$ and $\mu_{N,p} = \mu_{L_pD_N,\beta_{N,p}}$.

The family is regular if:

Reg1. there is $b_* > 0$ such that, for every $\varepsilon > 0$, eventually in N, $|\beta_{N,p}/\log(1/a_N) - b_*| < \varepsilon$ for every p;

Reg2. every $U \mapsto I_{N,p}(v,y;U)$ is continuous;

Reg3. every such insertion is gauge invariant;

Reg4. for each v there is a finite $\Sigma_v \subset \mathbb{Z}^4$, independent of N,p,y, such that the insertion depends only on links with both ends in $y + (\Sigma_v \bmod L_pD_N)$.

Definition 2.5 (Moments and ordered convergence)**.** For $v \in \mathcal{T}^n$, set

$$M^{\mathrm{lat}}_{N,p}(v;y) = \int \prod_i I_{N,p}(v_i,y_i;U)\,d\mu_{N,p}(U), \qquad C^{\mathrm{lat}}_{N,p}(v;f) = a_N^{4n} \sum_{y\in\Lambda^n_{L_pD_N}} f(\hat{y}) M^{\mathrm{lat}}_{N,p}(v;y)\,. \tag{2.8}$$

A separated test f is p-admissible if its support is contained in $((-L_p/2, L_p/2)^4)^n$. For $\mathsf{v} : \mathbb{N} \to \mathbb{N}$, ordered convergence to kernels $K_n(v;x)$ along v means that for every n, token list v and separated test f: f is $\mathsf{v}(k)$-admissible eventually in k, and there are numbers $C^\infty_p(v;f)$ such that $C^{\mathrm{lat}}_{N,p}(v;f) \to C^\infty_p(v;f)$ along the full sequence $N \to \infty$ at every f-admissible p, and $C^\infty_{\mathsf{v}(k)}(v;f) \to \int f K_n(v;\cdot)$. No condition is imposed on $C^\infty_p(v;f)$ at inadmissible volumes.

Local observables and blocking. Taylor normalization and transfer to a reference level fix the insertions.

Definition 2.6 (Polynomial labels and local sections)**.** Let $\mathcal{J}$ be the space of $\mathfrak{g}$-valued connection jets in radial gauge, written $\mathcal{A}_a(x) = \sum_{k\geq 0} \mathcal{A}_{k,a}(x)$, where $\mathcal{A}_{k,a}$ is homogeneous of polynomial degree $k+1$ and $\sum_a x^a \mathcal{A}_{k,a}(x) = 0$. Give the coordinates of $\mathcal{A}_k$ bidegree $(k+2,1)$: the first entry is engineering weight and the second counts polynomial factors. Let V_d be the complex polynomials in these coordinates of engineering weight at most d invariant under simultaneous $\mathrm{Ad}(G)$. Fix a

basis homogeneous in both degrees, the supremum coefficient norm, conjugation and the induced H_4 action. Equivalently these are invariant polynomials in curvature jets, with curvature weight 2 and covariant differentiation adding 1.

A local section in degree d is a linear map $\mathbf{c} \mapsto P_d^{\mathrm{loc}}[\mathbf{c}]$ to polynomials in the real and imaginary parts of link matrix entries on a fixed finite stencil $\Sigma_d \subset \mathbb{Z}^4$. Its placement $P_d^{\mathrm{loc}}[\mathbf{c}](y;U)$ evaluates it on the links of $y + \Sigma_d$. It is gauge invariant, commutes with conjugation, is H_4 covariant and sends the polynomial unit to 1. Its Taylor normalization is the following equality for every radial jet j and $0 \le k \le d$:

$$\frac{d^k}{dh_{\mathrm{lat}}^k} P_d^{\mathrm{loc}}[\mathbf{c}](U(h_{\mathrm{lat}},\mathsf{j}))\Big|_{h_{\mathrm{lat}}=0} = \frac{d^k}{dh_{\mathrm{lat}}^k} \mathbf{c}((h_{\mathrm{lat}}^{k'+2}\mathsf{j}_{k'})_{k'\ge 0})\Big|_{h_{\mathrm{lat}}=0} . \tag{2.9}$$

Here $U(h_{\mathrm{lat}},\mathsf{j})$ is parallel transport along the spacing-h_{lat} links of the radial polynomial potential through the required order, with transport equation $u'(t) = u(t)\mathcal{A}(t)$, $u(0) = 1_G$.

Definition 2.7 (Constants and level data)**.** The fixed numerical data are kernel parameters; stage data indexed by $j \ge 4$ (a derivative order $\mathfrak{q}_j \ge 6$, a spatial power $\varpi_j \ge 2$, support exponent $\sigma_j = 100$, path width $w_j^{\mathrm{path}} > 0$, path bound $A_j^{\mathrm{path}} > 0$, column bound $C_j \ge 1$, drift allowance $\omega_j^{\mathrm{RG}} \ge 0$ and positive clipping constants); odd stage ratios ℓ_j^{st}; exponents θ_j with $0 < \theta_{J_m} \le J_m^{-1}$; entry bounds; a reference coupling $\bar\beta > 1$; and entry levels and minimum periods. The orders $\mathfrak{q}_j$ and powers ϖ_j tend to infinity. Section 6 specifies the kernel parameters, column bounds, minimum periods and order of choice. The requirements of Sections 8–11, 13, 15 and 16 give lower bounds on stage ratios or entry levels using earlier data; raising entry levels preserves earlier requirements. Only the constants of the construction in Proposition 4.1, with its densities, marked transfers and lattice estimates, are called admissible; no assertion is made for other choices.

A level system has $J_m \ge 4$, $J_m \le J_{m+1} \le J_m + 1$, $J_m \to \infty$, $\ell_m = \ell_{J_{m+1}}^{\mathrm{st}}$, and the spacings fixed above. The stage orders, powers and support exponents are nondecreasing along the levels; θ_j is antitone. Put

$$\hat\beta_m = \bar\beta + \kappa_G s_m, \qquad M_m^{\mathrm{cut}} = \exp((\log\log\hat\beta_m)^2), \qquad t_m = \hat\beta_m^{-1/2} M_m^{\mathrm{cut}}, \tag{2.10}$$

$$B_m = \lceil (\log\hat\beta_m)^2 \rceil, \qquad \epsilon_m^{\mathrm{reg}} = \hat\beta_m^{-11/10}, \qquad \epsilon_m^{\mathrm{for}} = e^{-(M_m^{\mathrm{cut}})^{1/5}}, \tag{2.11}$$

$$\Lambda_m^{\mathrm{loop}} = 2(M_m^{\mathrm{cut}})^{\theta_{J_m}}, \qquad \mathfrak{B}_m = [\hat\beta_m/2, 2\hat\beta_m]\,. \tag{2.12}$$

The entrance conditions are $\hat\beta_m > 1$, $t_m \le 1$, $L_p D_m \ge 100(B_m + 1)$, and $L_p D_{m+1} \ge 100((J_{m+1} + 3)\ell_m + 1)$ for all m, p, together with

$$\frac{\omega_{J_{m+1}}^{\mathrm{RG}}}{\log \ell_m} \le J_{m+1}^{-2}, \qquad \frac{\log(1 + A_j^{\mathrm{path}} + (w_j^{\mathrm{path}})^{-1} + C_j + R_j^{\mathrm{rel}} + \sigma_j)}{\log \ell_m} \le j^{-2}, \quad j = J_{m+1}\,, \tag{2.13}$$

where R_j^{rel} is the clipping relaxation length. The stage ratios are nondecreasing. The conditions $\log \ell_m \to \infty$ and $\log \ell_m / s_m \to 0$ are required only as limits; the quotient at $m = 0$ plays no role.

Definition 2.8 (Block-spin transformation)**.** For a step of the fixed level system, U, V are link configurations on $\Lambda_{L_p D_{m+1}}, \Lambda_{L_p D_m}$. Let α_{tag} range over the finite frame and predictor tags of Section 8, $\lambda^{\mathrm{tag}}(U, \alpha_{\mathrm{tag}})$ be their product probability mass and $Q_e^{\mathrm{pred}}(U, \alpha_{\mathrm{tag}}) \in G$ the predictor on coarse link e. The cutoffs ensure unique minimizing branches and nonempty eligible axial tags.

These choices are fixed for the level system. Define

$$k_\beta^{(m)}(U,V)=\sum_{\alpha_{\mathrm{tag}}}\lambda^{\mathrm{tag}}(U,\alpha_{\mathrm{tag}})\prod_{e\ \mathrm{coarse}}\frac{e^{-\mathfrak{b}\beta\,\mathrm{pr}(V_eQ_e^{\mathrm{pred}}(U,\alpha_{\mathrm{tag}})^{-1})}}{\int_G e^{-\mathfrak{b}\beta\,\mathrm{pr}(u)}\,dh(u)},\tag{2.14}$$

$$(\mathcal{B}_\beta^{(m)}f)(V)=\int k_\beta^{(m)}(U,V)f(U)\,dU\,.\tag{2.15}$$

Here dU, dV are product probability Haar measures. Since the tag masses sum to one and dh is invariant, $\int k_\beta^{(m)}(U,V)\,dV=1$.

Definition 2.9 (Representations and their admissible class)**.** Distances in this definition use the unit-lattice sup metric. At level m, flag every site incident to a plaquette whose chord exceeds t_m, and let $\mathrm{Flag}_m(U)$ be the set of flagged sites. A support X is healthy when it contains no flagged site. A regular activity has a support X_i, anchor $o_i\in X_i$ and a Borel gauge-invariant real function E_i^{reg}, local on X_i, smooth through order $\mathfrak{q}_{J_m}$ in exponential left-link increments on healthy configurations, with the corresponding ordered derivatives existing there. It vanishes at the trivial configuration. A support is nonwrapping at o_i if $10\,\mathrm{dist}(x,o_i)$ is less than the torus side for every $x\in X_i$. On such a support the Hessian of $\mathcal{A}\mapsto E_i^{\mathrm{reg}}(\exp\mathcal{A})$ at zero vanishes on every pair of affine directions $\dot{\mathcal{A}}_{(y,a)}=u_a+\sum_c r_{L_pD_m}(y-o_i)^c v_{ac}$, using the centered lift $r_{L_pD_m}$. A forced activity has a support Y_i, nonempty flag set $I_i\subset Y_i$ and a Borel gauge-invariant function K_i^{for}, vanishing unless $Y_i\cap\mathrm{Flag}_m(U)=I_i$ and a fixed Borel gauge-invariant test holds. Both K_i^{for} and that test take equal values on configurations agreeing on the links inside Y_i and the flags in Y_i. Both activity families are countable and equivariant under translations and H_4, by permutations of their indices, including supports, anchors, flag sets, tests and values.

For a support X, let $\mathrm{cov}_{B_m}(X)$ be the minimum number of radius-B_m lattice balls with connected union covering X. Let $\|E_i^{\mathrm{reg}}\|_m$ be the supremum of all ordered left derivatives through order $\mathfrak{q}_{J_m}$, over healthy configurations and directions satisfying $|\dot{\mathcal{A}}_e|\le t_m(1+\mathrm{dist}(e^-,o_i))^{\varpi_{J_m}}$. The activity bounds mean the existence of nonnegative majorants $e_i^{\mathrm{maj}}\ge\|E_i^{\mathrm{reg}}\|_m$, $k_i^{\mathrm{maj}}\ge\sup_U|K_i^{\mathrm{for}}(U)|$ such that, for every site x, the following series converge and satisfy

$$\sum_{o_i=x}e^{\sigma_{J_m}\mathrm{cov}_{B_m}(X_i)}e_i^{\mathrm{maj}}\le\epsilon_m^{\mathrm{reg}},\qquad\sum_{x\in Y_i}e^{\sigma_{J_m}\mathrm{cov}_{B_m}(Y_i)}k_i^{\mathrm{maj}}\le\epsilon_m^{\mathrm{for}}\,.\tag{2.16}$$

A representation $\rho=(\beta,v_{\mathrm{vac}},P^{\mathrm{path}},E^{\mathrm{reg}},K^{\mathrm{for}})$ has the fixed path action $P^{\mathrm{path}}=P_{N,m}^{\mathrm{path}}$ and density

$$\mathrm{dens}\,\rho(U)=\exp\left\{v_{\mathrm{vac}}|\Lambda_{L_pD_m}|-P^{\mathrm{path}}(\beta,U)-\sum_i\mathbf{1}_{X_i\ \mathrm{healthy}}E_i^{\mathrm{reg}}(U)\right\}\cdot\sum_{\substack{\mathcal{A}_{\mathrm{for}}\subset\mathbb{N}\ \mathrm{finite}\\ Y_i\ (i\in\mathcal{A}_{\mathrm{for}})\ \mathrm{pairwise\ disjoint}\\ \bigcup_{i\in\mathcal{A}_{\mathrm{for}}}I_i=\mathrm{Flag}_m(U)}}\prod_{i\in\mathcal{A}_{\mathrm{for}}}K_i^{\mathrm{for}}(U)\,.\tag{2.17}$$

The path action and its arrays, with $c_{\mathrm{dir}}>16c_{\mathrm{lin}}>0$, are defined in Sections 7–10, independently of p. A slot is a finite lattice path followed by a plaquette, evaluated as $\mathrm{proj}_{\mathfrak{g}}(U_\square-U_\square^{-1})/2$ transported to the path origin. For these slots $\mathcal{F}_{\mathfrak{p}i}$, rows $\mathcal{L}_{\mathfrak{p}}^{\mathrm{path}}=\sum_i r_{\mathfrak{p}i}\mathcal{F}_{\mathfrak{p}i}$ and invariant k-linear atoms $\mathfrak{a}_i^{(k)}$, the bounds are

$$\sum_i|r_{\mathfrak{p}i}|e^{w_{J_m}^{\mathrm{path}}\,\mathrm{length}(\mathcal{F}_{\mathfrak{p}i})}\le A_{J_m}^{\mathrm{path}},\qquad\sum_i|\mathfrak{a}_i^{(k)}|e^{w_{J_m}^{\mathrm{path}}\,\mathrm{length}(\mathfrak{a}_i^{(k)})}\le\begin{cases}\Lambda_m^{\mathrm{loop}},&k=2,\\ A_{J_m}^{\mathrm{path}},&k=3,4.\end{cases}\tag{2.18}$$

Slot length is path length plus four; atom length is the sum of its slot lengths, and atom magnitude is the sum of absolute coefficients in the fixed colour basis. If $s_{\mathfrak{pq}}$ sums the row coefficients of orientation $\mathfrak{q}$, kinetic normalization means $c_{\mathrm{dir}}\delta_{\mathfrak{qr}} + \sum_{\mathfrak{p}} s_{\mathfrak{pq}} s_{\mathfrak{pr}} = \delta_{\mathfrak{qr}}$. For each ordered pair of orientations and colour components, the sum of quadratic-atom coefficients and its first moments in the difference of slot offsets vanish, with absolute convergence.

An admissibility datum consists of subsets $\mathcal{A}_{N,m,p}$ of these representations, initial subsets $\mathcal{I}_{N,p} \subset \mathcal{A}_{N,N,p}$ and relations $\mathcal{R}_{N,m,p} \subset \mathcal{A}_{N,m,p} \times \mathcal{A}_{N,m+1,p}$. A representation is admissible when it belongs to $\mathcal{A}_{N,m,p}$. A trajectory is a tuple $(\rho_m)_{0\le m\le N}$ with $\rho_N \in \mathcal{I}_{N,p}$ and $(\rho_m, \rho_{m+1}) \in \mathcal{R}_{N,m,p}$ for every $m < N$. Proposition 4.1 supplies these subsets and relations, with its continuation property on the chosen subsets; the activity bounds alone do not define membership.

Along a curve the index sets, supports, anchors and flag sets are fixed. Its β, v_{vac} are continuous; differences of regular derivatives and forced functions admit majorants whose sums with the support weights of (2.16) tend to zero as the parameter approaches any point. No continuity across flag boundaries is required.

Definition 2.10 (Renormalized insertions)**.** For each d choose a reference level $m_{\mathrm{ref}}(d)$ with $J_{m_{\mathrm{ref}}(d)} \ge d+3$ and $\mathfrak{q}_{J_m} \ge d+1$, $\varpi_{J_m} \ge d$ for every $m \ge m_{\mathrm{ref}}(d)$. Let $\mathsf{D}_t^{\mathrm{fac}}$ multiply a basis vector of bidegree (w,k) by t^k, counting polynomial factors. For $N \ge m_{\mathrm{ref}}(d)$, use the trajectory $\rho_{N,p,m}$ and couplings $\beta_{N,p,m}$ of Proposition 4.2. The marked densities, linear in terminal input $\mathbf{v} \in V_d$, are defined at the origin by

$$H^{\mathrm{ins}}_{N,p,N;d}(\mathbf{v};U) = \mathrm{dens}\,\rho_{N,p,N}(U)\, P^{\mathrm{loc}}_d[\mathsf{D}^{\mathrm{fac}}_{t_N^{-1}}\mathbf{v}](0;U), \tag{2.19}$$

$$H^{\mathrm{ins}}_{N,p,m;d}(\mathbf{v};\cdot) = \ell_m^d \mathcal{B}^{(m)}_{\beta_{N,p,m+1}} H^{\mathrm{ins}}_{N,p,m+1;d}(\mathbf{v};\cdot)\,. \tag{2.20}$$

The terminal factors a_N^d and a_N^{-d} cancel. The coefficient at level $m_{\mathrm{ref}}(d)$ is $\mathsf{Z}_{N,p,d}\mathbf{v}$, defining an invertible map $\mathsf{Z}_{N,p,d} : V_d \to V_d$. At $N < m_{\mathrm{ref}}(d)$ choose any invertible map. For every cutoff and site y set

$$O^{(d)}_{N,p}(\mathbf{c}, y; U) = a_N^{-d} P^{\mathrm{loc}}_d[\mathsf{D}^{\mathrm{fac}}_{t_N^{-1}} \mathsf{Z}^{-1}_{N,p,d}\mathbf{c}](y;U)\,. \tag{2.21}$$

The distinguished unit token $\mathbf{u}_d$ is $a^d_{m_{\mathrm{ref}}(d)}$ times the polynomial unit in coefficient coordinates; its insertion is 1 for $N \ge m_{\mathrm{ref}}(d)$. The quotient unit of Definition 2.12 is $[\mathbf{u}_d]$. The marked-density remainder norms are given in Section 11.

Definition 2.11 (Chessboard roots and differences)**.** For a tiling $\mathcal{C}^{\mathrm{tile}}$ of the fine torus into congruent cells, reflect a base-cell observable F_{cell} into every cell, applying conjugation on odd reflections, and denote these copies by $(F_{\mathrm{cell}})_C$. Its chessboard root is

$$\|F_{\mathrm{cell}}\|_{\mathcal{C}^{\mathrm{tile}}} = \left| \left\langle \prod_{C\in\mathcal{C}^{\mathrm{tile}}} (F_{\mathrm{cell}})_C \right\rangle_{N,p} \right|^{1/|\mathcal{C}^{\mathrm{tile}}|}. \tag{2.22}$$

Interior means dependence only on links with both ends strictly inside the base cell. The centered tiling $\mathcal{C}^{\mathrm{tile}}_m(y)$ has physical cell side a_m and base cell centered at y. For a list $\boldsymbol{l} = ((i_1,a_1),\dots,(i_k,a_k))$ of slot-axis pairs put $\Delta_{\boldsymbol{l}} = \prod_{\alpha=1}^k a_N^{-1}(\tau_{i_\alpha,a_\alpha} - 1)$, where $\tau_{i,a}$ shifts lattice slot i forward by one link. Let $\partial_{\boldsymbol{l}}$ be the corresponding continuum derivative. Physical torus distance is $\min_{z\in L_p\mathbb{Z}^4} |\hat{y} - \hat{y}' + z|_\infty$.

Euclidean fields and stress. The field quotient uses all spectators; shell fluxes normalize stress.

Definition 2.12 (Euclidean tables and field spaces)**.** A table S assigns a complex multilinear function of $\mathbf{c}_i \in V_{d_i}$ to each list of degrees and points. A word is a finite list $w = ((x_1, \mathbf{c}_1), \dots, (x_n, \mathbf{c}_n))$ with distinct positions, and $S(w) = S(\mathbf{c}_1(x_1), \dots, \mathbf{c}_n(x_n))$. Write ww' for concatenation, $hw = ((hx_i, h\mathbf{c}_i))_i$ for $h \in H_4$, and $w^* = ((x_n, \overline{\mathbf{c}_n}), \dots, (x_1, \overline{\mathbf{c}_1}))$. A word is positive in axis a if every $x_i^a > 0$. The Euclidean properties of S are:

E1. simultaneous permutation symmetry, $S_0 = 1$, and deletion of a distinguished unit slot $\mathbf{u}_d$.
E2. $S(h\mathbf{c}; hx) = S(\mathbf{c}; x)$ for $h \in H_4$, and $S(\overline{\mathbf{c}}; x) = \overline{S(\mathbf{c}; x)}$.
E3. $S(\mathbf{c}; x + z) = S(\mathbf{c}; x)$ for every $z \in \mathbb{R}^4$.
E4. for positive words w_i and coefficients a_i^{RP}, $\sum_{i,j} \overline{a_i^{\mathrm{RP}}} a_j^{\mathrm{RP}} S((\mathsf{s}_a^{\mathrm{refl}} w_i^*)\, w_j) \geq 0$ for each a.
E5. the same inequality holds for finite sums of words smeared against joint smooth compactly supported tests on positive-side separated configurations. Every cross-term integral is absolutely convergent.
E6. every kernel is smooth on Conf_n.
E7. for every $n, d_i, \mathbf{c}_i$ and positional derivative order k, there are $C \geq 0$ and an integer $q_{\mathrm{grow}} \geq 0$ such that $\|D^k S(\mathbf{c}; x)\| \leq C r^{-q_{\mathrm{grow}}}$ whenever $0 < r \leq 1$ and $\mathrm{sep}_\infty(x) \geq r$.

For such a table, with $n \geq 0$ and $\boldsymbol{v}$ any list of n spectator tokens, define

$$\begin{aligned} \mathcal{N}_d(S) &= \{\mathbf{c} \in V_d : S(\mathbf{c}(x_0), \boldsymbol{v}(x)) = 0 \text{ for every } n, \boldsymbol{v} \text{ and } (x_0, x) \in \mathrm{Conf}_{n+1}\}, \\ F_d &= V_d / \mathcal{N}_d(S)\,. \end{aligned} \tag{2.23}$$

Equip each finite-dimensional quotient with a norm. The response of $[\mathbf{c}]$ is its function of all spectators and separated configurations in (2.23); their linear span is the response label space $\mathcal{L}$. Inclusions $\jmath_{d,e}$, derivatives ∂_a, conjugation and units are characterized by equality, differentiation, conjugation and deletion of slots in all correlations. Proposition 4.6 proves their existence and the directed-union property of $\mathcal{L}$. The quotient unit is $\mathbf{1}_d = [\mathbf{u}_d]$.

Definition 2.13 (Free Maxwell fields and frames)**.** Let $d_{\mathfrak{g}}$ independent real Euclidean Maxwell curvatures have potential covariance $\delta_{\alpha\beta}\delta_{ac}\Gamma(x-y)$, $\Gamma(x) = 1/(4\pi^2|x|^2)$. Their curvature covariance is obtained by applying $\partial_a \mathcal{A}_c - \partial_c \mathcal{A}_a$ in each argument. The on-shell curvature jets satisfy antisymmetry, the linear Bianchi equations and $\sum_a \partial_a F_{ac} = 0$. Let M_d^{Max} be the space of $\mathrm{Ad}(G)$-invariant complex Wick polynomials in these jets of weight at most d, modulo these equations; S^{Max} is their separated Wick table. This defines its units, conjugation, inclusions, derivatives and $O(4)$ actions R_d^{Max}. Put $\mathcal{O}_{F^2} = \frac{1}{4}\sum_{a,c,\alpha} :F_{ac}^\alpha F_{ac}^\alpha:$. Let $T_{ab}^{\mathrm{Max}} \in M_4^{\mathrm{Max}}$ be the multiple of $T_{ab}^{\mathrm{can}} = \sum_{c,\alpha} :F_{ac}^\alpha F_{bc}^\alpha : - \delta_{ab}\mathcal{O}_{F^2}$ fixed by the translation shell law of Definition 2.15 with $\mathsf{p}_b = \partial_b$.

A scaling sequence consists of positive radii $r_k \to 0$ and complex linear isomorphisms $\mathsf{U}_{k,d}^{\mathrm{sc}} : M_d^{\mathrm{Max}} \to F_d$. It is stable if, at each fixed degree horizon, eventually these maps preserve conjugation, H_4, units and inclusions exactly. Scaling dimensions are included in $\mathsf{U}_{k,d}^{\mathrm{sc}}$; an additional factor r_k^d is not put on ordinary comparison slots.

Definition 2.14 (Words and slab operators)**.** For a table S with E1–E4, quotient the complex span of positive words by the null space of $\langle w, w' \rangle = S((\mathsf{s}_a^{\mathrm{refl}} w^*)\, w')$ and complete, as in [3, Section 4.1, p. 91]. This is $\mathcal{H}_a^{\mathrm{OS}}$. The empty word is Ω_a^{OS}. Positive translation in direction a induces the contraction semigroup $e^{-t_{\mathrm{heat}} H_a^{\mathrm{OS}}}$; put $\mathcal{H}_a^0 = \ker H_a^{\mathrm{OS}}$ and let P_a^0 be its orthogonal projection.

A slab word of width $h_{\mathrm{sl}} > 0$ has $0 < x_i^4 < h_{\mathrm{sl}}$. It has buffer r if $\mathrm{sep}_\infty(x) \geq r$ and $r \leq x_i^4 \leq h_{\mathrm{sl}} - r$. For such a word w, let $A_w w' = w\, \tau_{h_{\mathrm{sl}} e_4} w'$ on positive words, where τ_z translates every position by z, and let $w^\dagger = \tau_{h_{\mathrm{sl}} e_4} \mathsf{s}_4^{\mathrm{refl}} w^*$. The repeated reflected word with exterior w_{ext} is $(A_{w^\dagger} A_w)^{n_{\mathrm{rep}}} w_{\mathrm{ext}}$. A slab operator is the bounded extension $\mathsf{B}_{h_{\mathrm{sl}}}(w)$ of A_w, when it exists. Its compression is $M^0(w) = P_4^0 \mathsf{B}_{h_{\mathrm{sl}}}(w)|_{\mathcal{H}_4^0}$, after any translation placing the word in a slab. Proposition 4.11 proves existence and independence.

Definition 2.15 (Shell fluxes and stress laws)**.** For separated $x_0, x_1, \dots, x_n$, let $\chi \in C_c^\infty(\mathbb{R}^4;\mathbb{R})$ equal 1 near x_0, with $x_i \notin \operatorname{supp}\chi$ for $i>0$. For a tensor $J_{ab} \in F_4$, $A \in F_d$, spectator list $\boldsymbol{B}$, and $K_{\mathrm{rot}} \in \mathfrak{so}(4)$, define

$$\mathcal{P}_b^J(\chi; A, \boldsymbol{B}; x) = -\sum_a \int \partial_a\chi(y) S(J_{ab}(y), A(x_0), \boldsymbol{B}(x))\, dy, \tag{2.24}$$

$$\mathcal{Y}_{K_{\mathrm{rot}}}^J(\chi; A, \boldsymbol{B}; x) = -\sum_{a,b} \int \partial_a\chi(y)(K_{\mathrm{rot}}(y-x_0))_b S(J_{ab}(y), A(x_0), \boldsymbol{B}(x))\, dy\,. \tag{2.25}$$

A conserved tensor is real, symmetric and satisfies $\sum_a \partial_a J_{ab} = 0$ in F_5. It is centered when $S(J_{ab}(x)) = 0$ for all a, b, x. It is H_4 tensorial when $hJ_{ab} = \sum_{i,j} h_{ia}h_{jb}J_{ij}$. Raw shell laws for $\mathsf{p}_b, \mathsf{Y}_{K_{\mathrm{rot}}}$ mean $\mathcal{P}_b^J = S((\mathsf{p}_b A)(x_0), \boldsymbol{B}(x))$ and $\mathcal{Y}_{K_{\mathrm{rot}}}^J = S((\mathsf{Y}_{K_{\mathrm{rot}}} A)(x_0), \boldsymbol{B}(x))$ for every such choice. For an $O(4)$ representation R_d, write $\mathsf{L}_d(K_{\mathrm{rot}}) = \frac{d}{dt} R_d(e^{tK_{\mathrm{rot}}})\big|_{t=0}$. The normalized stress laws use $\mathsf{p}_b = \partial_b$ and $\mathsf{Y}_{K_{\mathrm{rot}}} = \mathsf{L}_d(K_{\mathrm{rot}})$. These integrals avoid all insertion points.

Relativistic reconstruction. Phase correlations determine relativistic fields.

Definition 2.16 (Relativistic data)**.** A relativistic datum is a complex Hilbert space $\mathcal{H}$, a complex vector space $\mathcal{L}$ (the response space of Definition 2.12 in Theorem B), a complex linear subspace $\mathcal{D} \subset \mathcal{H}$, $\Omega \in \mathcal{D}$, fields $\Phi(\ell_{\mathrm{lab}}, f) : \mathcal{D} \to \mathcal{D}$ complex linear in both $\ell_{\mathrm{lab}} \in \mathcal{L}$ and $f \in \mathcal{S}$, an antilinear label map $\ell_{\mathrm{lab}} \mapsto \ell_{\mathrm{lab}}^*$, and strongly continuous unitary translations $\mathsf{T}(z_{\mathrm{tr}})$. It includes a Borel projection-valued measure E on $\mathbb{R}^4$, extended by zero to non-Borel subsets, with diagonal finite measures $\mu_\psi(A) = \langle \psi, \mathsf{E}(A)\psi\rangle$ for Borel A.

It also includes a topological group $\mathcal{P}$ identified by a homeomorphism κ_{P} with $SO^+(1,3) \ltimes \mathbb{R}^4$, respecting identity and multiplication; write $\kappa_{\mathrm{P}}(\gamma) = (\Lambda_\gamma, z_\gamma)$. Here Λ preserves m, has determinant 1, and maps $\overline{V}_+$ onto itself. There are strongly continuous unitaries $\mathsf{U}(\gamma)$ forming a representation, with $\mathsf{U}(1, z_{\mathrm{tr}}) = \mathsf{T}(z_{\mathrm{tr}})$, and linear bijections $\rho_{\mathrm{lab}}(\gamma)$ of $\mathcal{L}$. Put $(\gamma \cdot f)(x) = f(\Lambda_\gamma^{-1}(x - z_\gamma))$. No continuity, finite-dimensionality, translation triviality or involution property is imposed on the label data in this definition.

Only Borel sets enter the spectral axioms below; assigning zero elsewhere fixes a convention for the same Borel projection-valued measure.

Definition 2.17 (Wightman axioms)**.** The datum in Definition 2.16 satisfies the following axioms.

W0a. $\langle \psi, \mathsf{T}(z_{\mathrm{tr}})\psi\rangle = \int e^{i\mathbf{p}\cdot z_{\mathrm{tr}}}\, d\mu_\psi(\mathbf{p})$ for every ψ, z_{tr}.

W0b. $\mathsf{E}(\mathbb{R}^4 \setminus \overline{V}_+) = 0$.

W0c. every vector fixed by all translations belongs to $\mathbb{C}\Omega$.

W1a–c. $\|\Omega\| = 1$, $\mathcal{D}$ is dense, and $\mathcal{D}$ equals the complex span of all $\Phi(\ell_{\mathrm{lab},1}, f_1)\cdots\Phi(\ell_{\mathrm{lab},k}, f_k)\Omega$, including $k = 0$.

W1d. for each ℓ_{lab} and $\psi \in \mathcal{D}$, $f \mapsto \Phi(\ell_{\mathrm{lab}}, f)\psi$ is continuous from $\mathcal{S}$ to $\mathcal{H}$ in norm.

W1e. $\langle \Phi(\ell_{\mathrm{lab}}, f)\psi, \psi'\rangle = \langle \psi, \Phi(\ell_{\mathrm{lab}}^*, \bar{f})\psi'\rangle$ on $\mathcal{D}$.

W2a. $\rho_{\mathrm{lab}}(1) = 1$ and $\rho_{\mathrm{lab}}(\gamma\gamma') = \rho_{\mathrm{lab}}(\gamma)\rho_{\mathrm{lab}}(\gamma')$.

W2b–d. $\mathsf{U}(\gamma)\mathcal{D} \subset \mathcal{D}$, $\mathsf{U}(\gamma)\Omega = \Omega$, and $\mathsf{U}(\gamma)\Phi(\ell_{\mathrm{lab}}, f)\psi = \Phi(\rho_{\mathrm{lab}}(\gamma)\ell_{\mathrm{lab}}, \gamma\cdot f)\mathsf{U}(\gamma)\psi$.

W3. if every $x \in \operatorname{supp} f$ and $y \in \operatorname{supp} f'$ have $\mathsf{m}(x-y) < 0$, then $[\Phi(\ell_{\mathrm{lab}}, f), \Phi(\ell'_{\mathrm{lab}}, f')]\psi = 0$ for every $\ell_{\mathrm{lab}}, \ell'_{\mathrm{lab}}$ and $\psi \in \mathcal{D}$.

Definition 2.18 (Hierarchy data)**.** A hierarchy over a Wilson family is the following data, with no further conditions implicit:

H1. a relativistic datum as in Definition 2.16;
H2. finite-dimensional complex normed spaces F_d and linear maps $\iota_d : F_d \to \mathcal{L}$; put $\mathbb{F} = \bigsqcup_{d\ge 0} F_d$, $\deg(d, A) = d$, $\iota(d, A) = \iota_d A$;
H3. a map $\pi_{\mathrm{quo}} : \mathcal{T} \to \mathbb{F}$;
H4. a compact metric space X, a Borel measure ν and a point $\xi_* \in \mathsf{X}$;
H5. functions $S_n^\zeta : \mathbb{F}^n \times (\mathbb{R}^4)^n \to \mathbb{C}$ for $\zeta \in \mathsf{X} \sqcup \{\mathrm{mix}\}$;
H6. functions $\mathsf{F}_n : \mathbb{F}^n \times \mathbb{C}^n \times (\mathbb{R}^3)^n \to \mathbb{C}$;
H7. finite sets Ch_q, fields $O_{q,\lambda} \in F_q$ and functions $c_{q,\lambda}(\boldsymbol{A}; x, o)$ for every finite index set I_{cl}, $\boldsymbol{A} \in \mathbb{F}^{I_{\mathrm{cl}}}$, $x \in (\mathbb{R}^4)^{I_{\mathrm{cl}}}$ and $o \in \mathbb{R}^4$;
H8. a field $\Theta \in F_4$.

Definition 2.19 (Reconstruction properties)**.** A hierarchy satisfies the reconstruction properties when all the following hold:

R1. Its relativistic datum satisfies W0–W3.
R2. For every nonzero $\ell_{\mathrm{lab}} \in \mathcal{L}$, $\Phi(\ell_{\mathrm{lab}}, f) \neq 0$ for some $f \in \mathcal{S}$.
R3. Each ι_d is injective.
R4. $\Phi(\iota(A), f) \neq 0$ for some $A \in \mathbb{F}$, $f \in \mathcal{S}$.
R5. Every $\ell_{\mathrm{lab}} \in \mathcal{L}$ equals $\iota_d A$ for some d, A.
R6. π_{quo} is surjective onto $\mathbb{F}$.
R7. $\nu(\mathsf{X}) = 1$.
R8. $\nu(O) > 0$ for every nonempty open $O \subset \mathsf{X}$.
R9. For every state ζ, simultaneous permutation of field and position slots preserves S_n^ζ on Conf_n.
R10. For every state ζ and field list $\boldsymbol{A}$, $S_n^\zeta(\boldsymbol{A}; \cdot)$ is smooth on Conf_n.
R11. For every fixed n, $\boldsymbol{A} \in \mathbb{F}^n$ and $x \in \mathrm{Conf}_n$, $\xi \mapsto S_n^\xi(\boldsymbol{A}; x)$ is continuous.
R12. $\int_{\mathsf{X}} S_n^\xi(\boldsymbol{A}; x)\, d\nu(\xi) = S_n^{\mathrm{mix}}(\boldsymbol{A}; x)$ for all n, $\boldsymbol{A} \in \mathbb{F}^n$ and $x \in \mathrm{Conf}_n$.
R13. At each fixed real spatial configuration, $\boldsymbol{z}_{\mathrm{time}} \mapsto \mathsf{F}_n(\boldsymbol{A}; \boldsymbol{z}_{\mathrm{time}}, \vec{x})$ is holomorphic on $\mathcal{Z}_n = \{\boldsymbol{z}_{\mathrm{time}} \in \mathbb{C}^n : \mathrm{Im}(\boldsymbol{z}_{\mathrm{time}})_1 < \cdots < \mathrm{Im}(\boldsymbol{z}_{\mathrm{time}})_n\}$.
R14. If $t(x_1) < \cdots < t(x_n)$, then $\mathsf{F}_n(\boldsymbol{A}; (it(x_i))_i, (\vec{x}_i)_i) = S_n^{\xi_*}(\boldsymbol{A}; x)$.
R15. For every n, $\boldsymbol{A} \in \mathbb{F}^n$, real $h_1 < \cdots < h_n$ and $f_i \in \mathcal{S}$, the limit below holds.

The limit in R15 is

$$\lim_{r\downarrow 0} \int \mathsf{F}_n(\boldsymbol{A}; (t(x_i) + irh_i)_i, (\vec{x}_i)_i) \prod_i f_i(x_i)\, dx = \langle \Omega, \Phi(\iota(A_1), f_1) \cdots \Phi(\iota(A_n), f_n)\Omega \rangle\,. \tag{2.26}$$

For this clause alone, use the convention that the integral of a nonintegrable function is zero; no separate integrability assertion at positive height is included.

Products and comparison fields. The OPE replaces a cluster by single fields, independently of spectators and state.

Definition 2.20 (OPE geometry and remainders)**.** For finite $I_{\mathrm{cl}}, I_{\mathrm{sp}}$, $\boldsymbol{A} \in \mathbb{F}^{I_{\mathrm{cl}}}$, $\boldsymbol{B} \in \mathbb{F}^{I_{\mathrm{sp}}}$, write $|\boldsymbol{A}|_{\deg} = \sum_{i \in I_{\mathrm{cl}}} \deg A_i$ and concatenate the two lists in a fixed order in $S^\zeta(\boldsymbol{A}, x; \boldsymbol{B}, y)$. For $r > 0$, $M_{\mathrm{geo}} \ge 0$, $\eta > 0$, a cluster (x, o) satisfies $|x_i - o| \le M_{\mathrm{geo}} r$ and $|x_i - x_{i'}| \ge \eta r$ for $i \neq i'$. A spectator configuration is δ_{sp}-clear if $|o - y_j| \ge \delta_{\mathrm{sp}}$ and $|y_j - y_{j'}| \ge \delta_{\mathrm{sp}}$ for $j \neq j'$. Define

$$\mathcal{R}_D^\zeta(\boldsymbol{A}, x; \boldsymbol{B}, y; o) = S^\zeta(\boldsymbol{A}, x; \boldsymbol{B}, y) - \sum_{q=0}^{D} \sum_{\lambda \in \mathrm{Ch}_q} c_{q,\lambda}(\boldsymbol{A}; x, o) S^\zeta(O_{q,\lambda}(o), \boldsymbol{B}(y))\,. \tag{2.27}$$

The OPE property is precisely:

O1. for every $I_{\mathrm{cl}}, \boldsymbol{A}, M_{\mathrm{geo}}, \eta, \varepsilon, q, \lambda$ with $M_{\mathrm{geo}} \geq 0$, $\eta, \varepsilon > 0$, there is $r_0 > 0$ such that $|c_{q,\lambda}(\boldsymbol{A}; x, o)| \leq r^{q-|\boldsymbol{A}|_{\deg}-\varepsilon}$ for every $0 < r < r_0$ and every such cluster;
O2. for every $I_{\mathrm{cl}}, I_{\mathrm{sp}}, \boldsymbol{A}, \boldsymbol{B}, M_{\mathrm{geo}}, \eta, \delta_{\mathrm{sp}}, \varepsilon, D$ with $M_{\mathrm{geo}} \geq 0$, $\eta, \delta_{\mathrm{sp}}, \varepsilon > 0$, there is $r_1 > 0$ such that $|\mathcal{R}_D^{\zeta}| \leq r^{D+1-|\boldsymbol{A}|_{\deg}-\varepsilon}$ for all $0 < r < r_1$, every such cluster, every δ_{sp}-clear spectator configuration and every state ζ.

The coefficients in H7 do not depend on spectators, state or truncation.

Definition 2.21 (Trace test and finite Wick profiles)**.** A mode is $(m_{\mathrm{sh}}, j_h, k_E, h_{\mathrm{mom}})$ with $m_{\mathrm{sh}} \geq 0$, $j_h, k_E \in \mathbb{N}$ and continuous real h_{mom} on $\mathbb{R}^3$ satisfying $h_{\mathrm{mom}}(\lambda_{\mathrm{dil}}\mathbf{k}) = \lambda_{\mathrm{dil}}^{j_h} h_{\mathrm{mom}}(\mathbf{k})$ for $\lambda_{\mathrm{dil}} > 0$. Put

$$K_{m_{\mathrm{sh}},k_E,h_{\mathrm{mom}}}(r) = \int_{\mathbb{R}^3} \frac{\omega_{m_{\mathrm{sh}}}(\mathbf{k})^{k_E} h_{\mathrm{mom}}(\mathbf{k})}{2\omega_{m_{\mathrm{sh}}}(\mathbf{k})} e^{-r\omega_{m_{\mathrm{sh}}}(\mathbf{k})}\, d\mathbf{k}, \qquad \omega_{m_{\mathrm{sh}}}(\mathbf{k}) = \sqrt{m_{\mathrm{sh}}^2 + |\mathbf{k}|^2}\,. \tag{2.28}$$

A finite Wick profile is any finite complex linear combination of finite products of these kernels. A mass-shell integrand is assigned value zero at $\mathbf{k} = 0$ if its denominator vanishes; this changes no integral. Define $G_\Theta(r) = S_2^{\xi*}(\Theta, \Theta; -re_4/2, re_4/2)$. The trace test is:
T1. $\langle \Omega, \Phi(\iota_4\Theta, f)\Omega\rangle = 0$ for every $f \in \mathcal{S}$;
T2. $A_G = 3d_{\mathfrak{g}}/\pi^4 > 0$;
T3. $r^8 \log^2(1/r) G_\Theta(r) \to A_G$ as $r \downarrow 0$, in $\mathbb{C}$;
T4. for every finite Wick profile W_{Wick} and every $\delta > 0$, some $0 < r < \delta$ has $G_\Theta(r) \neq W_{\mathrm{Wick}}(r)$.

Definition 2.22 (Polynomial mass-shell comparisons)**.** For $m_{\mathrm{sh}} \geq 0$ and a complex polynomial P_{mom} on $\mathbb{R}^4$, define the elementary momentum functional and its density by

$$L_{m_{\mathrm{sh}},P_{\mathrm{mom}}}(\varphi) = \int_{\mathbb{R}^3} v_{m_{\mathrm{sh}},P_{\mathrm{mom}}}(\mathbf{k})\varphi(\mathbf{k}, \omega_{m_{\mathrm{sh}}}(\mathbf{k}))\, d\mathbf{k}, \qquad v_{m_{\mathrm{sh}},P_{\mathrm{mom}}}(\mathbf{k}) = \frac{P_{\mathrm{mom}}(\mathbf{k}, \omega_{m_{\mathrm{sh}}}(\mathbf{k}))}{2\omega_{m_{\mathrm{sh}}}(\mathbf{k})}\,. \tag{2.29}$$

Finite polynomial comparison data on a set of generator labels assign a finite list $\mathcal{M}_{\mathrm{pair}}(\mathsf{x}, \mathsf{y})$ of mass–polynomial pairs to each ordered pair of labels, and a complex number $c_{\mathrm{pair}}(u, u', \sigma)$ to each pair of words $u = (\mathsf{x}_1, \dots, \mathsf{x}_{n_{\mathrm{word}}})$, $u' = (\mathsf{y}_1, \dots, \mathsf{y}_{n_{\mathrm{word}}})$ and each permutation σ. Put

$$\Pi^{\mathrm{pair}}(u, u'; \varphi) = \sum_{\sigma} c_{\mathrm{pair}}(u, u', \sigma) \sum_{\substack{(m_i,P_i)\in\mathcal{M}_{\mathrm{pair}}(\mathsf{x}_i,\mathsf{y}_{\sigma i}) \\ 1\leq i\leq n_{\mathrm{word}}}} \int_{(\mathbb{R}^3)^{n_{\mathrm{word}}}} \prod_i v_{m_i,P_i}(\mathbf{k}_i) \cdot \varphi\Big(\sum_i (\mathbf{k}_i, \omega_{m_i}(\mathbf{k}_i))\Big)\, d\boldsymbol{k}\,. \tag{2.30}$$

For words of different lengths put $\Pi^{\mathrm{pair}} = 0$. A polynomial field Q is a finite list of words u_i with complex amplitudes α_i^{pol}. Empty words are allowed. For two polynomial fields put

$$W^{\mathrm{pol}}(\mathsf{Q}, \mathsf{Q}'; \varphi) = \sum_{i,j} \alpha_i^{\mathrm{pol}} \alpha_j'^{\mathrm{pol}} \Pi^{\mathrm{pair}}(u_i, u_j'; \varphi)\,. \tag{2.31}$$

With $\widehat{f}_+(\mathbf{p}) = \int e^{i\mathbf{p}\cdot x} f(x)\, dx$, the relative transform is

$$(\mathcal{W}_{\mathrm{rel}}\psi)(\mathbf{p}) = \int e^{i\mathbf{p}\cdot(y-x)} \psi(x, y)\, dx\, dy, \qquad \mathcal{W}_{\mathrm{rel}}(\bar{f} \otimes h) = \overline{\widehat{f}_+}\, \widehat{h}_+\,. \tag{2.32}$$

Definition 2.23 (Maxwell Fock comparisons)**.** Put $d\sigma(\mathbf{k}) = d^3\mathbf{k}/(2|\mathbf{k}|)$ and $\mathsf{P}^{\perp}_{ij}(\mathbf{k}) = \delta_{ij} - \mathbf{k}_i\mathbf{k}_j/|\mathbf{k}|^2$, assigning the density and projector value zero at 0. For n_{col} colours put

$$\mathcal{K}_{n_{\mathrm{col}}} = L^2(\mathbb{R}^3, d\sigma; \mathbb{C}^{n_{\mathrm{col}}} \otimes \mathbb{C}^3), \qquad \mathcal{F}_{n_{\mathrm{col}}} = \bigoplus_{k\geq 0} \mathcal{K}_{n_{\mathrm{col}}}^{\otimes k}. \tag{2.33}$$

Let $\mathcal{F}^{\mathrm{phys}}_{n_{\mathrm{col}}}$ be the symmetric subspace transverse in every momentum slot. Translations multiply a k-particle wave by $e^{iz_{\mathrm{tr}}\cdot\sum_j(\mathbf{k}_j,|\mathbf{k}_j|)}$. The vacuum Ω_0 lies in sector zero. For a list $\boldsymbol{h} = (h_{\alpha,i})$ of Schwartz tests put

$$v_{\boldsymbol{h}}(\alpha, i; \mathbf{k}) = |\mathbf{k}| \sum_j \mathsf{P}^{\perp}_{ij}(\mathbf{k}) \widehat{h}_{\alpha,j,+}(\mathbf{k}, |\mathbf{k}|), \qquad \mathcal{E}(\boldsymbol{h}) = e^{-\|v_{\boldsymbol{h}}\|^2/2} \bigoplus_{k\geq 0} \frac{v_{\boldsymbol{h}}^{\otimes k}}{\sqrt{k!}}. \tag{2.34}$$

A Fock comparison is a complex linear isometry $\mathsf{I}_{\mathrm{F}} : \mathcal{H} \to \mathcal{F}_{n_{\mathrm{col}}}$ with image in $\mathcal{F}^{\mathrm{phys}}_{n_{\mathrm{col}}}$, $\mathsf{I}_{\mathrm{F}}\Omega = \Omega_0$, and intertwining translations. Write $\psi_{\mathsf{I}_{\mathrm{F}}}(f) = \mathsf{I}_{\mathrm{F}}\Phi(\iota_4\Theta, f)\Omega$. For the averaged comparison, a probability space (Z, λ_Z) and maps $P^{\mathrm{av}}_z : \mathcal{F}_{n_{\mathrm{col}}} \to \mathcal{F}_{n_{\mathrm{col}}}$ preserve inner products on pairs of physical vectors and fix $\mathsf{I}_{\mathrm{F}}\mathcal{H}$ pointwise. Linearity of P^{av}_z is not required. The scalar integrals use the convention of Definition 2.19.

Nested products and contacts. Successive OPE expansions use relative scales.

Definition 2.24 (Nested clusters and their rates)**.** Let a finite rooted tree have leaves carrying fields A_i and an internal root o_{root}. Assign a position z_v to every vertex and a positive scale ϱ_v to every internal vertex, with $\varrho_{o_{\mathrm{root}}} = r$ and $\varrho_v \leq \varrho_{\mathrm{par}(v)}$. At each internal v, every descendant position lies within $M_{\mathrm{geo}}\varrho_v$ of z_v, and positions belonging to different immediate child subtrees have distance at least $\eta\varrho_v$. These conditions apply recursively. Spectator clearance δ_{sp} means distance at least δ_{sp} from every tree position and between distinct spectators. Put $W_v = \sum_{i \text{ leaf below } v} \deg A_i$ and $q_v = \varrho_v/\varrho_{\mathrm{par}(v)}$. For truncations $\boldsymbol{D} = (D_v)$ define

$$\mathcal{Q}_{\varepsilon,q}(\varrho) = r^{q - W_{o_{\mathrm{root}}} - \varepsilon} \prod_{\substack{v \neq o_{\mathrm{root}} \\ v \text{ internal}}} q_v^{-W_v - \varepsilon}, \qquad \mathsf{e}_{\boldsymbol{D}}(\varrho) = \sum_{\substack{v \neq o_{\mathrm{root}} \\ v \text{ internal}}} q_v^{D_v+1}. \tag{2.35}$$

The direct coefficient $C^{\mathrm{dir}}_{q,\lambda}$ is $c_{q,\lambda}$ of all original leaves at $z_{o_{\mathrm{root}}}$. Define $C^{\mathrm{it},\boldsymbol{D}}_{q,\lambda}$ recursively. A leaf i carries $(A_i, 1)$. A proper internal vertex v with children $c_1, \ldots, c_s$ carries $(O_{q',\lambda'}, \kappa_v(q', \lambda'))$ for $q' \leq D_v$, $\lambda' \in \mathrm{Ch}_{q'}$, where

$$\kappa_v(q', \lambda') = \sum_{(B_j,\kappa_j)_j} \Big(\prod_{j=1}^{s} \kappa_j\Big) c_{q',\lambda'}(B_1, \ldots, B_s; z_{c_1}, \ldots, z_{c_s}, z_v). \tag{2.36}$$

The sum chooses one pair (B_j, κ_j) from each child. The same formula at the root gives $C^{\mathrm{it},\boldsymbol{D}}_{q,\lambda} = \kappa_{o_{\mathrm{root}}}(q, \lambda)$, without using the root truncation. A tested root remainder in state ζ is $\mathcal{R}^{\zeta}_q$ for the original leaves at $z_{o_{\mathrm{root}}}$.

Definition 2.25 (Contact distributions)**.** A distribution on $\mathbb{R}^{4n}$ is a complex linear continuous functional on $C^\infty_c(\mathbb{R}^{4n})$. Locally finite order means that each compact K admits $q_K \in \mathbb{N}$, $C_K < \infty$ with

$$|\mathfrak{D}(f)| \leq C_K \max_{|\alpha|\leq q_K} \|\partial^\alpha f\|_\infty \tag{2.37}$$

for tests supported in K. A contact extension $\mathfrak{D}^{\zeta}_{\boldsymbol{A}}$ of $S^{\zeta}(\boldsymbol{A};\cdot)$ is such a distribution equal to $\int f S^{\zeta}(\boldsymbol{A};\cdot)$ on separated tests. The subtraction prescription is defined in Section 16, using the field projections of the OPE.

Continuity gives locally finite order for every distribution; Proposition 4.20 records the dependence of q_K, C_K.

3 The main theorem

One family of local Wilson observables gives continuum correlations after the full ultraviolet limit at fixed volume and a refinement of any prescribed increasing volume sequence. The limiting mixture has a phase decomposition, a selected phase has a relativistic reconstruction, and one system of OPE coefficients works in every state. A degree-four field distinguishes the selected theory from every finite Wick profile of Definition 2.21.

Theorem A (Existence)**.** *For every gauge group of Definition 2.1 with simple nonabelian Lie algebra, there exists a regular Wilson family $\mathcal{W}$ such that, for every strictly increasing $\mathsf{p} : \mathbb{N} \to \mathbb{N}$, there exist a strictly increasing $\mathsf{q} : \mathbb{N} \to \mathbb{N}$ and hierarchy data H1–H8 over $\mathcal{W}$ with the following properties, for $\mathsf{v} = \mathsf{p} \circ \mathsf{q}$:*

A-P1. *$\mathcal{W}$ converges in the sense of Definition 2.5 along v to*

$$K_n(v;x) = S_n^{\mathrm{mix}}(\pi_{\mathrm{quo}}(v_1),\dots,\pi_{\mathrm{quo}}(v_n);x)\,. \tag{3.1}$$

A-P2. *all R1–R15 of Definition 2.19 hold.*

A-P3. *O1 and O2 of Definition 2.20 hold for the single system H7.*

A-P4. *T1–T4 of Definition 2.21 hold for the field H8.*

Remark 3.1. All lattice data precede the quantifier over p; the refinement and every hierarchy datum follow it. Theorem A has no additional hypotheses or conclusions implicit in the words "hierarchy", "trace" or "existence". In particular, it asserts neither a stress tensor nor an identification of polynomial field content. The integral convention in Definition 2.19 is part of its formulation. The relativistic theory belongs to ξ_*, whereas the lattice limit is S^{mix}. The phase space need not be a point, distinct phases need not agree, and an individual phase need not arise as a limit of the lattice states. No mass gap is asserted.

Remark 3.2. The volume refinement is common to every arity, label list and separated test. The ultraviolet limit uses the full sequence at every admissible fixed volume, before the volume tends to infinity. Uniqueness of the infinite-volume limit and independence of the chosen refinement are not asserted; neither an interchange of the limits nor convergence along a diagonal is claimed. If the ultraviolet index is restricted to a prescribed subsequence, the same refinement and hierarchy satisfy A-P1, since each full fixed-volume limit is also the limit along that subsequence.

The construction also identifies the local field spaces, supplies a conserved stress tensor, reconstructs a theory in every phase, and gives nested products and contact distributions. The next theorem states these conclusions for every positive block-spin penalty; the principal propositions below give their quantitative content.

Theorem B (The construction)**.** *For every gauge group of Definition 2.1 with simple nonabelian Lie algebra and every real $\mathsf{b} > 0$, there exist admissible constants, level data, local sections, reference levels and one regular Wilson family with the following properties.*

B1. **Lattice structure.** *The block-spin transformation is Definition 2.8 with penalty* b*. The spacings and volumes satisfy Proposition 4.1 and Proposition 4.2; the tokens are* $(d, \mathbf{c})$*,* $\mathbf{c} \in V_d$*, and their insertions are Definition 2.10. Local sections have the Taylor normalization of Definition 2.6, and all marked transfers have the reference normalization of Definition 2.10. The bare couplings belong to one specified recursion of represented densities reaching* $\bar{\beta}$ *at level zero, with the exact equations and bounds of Proposition 4.2. In particular,*

$$\lim_{m\to\infty} \sup_{N\geq m,\ p} \left| \frac{\beta_{N,p,m}}{s_m} - \kappa_G \right| = 0\,. \tag{3.2}$$

B2. **Fixed volume.** *Proposition 4.5 holds along the full ultraviolet sequence at every volume, for every token list and every finite positional derivative order.*

For every strictly increasing volume sequence p*, there are a strictly increasing* q*, a mixed table* S *and one hierarchy over this same family, with all properties A-P1–A-P4 along* $\mathsf{p} \circ \mathsf{q}$*, and with the following further properties. All objects in B3–B6 refer to the table* S*, the conserved tensor* T_{ab} *of Proposition 4.10, the phase decomposition of Proposition 4.11 and the selected theory.*

B3. **Euclidean fields and stress.** *The fields are* $F_d = V_d/\mathcal{N}_d(S)$ *with the inclusions, derivatives and response realization in Proposition 4.6, and* $\dim F_d = \dim M_d^{\mathrm{Max}}$*. The mixed table and every phase obey E1–E7 and the* $O(4)$ *covariance of Proposition 4.10. For its conserved tensor* $T_{ab} \in F_4$ *they satisfy the normalized stress laws of Definition 2.15 (Propositions 4.10 and 4.11). The field* $\Theta = \tau - S^{\xi*}(\tau)\mathbf{1}_4$*,* $\tau = \sum_a T_{aa}$*, is the selected-phase centered trace of* T*. The comparison in Proposition 4.10(e), including all derivatives, arbitrary stress slots, transported derivatives and rotation generators, holds along one common scaling sequence. Its ordinary field frames are eventually exactly compatible with the structures specified in Proposition 4.8.*

B4. **Every phase.** *The theories, continuations, surjective OS comparison maps, one-dimensional zero-energy spaces, trivial bounded weak commutants, phase-uniform low-energy and heat bounds, continued stress tensors, continuity and joint-test averages of Proposition 4.13 hold simultaneously for every* $\xi \in \mathsf{X}$*.*

B5. **Trace and non-freeness.** *Proposition 4.15 holds, including the operator identification, the full-radius limit and the unique positive spectral measure with its compact-test and Schwartz-test high-energy limit. All three comparison conclusions of Proposition 4.16 hold for the selected theory, including its finite-profile conclusion and its localized averaged Fock countertest.*

B6. **Products and contacts.** *There is one OPE prescription with all separated, integrated and nested estimates of Proposition 4.18, with the rates in Definition 2.24. With exactly the same field projections there are the real bases, subtraction scales and contact distributions of Proposition 4.20.*

Remark 3.3. The gauge group and penalty are fixed first. The constants, level data, local sections, reference levels, represented trajectories and Wilson family are one choice satisfying Propositions 4.1 and 4.2, made in the order of Section 6. The assertion does not quantify over every choice satisfying the numerical inequalities. Only afterwards is p prescribed; one refinement serves all later conclusions. Setting $\mathsf{b} = 1$ and retaining A-P1–A-P4 gives Theorem A. Theorem B is a weakening of the construction: the degree-four position normalization of the trace and the contact estimates on general collision subspaces are proved in Part II but are not clauses of the theorem.

Remark 3.4. The selected phase ξ_* is taken from the full-measure set on which the zero-energy space is one-dimensional (Proposition 4.11). Proposition 4.13 gives this property for every phase, so the choice imposes no further condition. The contact distributions extend separated correlations; uniqueness is not asserted.

4 Principal propositions

The construction begins with a block transformation that preserves the partition integral exactly, transports marked insertions, and separates the running curvature-square coefficient from the other interactions. Propositions 4.1–4.5 carry this transformation through all ultraviolet levels: they choose the bare coupling, control every fixed field label uniformly in volume, and remove the cutoff at fixed volume. Proposition 4.6 then takes an infinite-volume refinement and constructs the field spaces and their derivatives.

The continuum argument has two parts. Maxwell comparison identifies a conserved current, and its shell identities determine the stress tensor and continuous rotations in Propositions 4.8–4.10. Bounded slab insertions then give the phase decomposition and reconstruction in Propositions 4.11–4.13. Finally, Propositions 4.15–4.20 identify the trace law, exclude the specified free-field comparisons, and construct operator product expansions and contact extensions. The principal order of the argument is

$$\begin{aligned} &\text{U1–U4} \longrightarrow \text{P5} \longrightarrow \text{P6–P7} \longrightarrow \text{P8} \\ &\text{P8} \longrightarrow \text{P9} \longrightarrow \text{P10–P11} \\ &\text{P8} \longrightarrow \text{P12} \xrightarrow{\text{prescription}} \text{P13} \end{aligned}$$

This diagram displays the order of construction; direct lattice estimates for the chosen construction also enter the field calculus, the Maxwell comparison, the trace law, and the product and contact estimates. The product and contact arguments use the Euclidean phase tables independently of reconstruction; the last arrow refers only to the common field projections.

All thirteen propositions concern the constants and construction chosen in Proposition 4.1; these constants are called admissible. The further lattice estimates for this same construction are proved in Part II: corrected coefficient and finite Maxwell comparison estimates for Proposition 4.6 in Section 12; stopped marked comparisons for Propositions 4.8 and 4.15 in Sections 13 and 15; and marked product and contact estimates for Propositions 4.18–4.20 in Section 16. They do not use the continuum propositions. Their numerical requirements enter the lower bounds on block ratios and entry levels in Section 6. Proposition 4.6 chooses one refinement and table; nothing later changes the Wilson family or requires another volume refinement.

The Wilson family consists of the sides L_p, scale factors D_N and bare couplings $\beta_{N,p}$ of Proposition 4.2, with label set $\mathcal{T} = \bigsqcup_d V_d$ and insertions $I_{N,p}((d,\mathbf{c}),y;U) = O^{(d)}_{N,p}(\mathbf{c},y;U)$ of Definition 2.10.

4.1 The ultraviolet construction

The one-step result retains both the density and marked insertions. Its coupling estimate permits iteration through the bands $\mathfrak{B}_m$, while its marked bounds control the renormalization of local fields.

Proposition 4.1 (U1. One renormalization step)**.** *For each G as above and $\mathsf{b} > 0$, there are constants chosen in the order of Section 6, a level system satisfying Definition 2.7 whose sides L_p*

are strictly increasing multiples of 4 *with every positive integer dividing* L_p *for infinitely many* p*, and one construction with kernel choices satisfying the branch conditions of Definition 2.8, path actions* $P^{\mathrm{path}}_{N,m}$ *independent of the volume index, admissible classes* $\mathcal{A}_{N,m,p}$ *and relations specifying the recursion as in Definition 2.9. The construction also includes the local sections, reference levels and marked transfers of Definitions 2.6 and 2.10 along the trajectories of Proposition 4.2. For this chosen construction the following holds for all* N, p*.*

(a) There is a continuous curve $\beta \mapsto \rho^{\mathrm{W}}_{\beta} \in \mathcal{I}_{N,p} \subset \mathcal{A}_{N,N,p}$ *on* $\mathfrak{B}_N$*, with coupling* β*, vacuum scalar* $v_{\mathrm{vac}} = 0$ *and density* $e^{-\beta E^{\mathrm{W}}_{L_p D_N}}$*.*

(b) For every $m < N$*, every real* $u_- \le u_+$ *and every continuous curve* $u_{\mathrm{curv}} \mapsto \rho^{\mathrm{in}}_{u_{\mathrm{curv}}} \in \mathcal{A}_{N,m+1,p}$ *on* $[u_-, u_+]$ *with* $\beta^{\mathrm{in}}(u_{\mathrm{curv}}) = \beta(\rho^{\mathrm{in}}_{u_{\mathrm{curv}}}) \in \mathfrak{B}_{m+1}$*, there is a continuous curve* $u_{\mathrm{curv}} \mapsto \rho^{\mathrm{out}}_{u_{\mathrm{curv}}} \in \mathcal{A}_{N,m,p}$ *obeying the specified adjacent-level relation and, for all* $u_{\mathrm{curv}} \in [u_-, u_+]$ *and every coarse configuration* V*,*

$$\begin{aligned} \operatorname{dens} \rho^{\mathrm{out}}_{u_{\mathrm{curv}}}(V) &= \mathcal{B}^{(m)}_{\beta^{\mathrm{in}}(u_{\mathrm{curv}})}(\operatorname{dens} \rho^{\mathrm{in}}_{u_{\mathrm{curv}}})(V), \\ |\beta^{\mathrm{in}}(u_{\mathrm{curv}}) - \beta(\rho^{\mathrm{out}}_{u_{\mathrm{curv}}}) - \kappa_G \log \ell_m| &\le \omega^{\mathrm{RG}}_{J_{m+1}} . \end{aligned} \tag{4.1}$$

All representations have the activity and path bounds of Definition 2.9. The output coupling in (b) need not belong to $\mathfrak{B}_m$*.*

(c) For every degree d *and* $N \ge m_{\mathrm{ref}}(d)$*, fix* N, p, d *in the following marked quantities. The marked densities* $H^{\mathrm{ins}}_{N,p,m;d}$ *of Definition 2.10 along the chosen trajectory of Proposition 4.2 have, at each level* $m_{\mathrm{ref}}(d) \le m \le N$*, a coefficient vector* $\mathbf{x}_m(\mathbf{v})$ *and regular and forced remainder functions, all linear in the terminal input* $\mathbf{v} \in V_d$*, with supports independent of* $\mathbf{v}$*. The marked density is the unmarked density times the sum of its constant coefficient, the masked local section of* $\mathsf{D}^{\mathrm{fac}}_{t_m^{-1}}$ *applied to its nonconstant coefficient part, with* $\mathsf{D}^{\mathrm{fac}}_t$ *the factor-count scaling, and the masked regular remainders, together with the terms obtained by marking one forced activity in the disjoint-support sum defining the unmarked density. Here masking means multiplication by the indicator* $\mathbf{1}_{\{X \text{ healthy}\}}$ *for the support* X*, and the insertion anchor is the site at which the marked local section is placed. Every regular support contains this anchor; the regular remainders vanish at the trivial configuration and, on nonwrapping supports, have zero radial-jet Taylor coefficients through order* d*. Forced remainders obey the flag-inventory condition and the gauge invariance, locality and measurability conditions in Definition 2.9 and vanish unless their support contains the anchor.*

Write $|\mathbf{x}|_1$ *for the sum of the absolute values of its coefficients in the fixed homogeneous basis, and* $\mathbf{x}^\circ$ *for* $\mathbf{x}$ *with its unit coordinate set to zero. In the derivative norms and exponential support weights of Definition 2.9, the remainders admit nonnegative majorants with total regular and forced masses* $\mathcal{E}_m(\mathbf{v})$ *and* $\mathcal{K}_m(\mathbf{v})$ *satisfying*

$$\mathcal{E}_m(\mathbf{v}) + (\epsilon^{\mathrm{for}}_m)^{-1/2} \mathcal{K}_m(\mathbf{v}) \le C_{J_m} |\mathbf{x}_m(\mathbf{v})^\circ|_1 . \tag{4.2}$$

For $m_{\mathrm{ref}}(d) \le m < N$ *the coefficient transport satisfies*

$$|\mathbf{x}_{m+1}(\mathbf{v})|_1 \le C_{J_{m+1}} |\mathbf{x}_m(\mathbf{v})|_1, \qquad |\mathbf{x}_m(\mathbf{v})|_1 \le C_{J_{m+1}} \ell_m^d |\mathbf{x}_{m+1}(\mathbf{v})|_1 . \tag{4.3}$$

Its endpoints are $\mathbf{x}_N(\mathbf{v}) = \mathbf{v}$ *and* $\mathbf{x}_{m_{\mathrm{ref}}(d)}(\mathbf{v}) = \mathsf{Z}_{N,p,d}\mathbf{v}$*. The marked densities satisfy the exact step equation of Definition 2.10. The terminal maps commute with conjugation and* H_4*, and the unit coordinates are transported as* $\mathbf{x}_m(a_N^d \mathbf{1}) = a_m^d \mathbf{1}$*. The precise remainder norms are given in Section 11.*

Part (c) concerns the trajectories that Proposition 4.2 constructs from (a)–(b) alone; neither that construction nor the proofs of (a)–(b) use (c).

The normalization of the block kernel preserves the partition integral exactly. Symmetric gauge fixing and the stationary-background equation reduce the small-field integral to Gaussian integration with an inverse Hessian bounded uniformly in the torus. A large-field seed supplies quadratic energy of order $(M_m^{\mathrm{cut}})^2$, which dominates the sums over supports and paths. The connected expansion separates the marginal curvature-square term from the irrelevant activities. Its one-loop coefficient is $-\kappa_G \log \ell_m$, up to a bounded remainder; resetting the scalar coupling absorbs this term. The same bounds preserve admissibility and continuity, including the flag restrictions. Proved in Sections 8–10.

Proposition 4.2 (U2. Trajectories, running coupling and tuning)**.** *The ratios of Proposition 4.1 satisfy*

$$\ell_m \geq 3 \text{ odd}, \qquad \log \ell_m \to \infty, \qquad \frac{\log \ell_m}{s_m} \to 0\,. \tag{4.4}$$

The classes and specified recursion of the chosen construction of Proposition 4.1 have, for every N, p, a trajectory $\rho_{N,p,m}$, $0 \leq m \leq N$, reaching the reference coupling $\bar{\beta}$ of that system, such that

$$\begin{aligned} \rho_{N,p,N} &= \rho^{\mathrm{W}}_{\beta_{N,p}}, & \beta_{N,p,m} &\in \mathfrak{B}_m, \quad \beta_{N,p,0} = \bar{\beta}, \\ \operatorname{dens} \rho_{N,p,m} &= \mathcal{B}^{(m)}_{\beta_{N,p,m+1}} \operatorname{dens} \rho_{N,p,m+1}, & m &< N, \\ |\beta_{N,p,m+1} - \beta_{N,p,m} - \kappa_G \log \ell_m| &\leq \omega^{\mathrm{RG}}_{J_{m+1}}, & m &< N\,. \end{aligned} \tag{4.5}$$

The representations are admissible and the path actions are independent of p. Their integrals are all $Z_{L_p D_N, \beta_{N,p}}$. For every $\varepsilon > 0$ there is m_0, independent of N, p, such that

$$|\beta_{N,p,m} - \kappa_G s_m| \leq \varepsilon s_m \qquad (m_0 \leq m \leq N,\ p \in \mathbb{N})\,. \tag{4.6}$$

There is $w_{\mathrm{tun}} > 0$, independent of N, p, such that, for every N, p and $u_{\mathrm{tun}} \in [-1, 1]$, exactly one bare coupling $\beta_{N,p}(u_{\mathrm{tun}})$ starts a trajectory of this specified recursion reaching $\bar{\beta} + w_{\mathrm{tun}} u_{\mathrm{tun}}$ at level zero. It equals $\beta_{N,p}$ at $u_{\mathrm{tun}} = 0$. For each N, p there is a holomorphic function on $\{|z| < 1\}$ equal to $\beta_{N,p}(u_{\mathrm{tun}})$ on the real interval $(-1, 1)$.

Choose each block ratio after the constants that determine its lower bound, and enter the corresponding range of levels only when its coupling requirements hold. The resulting drift bounds make the accumulated one-step errors $o(s_m)$, uniformly in the terminal cutoff and volume. Continuous variation of the bare coupling through its band reaches the common level-zero endpoint. The complex extension of the specified recursion gives a local analytic inverse in the affine endpoint coordinate. Increasing the later coupling entry bounds meets the additional finite requirements without altering the earlier choices. Proved in Section 11.

Remark 4.3. Uniqueness refers to the specified recursion; the density identity alone does not imply uniqueness among all representations of a measure.

Proposition 4.4 (U3. Bounds uniform in volume)**.** *For the admissible constants of Proposition 4.1, the family of Proposition 4.2 with the renormalized insertions of Definition 2.10 satisfies:*

(a) For every $d \in \mathbb{N}$ and $R_{\mathrm{lab}}, \varepsilon > 0$, there is $C_{\mathrm{root}} \geq 0$ such that for every $m \geq m_{\mathrm{ref}}(d)$ there is N_0 for which, for all $N \geq N_0$, every p, every $\mathbf{c} \in V_d$ with $\|\mathbf{c}\|_\infty \leq R_{\mathrm{lab}}$ and every site y, $m \leq N$, the insertion is interior to the centered base cell and

$$\|O^{(d)}_{N,p}(\mathbf{c}, y)\|_{\mathcal{C}^{\mathrm{tile}}_m(y)} \leq C_{\mathrm{root}} a_m^{-d-\varepsilon}\,. \tag{4.7}$$

(b) For every arity n, fixed degree-labelled list $v_i = (d_i, \mathbf{c}_i)$, finite list $\boldsymbol{l}$ of slot-axis pairs and $r > 0$, there are $C_{\mathrm{diff}} \geq 0$ and N_0 such that, for every $N \geq N_0$, every p and every lattice tuple with physical torus distances at least r between distinct slots,

$$|\Delta_{\boldsymbol{l}} M^{\mathrm{lat}}_{N,p}(v; y)| \leq C_{\mathrm{diff}} \,. \tag{4.8}$$

Here C_{root} is chosen before m, whereas N_0 in (a) may depend on m. Neither C_{diff} nor N_0 in (b) depends on p. There is no common threshold for an unbounded family of labels or derivative orders.

Transport a marked coefficient along the unmarked density trajectory. Its low-degree part is measured at the fixed reference level; the remainder gains powers under further blocking. A determinant bound controls the inverse terminal maps $\mathsf{Z}^{-1}_{N,p,d}$ without a volume factor. The centered chessboard inequality [2, Theorem 4.1], applied to the interior insertions on the reflection-positive periodic lattice, converts these coefficient bounds into the stated physical powers of a_m. A lattice difference is represented by a marked coefficient of one higher degree and a controlled remainder. Taking the comparison scale below the prescribed torus separation gives the uniform difference bound. Proved in Section 11.

Proposition 4.5 (U4. Fixed-volume convergence for every label)**.** *For the admissible constants of Proposition 4.1, for every volume p, arity n and degree-labelled list $v = (d_i, \mathbf{c}_i)$, there is a function $f_{p,v}$ smooth on $\mathrm{NoHit}_{L_p,n}$ such that, for every p-admissible separated test t,*

$$\lim_{N\to\infty} C^{\mathrm{lat}}_{N,p}(v; t) = \int t(x) f_{p,v}(x)\, dx \,. \tag{4.9}$$

For every finite list $\boldsymbol{l}$ of slot-axis pairs, every compact $\mathcal{K}_{\mathrm{pos}} \subset \mathrm{NoHit}_{L_p,n}$ and every $\varepsilon > 0$, there is N_0 such that, for $N \geq N_0$, at every spacing-a_N point $x \in \mathcal{K}_{\mathrm{pos}}$ of the periodic lift,

$$|\Delta_{\boldsymbol{l}} M^{\mathrm{lat}}_{N,p}(v; x) - \partial_{\boldsymbol{l}} f_{p,v}(x)| < \varepsilon \,. \tag{4.10}$$

The multilinear lattice interpolants of these difference functions also converge to $\partial_{\boldsymbol{l}} f_{p,v}$ in the moving-point sense: if $x_N \to x \in \mathrm{NoHit}_{L_p,n}$, their values at x_N converge to $\partial_{\boldsymbol{l}} f_{p,v}(x)$. Moreover $t\partial_{\boldsymbol{l}} f_{p,v}$ is integrable and the corresponding smeared lattice difference sums converge to $\int t\partial_{\boldsymbol{l}} f_{p,v}$. All limits use the full sequence $N \to \infty$. Thresholds may depend on $p, n, v, \boldsymbol{l}, \mathcal{K}_{\mathrm{pos}}, t$; no uniformity in p is asserted.

Use the common level-zero endpoint coordinate $\bar{\beta} + w_{\mathrm{tun}} z$. At every fixed level m and volume p, the coupling and the normalized regular and forced activities are holomorphic in z on $|z| < 1$, for every cutoff $N \geq m$. The fixed-level comparison estimates make them Cauchy, uniformly on compact subdiscs, in their fixed weighted norms. The corresponding marked coefficients and remainders are Cauchy as well. Normalized correlations cancel the vacuum scalar v_{vac}, which need not converge.

For a fixed configuration on a common lattice, let $C^{(m)}_{N,p}$ be the scalar-normalized level-m correlation. It is Cauchy in N, and the marked comparison gives, for all sufficiently large N,

$$\left| M^{\mathrm{lat}}_{N,p}(v; x) - C^{(m)}_{N,p} \right| \leq 2F_0 e^{-B_m/64} \,, \tag{4.11}$$

where F_0 depends on the fixed request but not on m, N. First choose m large and then the cutoffs large. The triangle inequality bounds the limiting difference between two cutoffs by $4F_0 e^{-B_m/64} \to 0$. Proposition 4.4, with one further difference, extends convergence from these dense lattice configurations and identifies the limiting differences by summation by parts. Multilinear interpolation on the periodic lift also handles crossings of the fundamental-cell boundary away from periodic collisions. Proved in Section 11.

4.2 Infinite volume

The ultraviolet limits retain estimates independent of the torus. These estimates allow a single volume refinement for all labels and identify the derivatives before passage to the quotient field spaces.

Proposition 4.6 (P5. Infinite volume and field calculus)**.** *For the admissible constants of Proposition 4.1 and every strictly increasing* $\mathsf{p} : \mathbb{N} \to \mathbb{N}$*, there are a strictly increasing* q *and one multilinear table* S *with the following properties, with* $\mathsf{v} = \mathsf{p} \circ \mathsf{q}$*.*

(a) The Wilson family converges to S *along* v *in the sense of Definition 2.5.*

(b) For every volume p *and every* n, v *there is a function* $K_{p,v}$ *on the periodic generic set* $\{x : x_i^a - x_j^a \notin L_p\mathbb{Z} \ (i \neq j,\ a = 1, \dots, 4)\}$*, continuous there, such that for every compact* $\mathcal{K}_{\mathrm{pos}} \subset \mathrm{Gen}_n$*,* $K_{\mathsf{v}(k),v} \to S_n(v; \cdot)$ *uniformly on* $\mathcal{K}_{\mathrm{pos}}$*, which lies in that periodic generic set for all large* k*. On their common domain* $K_{p,v} = f_{p,v}$*.*

(c) The table satisfies E1–E7.

(d) The quotient fields F_d *have* $\dim_{\mathbb{C}} F_d = \dim_{\mathbb{C}} M_d^{\mathrm{Max}}$*. There are injective compatible inclusions* $\jmath_{d,e} : F_d \to F_e$ *for* $d \leq e$*, preserving units and correlations, and linear derivatives* $\partial_a : F_d \to F_{d+1}$ *with*

$$\partial_{x_i^a} S(\boldsymbol{A}; x) = S(A_1, \dots, \partial_a A_i, \dots; x)\,. \tag{4.12}$$

They commute with conjugation and satisfy $h\partial_a = \sum_b h_{ba}\partial_b h$ *for* $h \in H_4$*. Each* F_d *embeds in the response space* $\mathcal{L}$*, every element of* $\mathcal{L}$ *is in one such image, and every field has a degree-labelled representative. The class of the distinguished unit label is* $\mathbf{1}_d$*.*

(e) For every degree bound $d_{\max}$ *and coefficient norm bound* $R_{\mathrm{lab}} \in \mathbb{R}$*, there are* $C_{\mathrm{tr}} > 0$ *and* $q_{\mathrm{tr}} \in \mathbb{N}$ *such that, for every nonempty word* w *of length* n *with these bounds, every* $h_{\mathrm{sl}} > 0$ *and* $0 < r \leq 1$ *such that* w *is a slab word of width* h_{sl} *with buffer* r*, there are* $0 < h_{\mathrm{sl}}^{\mathrm{fit}} \leq h_{\mathrm{sl}}$ *and integers* $k, n_{\mathrm{lat}} \geq 0$ *with* $h_{\mathrm{sl}}^{\mathrm{fit}} = n_{\mathrm{lat}} a_k$*, such that* w *is a slab word of width* $h_{\mathrm{sl}}^{\mathrm{fit}}$ *and, for every positive exterior word* w_{ext}*, there is* $B_{w_{\mathrm{ext}}}^{\mathrm{ext}} < \infty$ *with*

$$\|[(A_{w^\dagger} A_w)^{n_{\mathrm{rep}}} w_{\mathrm{ext}}]\|_{\mathrm{OS}}^2 \leq B_{w_{\mathrm{ext}}}^{\mathrm{ext}} \, [C_{\mathrm{tr}}^n (n/r)^{q_{\mathrm{tr}} n}]^{4 n_{\mathrm{rep}}} \qquad (n_{\mathrm{rep}} \geq 0)\,. \tag{4.13}$$

The exterior constant is independent of n_{rep}*. The width and the constants precede the exterior word. The empty slab word is treated by the contraction semigroup.*

The volume-uniform difference bounds of Proposition 4.4(b), with one further difference, pass through the moving-point limits of Proposition 4.5. They make the fixed-volume kernels and all their derivatives uniformly bounded and equicontinuous on compact subsets of Conf_n. Apply Arzelà–Ascoli over countable label bases and compact exhaustions; the full inner limits of Proposition 4.5 leave only a volume refinement. Corrected finite-dimensional projections identify the null spaces. An equation count bounds $\dim F_d$ above by $\dim M_d^{\mathrm{Max}}$, and a finite comparison of corrected lattice correlations with free Maxwell correlations bounds it below. Corrected lattice differences and summation by parts give derivative representatives against all spectators. In a repeated reflected word a finer tiling bounds the exterior factor only once, giving (e). Proved in Section 12, Lemmas 12.1 and 12.5 and the subsequent field calculus.

4.3 Maxwell comparison and the stress tensor

Free Maxwell fields determine the dimensions and normalization used in the short-distance comparison. The following auxiliary proposition records the free facts needed for the conserved current, its shell laws and the trace amplitude.

Proposition 4.7 (Free Maxwell fields)**.** *The table S^{Max} satisfies E1–E7, its inclusions and derivatives obey the field calculus, and it is $O(4)$ covariant. For every nonzero on-shell invariant Wick polynomial A, its two-point reflected Gram value at a pair of distinct reflected points is strictly positive. For simple $\mathfrak{g}$, the real vector space of centered real symmetric H_4-tensorial degree-four Maxwell tensors of zero divergence is the line spanned by T^{Max}. That tensor has the normalized translation and angular shell laws of Definition 2.15, and*

$$S_2^{\mathrm{Max}}(\mathcal{O}_{F^2}, \mathcal{O}_{F^2}; x, y) = \frac{3d_{\mathfrak{g}}}{\pi^4 |x-y|^8} \qquad (x \neq y)\,. \tag{4.14}$$

Strict reflected positivity is proved directly for nonzero on-shell Wick polynomials. The degree-four invariant tensor calculation determines the conserved stress line, and the shell flux fixes its normalization. Wick contraction then gives the displayed scalar two-point constant. Proved in Section 13.

Proposition 4.8 (P6. Maxwell comparison and a conserved current)**.** *For the admissible constants of Proposition 4.1 and the table of Proposition 4.6 there is one stable scaling sequence $(r_k, \mathsf{U}^{\mathrm{sc}}_{k,d})_{k,d}$ such that for every arity n, free field list $u_i \in M^{\mathrm{Max}}_{d_i}$, compact $\mathcal{K}_{\mathrm{pos}} \subset \mathrm{Conf}_n$, $M_{\mathrm{test}} \geq 0$ and $\varepsilon > 0$, there is k_0 such that for every $k \geq k_0$ and every smooth compactly supported joint test f with support in $\mathcal{K}_{\mathrm{pos}}$ and $\|f\|_\infty \leq M_{\mathrm{test}}$,*

$$\left| \int f(x) \left[S_n(\mathsf{U}^{\mathrm{sc}}_{k,d_1} u_1, \dots, \mathsf{U}^{\mathrm{sc}}_{k,d_n} u_n; r_k x) - S_n^{\mathrm{Max}}(u; x) \right] dx \right| \leq \varepsilon M_{\mathrm{test}}\,. \tag{4.15}$$

The same sequence serves every fixed request; k_0 may depend on all the data before it.

The real space of centered real symmetric H_4-tensorial $J_{ab} \in F_4$ with $\sum_a \partial_a J_{ab} = 0$ is one-dimensional. One can choose a nonzero J, a refinement of the comparison sequence and positive numbers $\zeta_k^{\mathrm{stress}}$ such that, componentwise,

$$(\mathsf{U}^{\mathrm{sc}}_{k,4})^{-1}((\zeta_k^{\mathrm{stress}})^{-1} r_k^4 J_{ab}) \longrightarrow T^{\mathrm{Max}}_{ab}\,. \tag{4.16}$$

There are complex linear $\mathsf{p}_b : F_d \to F_{d+1}$ and maps $\mathsf{Y}_{K_{\mathrm{rot}}} : F_d \to F_d$, real linear in K_{rot} and complex linear in the field, obeying the raw shell laws of Definition 2.15 for this J, for every degree, field, spectator list, separated configuration and allowed cutoff.

Stop the marked lattice evolution near the requested physical scale and compare its Gaussian part with the free covariance. Strict reflected positivity gives finitely many correlation functionals separating the Maxwell labels. Their ranks and the upper rank bound from extraction give field-space isomorphisms. Finite-dimensional corrections impose exact compatibility with inclusions, conjugation and signed permutations eventually at each degree. Surjectivity of the free divergence is an open condition, so in the stable frames the interacting divergence is onto. Equal dimensions then give a one-dimensional kernel, and the free stress line fixes its sign and scale. Lattice integration by parts gives its raw shell maps. Proved in Section 13.

Remark 4.9. The comparison is weak and uses moving frames. Compatibility with derivatives and comparison with the normalized interacting stress are the further conclusions of Proposition 4.10(e).

Proposition 4.10 (P7. Ward identities, rotations and stress)**.** *For the admissible constants of Proposition 4.1 and the data of Propositions 4.6–4.8, there are a real $c_\star \neq 0$, continuous complex linear representations R_d of $O(4)$ on F_d, and fields $T_{ab} \in F_4$ such that:*

(a) $\mathsf{p}_b = c_\star \partial_b$ *and* $\mathsf{Y}_{K_{\mathrm{rot}}} = c_\star \mathsf{L}_d(K_{\mathrm{rot}})$ *in every degree. The representations extend the* H_4 *actions, fix units, commute with conjugation and inclusions, and satisfy*

$$R_{d+1}(h)\partial_a = \sum_i h_{ia}\partial_i R_d(h) \qquad (h \in O(4))\,. \tag{4.17}$$

For each d *there is an isomorphism* $\mathcal{V}_d : M_d^{\mathrm{Max}} \to F_d$ *intertwining* R_d^{Max} *and* R_d.

(b) For every n, $A_i \in F_{d_i}$, $x \in \mathrm{Conf}_n$ *and* $h \in O(4)$,

$$S_n(R_{d_i}(h)A_i; hx) = S_n(\boldsymbol{A}; x)\,. \tag{4.18}$$

(c) T *is real, symmetric, conserved and tensorial:*

$$R_4(h)T_{ab} = \sum_{i,j} h_{ia}h_{jb}T_{ij}\,. \tag{4.19}$$

(d) For every $n, d, A, \boldsymbol{B}, x, \chi$ *allowed in Definition 2.15, every* b *and* $K_{\mathrm{rot}} \in \mathfrak{so}(4)$,

$$\mathcal{P}_b^T(\chi; A, \boldsymbol{B}; x) = \partial_{x_0^b} S(A(x_0), \boldsymbol{B}(x)), \qquad \mathcal{Y}_{K_{\mathrm{rot}}}^T(\chi; A, \boldsymbol{B}; x) = S((\mathsf{L}_d(K_{\mathrm{rot}})A)(x_0), \boldsymbol{B}(x))\,. \tag{4.20}$$

(e) There is one stable scaling sequence such that, for every n, *fixed free field list* $u_i \in M_{d_i}^{\mathrm{Max}}$, *subset* I *of slots, component pairs* (a_i, b_i) *in those slots, derivative order* k_{der} *and compact* $\mathcal{K}_{\mathrm{pos}} \subset \mathrm{Conf}_n$,

$$D_x^{k_{\mathrm{der}}} S_n(A_{k,1}, \dots, A_{k,n}; r_k x) \longrightarrow D_x^{k_{\mathrm{der}}} S_n^{\mathrm{Max}}(A_1^0, \dots, A_n^0; x) \quad \textit{uniformly on } \mathcal{K}_{\mathrm{pos}}\,, \tag{4.21}$$

where $A_{k,i} = r_k^4 T_{a_i b_i}$ *and* $A_i^0 = T^{\mathrm{Max}}_{a_i b_i}$ *for* $i \in I$, *and* $A_{k,i} = \mathsf{U}^{\mathrm{sc}}_{k,d_i} u_i$, $A_i^0 = u_i$ *otherwise. In operator norm on the finite-dimensional free spaces,*

$$r_k(\mathsf{U}^{\mathrm{sc}}_{k,d+1})^{-1}\partial_a \mathsf{U}^{\mathrm{sc}}_{k,d} \to \partial_a^{\mathrm{Max}}, \qquad (\mathsf{U}^{\mathrm{sc}}_{k,d})^{-1}\mathsf{L}_d(K_{\mathrm{rot}})\mathsf{U}^{\mathrm{sc}}_{k,d} \to \mathsf{L}_d^{\mathrm{Max}}(K_{\mathrm{rot}})\,. \tag{4.22}$$

Thresholds may depend on the fixed list, slots, components, derivative order and compact set; one sequence precedes all these requests.

Differentiate the shell identities to obtain the commutation relations of the raw charges. Free comparison identifies the translation charge as a nonzero scalar multiple of the field derivative. Conservation and symmetry cancel the interior terms of the collinear angular identity, and exterior-flux estimates make its remaining term vanish. Analytic-vector bounds extend the identity to all separated configurations. Finite-dimensional comparison then identifies an $O(4)$ action; averaging the normalized tensor over $O(4)$ with probability Haar measure preserves its shell laws. The same normalization gives the joint field and stress comparison. Proved in Section 13.

4.4 Phases and reconstruction

Bounded slab insertions turn separated correlations into operators. Their zero-energy compressions form a commutative algebra; its characters determine the phases in which the relativistic fields will be reconstructed.

Proposition 4.11 (P8. Slab operators and phases)**.** *For the table* S *of Proposition 4.6 (the mixed table), with Propositions 4.8–4.10, let* $\mathcal{H}_4^0 = \ker H_4^{\mathrm{OS}}$ *in its time-axis OS space. Bounded slab operators exist. For every degree bound* $d_{\max}$ *and coefficient norm bound* $R_{\mathrm{lab}} \in \mathbb{R}$, *there are* $C_{\mathrm{tr}} > 0$

and $q_{\mathrm{tr}} \in \mathbb{N}$ *such that for every width* $h_{\mathrm{sl}} > 0$*, every slab word* w *of length* n *with these bounds and buffer* $0 < r \le 1$*,*

$$\|\mathsf{B}_{h_{\mathrm{sl}}}(w)\| \le C_{\mathrm{tr}}^{n}(n/r)^{q_{\mathrm{tr}}n}\,. \tag{4.23}$$

For slab words w, w' *of widths* $h_{\mathrm{sl}}, h'_{\mathrm{sl}}$ *they obey*

$$\begin{aligned} \mathsf{B}_{h_{\mathrm{sl}}}(w)^{*} &= \mathsf{B}_{h_{\mathrm{sl}}}(w^{\dagger}),\\ \mathsf{B}_{h_{\mathrm{sl}}+h'_{\mathrm{sl}}}(w\,\tau_{h_{\mathrm{sl}}e_4}w') &= \mathsf{B}_{h_{\mathrm{sl}}}(w)\mathsf{B}_{h'_{\mathrm{sl}}}(w'),\\ \partial_{x_i^a}\mathsf{B}_{h_{\mathrm{sl}}}(w) &= \mathsf{B}_{h_{\mathrm{sl}}}(w_{i,a})\,, \end{aligned} \tag{4.24}$$

where $w_{i,a}$ *replaces* A_i *in slot* i *by* $\partial_a A_i$*. Their compressions* $M^0(w)$ *are independent of the admissible slab width and placement and generate a separable commutative unital* C^**-algebra on* $\mathcal{H}_4^0$ *with cyclic and separating vector* Ω*.*

There are a compact metric spectrum X *and a full-support probability measure* ν *such that, for every separated word* w*,*

$$S^{\xi}(w) = \xi(M^0(w)), \qquad \int_{\mathsf{X}} S^{\xi}(w)\,d\nu(\xi) = S(w)\,. \tag{4.25}$$

Each S^{ξ} *satisfies E1–E7, has exactly the null spaces* $\mathcal{N}_d(S)$*, and has the same inclusions, derivatives, conjugation,* $O(4)$ *representations and stress laws as the mixed table. For every fixed word its value is continuous in* ξ*. The slab norm bound holds in every phase with the same* $C_{\mathrm{tr}}, q_{\mathrm{tr}}$*.*

For ν*-almost every* ξ*, the time-axis OS ground space of* S^{ξ} *is one-dimensional. Choose* ξ_* *from that set; this choice imposes no additional trace condition.*

Apply Proposition 4.6(e) to high powers of a slab insertion and its adjoint. Taking roots removes the exterior constant and proves the operator bound. Compression to zero energy removes dependence on the slab's translation. Four separated clusters, with time and space axes interchanged, allow the middle factors to be exchanged and prove commutativity. Characters give the phase values and inherit the algebra's identities and norm bounds. The variance argument on a countable family total in every phase gives ground-space uniqueness outside one null set. Proved in Section 14.

Remark 4.12. Proposition 4.13 gives ground-space uniqueness for every character. The choice of ξ_* here uses only the almost-everywhere conclusion.

Proposition 4.13 (P9. Reconstruction in every phase)**.** *For the data of Proposition 4.11 there are, simultaneously for every* $\xi \in \mathsf{X}$*, relativistic data* $\mathfrak{W}^{\xi}$ *on the common response label space* $\mathcal{L}$*, and continuations* F_n^{ξ}*, with the following properties.*

(a) Each datum satisfies W0–W3; each nonzero label acts nontrivially for some Schwartz test. R13–R15 hold with S^{ξ}*,* F^{ξ}*,* Ω_{ξ} *and* Φ_{ξ} *in place of the selected quantities. Its word-domain vacuum expectations extend to joint Schwartz distributions* $W_n^{\xi}(\boldsymbol{A};\psi)$*. Its Wightman functional is positive.*

(b) The bounded comparison map from its Wightman space to its phase OS space is onto. Both zero-energy spaces are the vacuum line. Its bounded weak commutant consists exactly of scalars: if B *is bounded and for every smeared field* Φ *and* u, v *in the word domain,* u *belongs to the domain of* Φ^* *and* $\langle \Phi^* u, Bv\rangle = \langle u, B\Phi v\rangle$*, then* $B = \lambda \mathrm{Id}_{\mathcal{H}_{\xi}}$*, and conversely.*

(c) For every fixed finite complex linear combination A *of smeared field words, there are* $M_A^{\mathrm{low}}, M_A^{\mathrm{heat}} \ge 0$*, independent of* ξ*, such that*

$$\begin{aligned} \mu_{A\Omega_{\xi},\xi}(\{\mathbf{p} : 0 < \mathbf{p}^4 \le u\}) &\le M_A^{\mathrm{low}} u^2 \qquad (0 < u \le 1),\\ 0 \le \langle A\Omega_{\xi}, (e^{-sH_{\xi}} - P_{\xi}^0)A\Omega_{\xi}\rangle &\le M_A^{\mathrm{heat}} s^{-2} \quad (s \ge 1)\,. \end{aligned} \tag{4.26}$$

Here P_ξ^0 projects onto $\mathbb{C}\Omega_\xi$ and H_ξ is the energy generator. The constants may depend on the fixed words, their labels and tests.

(d) For every fixed $n, \boldsymbol{A}$ and $\psi \in \mathcal{S}_n$, $\xi \mapsto W_n^\xi(\boldsymbol{A};\psi)$ is continuous and

$$\int_{\mathsf{X}} W_n^\xi(\boldsymbol{A};\psi)\, d\nu(\xi) = W_n^{\mathrm{mix}}(\boldsymbol{A};\psi)\,, \tag{4.27}$$

where the right side is the boundary distribution of the mixed Euclidean continuation.

(e) The tensor T continues in every phase to a symmetric Hermitian conserved Lorentz tensor of operator-valued distributions, with the translation and angular charge actions of its Ward laws. The chosen theory and continuation are $\mathfrak{W} = \mathfrak{W}^{\xi_*}$ and $\mathsf{F}_n = \mathsf{F}_n^{\xi_*}$.

Place bounded thin-slab insertions between heat evolutions and continue their time variables. Polynomial bounds in the slab widths give joint tempered boundary distributions. The Ward identities continue to boosts; agreement at spacelike configurations gives local commutativity. Field words define the fields on the prescribed domain and realize the response labels faithfully. The low-energy bound, uniform in characters, and continuity in the character extend ground-space uniqueness to every phase. The same construction yields the onto OS comparison, scalar weak commutant and continued stress tensor. Proved in Section 14.

Remark 4.14. The estimates in (c) are polynomial; no positive mass gap is asserted.

4.5 Identification of the theory

The centered stress trace detects the logarithmic correction to free scaling. Local products and collision subtractions use one common family of field projections, chosen before spectators or states.

Proposition 4.15 (P10. The centered trace and its spectral law)**.** *For the admissible constants of Proposition 4.1, let T be the tensor of Proposition 4.10 and ξ_* the phase chosen in Proposition 4.11, with its reconstruction in Proposition 4.13. Set*

$$\tau = \sum_a T_{aa}, \qquad e_{\mathrm{tr}} = S^{\xi_*}(\tau), \qquad \Theta = \tau - e_{\mathrm{tr}}\mathbf{1}_4, \qquad G_\Theta(r) = S_2^{\xi_*}(\Theta,\Theta;-re_4/2, re_4/2)\,. \tag{4.28}$$

The one-point function defining e_{tr} is independent of position. The field $\Theta \neq 0$ is real and $O(4)$ invariant; its relativistic field is a Hermitian Lorentz scalar. For every $f \in \mathcal{S}$, on the word domain,

$$\Phi(\iota_4\Theta, f) = \Phi(\iota_4\tau, f) - e_{\mathrm{tr}}\left(\int f\right)\mathrm{Id}_{\mathcal{H}}, \qquad \langle\Omega, \Phi(\iota_4\Theta, f)\Omega\rangle = 0\,. \tag{4.29}$$

With $A_G = 3d_{\mathfrak{g}}/\pi^4 > 0$, the limit over all positive radii is

$$\lim_{r\downarrow 0} r^8 \log^2(1/r) G_\Theta(r) = A_G\,. \tag{4.30}$$

There is exactly one positive Borel measure μ_Θ with all of the following properties: it has tempered growth, is supported in $\overline{V}_+$, its energy–momentum identification

$$d\mu_{\Phi(\iota_4\Theta,\varphi)\Omega}(\mathbf{p}) = |\widehat{\varphi}_+(\mathbf{p})|^2 d\mu_\Theta(\mathbf{p}) \qquad (\varphi \in \mathcal{S}) \tag{4.31}$$

holds, and it represents the trace two-point boundary distribution by

$$W_2^{\xi_*}(\Theta,\Theta;\psi) = \int \mathcal{W}_{\mathrm{rel}}\psi(\mathbf{p})\, d\mu_\Theta(\mathbf{p}) \qquad (\psi \in \mathcal{S}_2)\,. \tag{4.32}$$

It is Lorentz invariant in momentum variables and satisfies $G_\Theta(r) = \int e^{-r\mathbf{p}^4}\, d\mu_\Theta(\mathbf{p})$ *for* $r > 0$. *For every continuous compactly supported* $\varphi : \mathbb{R}^4 \to \mathbb{C}$, *and for every Schwartz* φ, *it obeys*

$$\lim_{R_{\mathrm{E}}\to\infty} R_{\mathrm{E}}^{-8}(\log R_{\mathrm{E}})^2 \int \varphi(\mathbf{p}/R_{\mathrm{E}})\, d\mu_\Theta(\mathbf{p}) = \frac{A_G}{1536\pi} \int_{\overline{V}_+} \mathsf{m}(\mathbf{p})^2 \varphi(\mathbf{p})\, d^4\mathbf{p}\,. \tag{4.33}$$

Uniqueness already holds among measures with the energy–momentum identification.

Compare the centered trace with the free $\mathcal{O}_{F^2}$ field in a fourth-order frame. The leading scalar coefficient is $4d_2/\beta$, where $d_2 = -b_0/2$ and $\log(1/r)/\beta \to \kappa_G^{-1}$; thus $(4d_2/\kappa_G)^2 = 1$ gives the free scalar amplitude. Along a subsequence of every prescribed sequence of radii, the comparison is uniform in phase at configurations near $(-re_4/2, re_4/2)$ with separations slightly above and below r. The Laplace representation by a positive measure supported in $\overline{V}_+$ makes G_Θ nonincreasing. Squeezing between those separations gives the limit at the exact radius. The subsequence principle then gives the limit over all positive radii.

For the momentum-space limit, rescale the positive measure by

$$\mu_\Theta^{(R_{\mathrm{E}})} = R_{\mathrm{E}}^{-8}(\log R_{\mathrm{E}})^2 (\mathbf{p} \mapsto \mathbf{p}/R_{\mathrm{E}})_* \mu_\Theta\,. \tag{4.34}$$

Its time-axis Laplace transform is $R_{\mathrm{E}}^{-8}(\log R_{\mathrm{E}})^2 G_\Theta(s/R_{\mathrm{E}}) \to A_G s^{-8}$ for $s > 0$. Lorentz invariance extends this convergence to every point inside the dual forward cone. The continuity theorem for Laplace transforms of positive measures gives convergence against compactly supported continuous tests. The limiting density is $A_G \mathsf{m}(\mathbf{p})^2/(1536\pi)$ on $\overline{V}_+$, since its time-axis Laplace transform is $A_G s^{-8}$. The short-distance upper bound also gives $\mu_\Theta^{(R_{\mathrm{E}})}(\{\mathbf{p} : \mathbf{p}^4 \le L\}) \le CL^8$ for $L \ge 1$, uniformly for large R_{E}. On $\overline{V}_+$, $|\mathbf{p}| \le \sqrt{2}\mathbf{p}^4$, so these bounds control the Schwartz tails uniformly. Finally, the energy–momentum identification determines the measure uniquely by testing compactly supported momentum functions. Proved in Section 15.

Proposition 4.16 (P11. Non-freeness)**.** *For the selected trace and theory of Proposition 4.15:*

(a) For every finite Wick profile W_{Wick} *of Definition 2.21 and every* $\delta > 0$, *some* $r \in (0, \delta)$ *satisfies* $G_\Theta(r) \ne W_{\mathrm{Wick}}(r)$.

(b) For every choice of finite polynomial comparison data and every pair of finite polynomial fields Q, Q' *of Definition 2.22, there are* $f, h \in \mathcal{S}$ *such that* $\mathcal{W}_{\mathrm{rel}}(\bar{f} \otimes h)$ *has compact support and*

$$\langle \Phi(\iota_4\Theta, f)\Omega, \Phi(\iota_4\Theta, h)\Omega\rangle \ne W^{\mathrm{pol}}(\mathsf{Q}, \mathsf{Q}'; \mathcal{W}_{\mathrm{rel}}(\bar{f} \otimes h))\,. \tag{4.35}$$

(c) For every $n_{\mathrm{col}} \ge 0$ *and every Fock comparison* l_{F} *of Definition 2.23, there are compactly supported Schwartz* f *and a list of real compactly supported Schwartz* $h_{\alpha,a_{\mathrm{sp}}}$, *each causally separated from* f, *such that*

$$\langle \mathcal{E}(-\boldsymbol{h}), \psi_{\mathsf{l}_{\mathrm{F}}}(f)\rangle \ne \langle \psi_{\mathsf{l}_{\mathrm{F}}}(\bar{f}), \mathcal{E}(\boldsymbol{h})\rangle\,. \tag{4.36}$$

More precisely, for every such comparison, every probability space (Z, λ_Z) *and family* P_z^{av} *of Definition 2.23, and every* $r > 0$, *there are open* $A_{\mathrm{loc}}, B_{\mathrm{loc}} \subset \mathbb{R}^4$ *with* $\overline{A_{\mathrm{loc}}} \subset B(0, r)$ *and* $\mathsf{m}(x-y) < 0$ *for every* $x \in \overline{A_{\mathrm{loc}}}$, $y \in \overline{B_{\mathrm{loc}}}$, *and tests as above supported in* A_{loc} *and* B_{loc}, *respectively, for which*

$$\int_Z \langle P_z^{\mathrm{av}} \mathcal{E}(-\boldsymbol{h}), \psi_{\mathsf{l}_{\mathrm{F}}}(f)\rangle\, d\lambda_Z(z) \ne \int_Z \langle \psi_{\mathsf{l}_{\mathrm{F}}}(\bar{f}), P_z^{\mathrm{av}} \mathcal{E}(\boldsymbol{h})\rangle\, d\lambda_Z(z)\,. \tag{4.37}$$

The integral convention in Definition 2.23 applies to these last integrals.

A finite mass-shell profile has a short-distance expansion in integer powers and nonnegative logarithmic powers. Multiplication by $r^8 \log^2(1/r)$ cannot give the nonzero finite limit of Proposition 4.15. Fourier and Laplace uniqueness convert agreement on the specified polynomial two-point tests into this forbidden Euclidean agreement. In Fock coordinates, the locality identity would restrict the particle-sector waves and their growth. Those restrictions exclude the trace's logarithmically corrected spectral tail. The localized argument uses causally separated tests in any prescribed ball and also applies to the averaged identities. Proved in Section 15.

Remark 4.17. The excluded polynomial comparisons have finitely many mass shells. The statement does not exclude every composite of a continuous-mass generalized free field.

Proposition 4.18 (P12. Operator product expansion)**.** *For the admissible constants of Proposition 4.1 and the mixed table and phases of Propositions 4.6–4.11, there are finite channel sets* Ch_q*, real fields* $O_{q,\lambda}$ *giving a complex basis of a conjugation-invariant complement of* F_{q-1} *in* F_q *(with* $F_{-1} = 0$*), and coefficients* $c_{q,\lambda}$ *with the following properties. One prescription is chosen before all spectators, states and truncation degrees. The degree-zero basis spans the unit line; its vector is not required to equal the unit.*

The prescription includes linear field projections $\mathsf{P}^{\mathrm{OPE}}_{\le q}(o)$*, assigning to finite combinations of separated words near* o *their components in* F_q*. Their stratum parts* $\mathsf{P}^{\mathrm{OPE}}_{=q}(o)$ *take values in* $\operatorname{span}\{O_{q,\lambda}\}$*. For* $w = \prod_i A_i(x_i)$*,*

$$\mathsf{P}^{\mathrm{OPE}}_{\le D}(o)w = \sum_{q=0}^{D} \mathsf{P}^{\mathrm{OPE}}_{=q}(o)w = \sum_{q=0}^{D} \sum_{\lambda\in \mathrm{Ch}_q} c_{q,\lambda}(\boldsymbol{A}; x, o) O_{q,\lambda}\,, \tag{4.38}$$

where inclusions into F_D *are understood. These projections are read by finitely many separating correlation functionals and their rotation averages. Choose a positive clearance at each degree, smaller than half the distance from the origin to every spectator in those functionals; rotations preserve these distances. Let* δ^{pres}_q *be the minimum of these clearances through degree* q*. Thus* $\delta^{\mathrm{pres}}_q > 0$ *is nonincreasing, and all spectators used through degree* q *are at distance at least* $2\delta^{\mathrm{pres}}_q$ *from* o*.*

(a) O1 and O2 hold exactly as in Definition 2.20, including the coefficient 1 *on both bounds and the stated order of quantifiers. The coefficients are independent of spectators, state and truncation. The thresholds may depend on every fixed datum preceding them, and are uniform in all positions and in the state.*

(b) For every finite $I_{\mathrm{cl}}, I_{\mathrm{sp}}$*, fixed* $\boldsymbol{A}, \boldsymbol{B}$*,* $M_{\mathrm{geo}} \ge 0$*,* $\eta, \delta_{\mathrm{sp}}, \varepsilon > 0$*,* $D \ge 0$*, measurable space* Z *and* $B_0 \in \mathbb{R}$*, there is* $r_0 > 0$ *such that for every* $0 < r < r_0$*, measure* λ_Z*, integrable complex weight* ϑ_Z *with* $\int |\vartheta_Z|\, d\lambda_Z \le B_0$*, almost-everywhere measurable configuration map* $z \mapsto (x(z), o(z), y(z))$ *satisfying the cluster and clearance conditions for* λ_Z*-almost every* z *with* $\vartheta_Z(z) \ne 0$*, and almost-everywhere measurable* $\xi : Z \to \mathsf{X}$*,*

$$\left| \int_Z \vartheta_Z(z) \mathcal{R}^{\xi(z)}_D(\boldsymbol{A}, x(z); \boldsymbol{B}, y(z); o(z))\, d\lambda_Z(z) \right| \le B_0 r^{D+1-|\boldsymbol{A}|_{\mathrm{deg}}-\varepsilon}\,. \tag{4.39}$$

The same holds for the mixed state. The full and truncated weighted expressions being compared are integrable. The threshold is independent of the measure, weight, configuration and phase maps.

(c) Fix any finite rooted tree with internal root, leaf fields $\boldsymbol{A}$*, output channel* (q, λ)*, proper-vertex truncations* $\boldsymbol{D} = (D_v)$*, and* $M_{\mathrm{geo}} \ge 0$*,* $\eta, \varepsilon > 0$*. There is* $r_0 > 0$ *such that for every* $0 < r < r_0$ *and every geometry and scale vector of Definition 2.24,*

$$|C^{\mathrm{dir}}_{q,\lambda}| \le \mathcal{Q}_{\varepsilon,q}(\varrho), \qquad |C^{\mathrm{dir}}_{q,\lambda} - C^{\mathrm{it},\boldsymbol{D}}_{q,\lambda}| \le \mathcal{Q}_{\varepsilon,q}(\varrho)\mathsf{e}_{\boldsymbol{D}}(\varrho)\,. \tag{4.40}$$

The coefficient expressions are smooth in neighborhoods of these geometries. For every additional fixed spectator list $\boldsymbol{B}$*, truncation* D *and clearance* $\delta_{\mathrm{sp}} > 0$*, there is such a threshold, uniform in every state, with*

$$|\mathcal{R}_D^{\zeta}(\boldsymbol{A}, z_{\mathrm{leaves}}; \boldsymbol{B}, y; z_{o_{\mathrm{root}}})| \le \mathcal{Q}_{\varepsilon,D+1}(\varrho)\,. \tag{4.41}$$

The same bound holds in the norm of the compressed commutative algebra before evaluation in a phase or in the mixed vacuum.

(d) The three bounds in (c) have integrated versions. Precisely, after fixing the data, a measurable parameter space and $B_0 \in \mathbb{R}$*, a threshold can be chosen before the measure, weight, scale vector and configuration maps, so that for any one fixed admissible scale vector and almost-everywhere measurable geometries satisfying the conditions for* λ_Z*-almost every point with* $\vartheta_Z \neq 0$*,* $\int |\vartheta_Z| \le B_0$ *implies bounds* $B_0\mathcal{Q}_{\varepsilon,q}$*,* $B_0\mathcal{Q}_{\varepsilon,q}\mathsf{e}_{\boldsymbol{D}}$ *and* $B_0\mathcal{Q}_{\varepsilon,D+1}$ *for the respective weighted expressions. The tested bound permits any almost-everywhere measurable phase map, and also the mixed state; the compared weighted expressions are integrable.*

Block a local product from its cluster scale towards its spectator scale. Retain the low-degree fields and bound the discarded part in the centered chessboard norm. One coherent family of separating correlation functionals extracts the new field component in each degree; reality and rotation averaging preserve this choice. An estimate in the compressed algebra bounds the remainder in every phase and the mixture at once. Integration preserves the bound for bounded total variation and measurable phase choices. Recursive expansion over the tree adds the separate proper-cluster errors, producing the two rates in (c). Proved in Section 16.

Remark 4.19. The nested coefficient error contains the sum $\mathsf{e}_{\boldsymbol{D}}$ of the proper-vertex gains. Its geometry controls all descendants and separation across different child subtrees.

Proposition 4.20 (P13. Contact extensions)**.** *For the admissible constants of Proposition 4.1 and the table and phases of Propositions 4.6–4.11, the contact construction uses exactly the field projections of the prescription chosen in Proposition 4.18, and:*

(i) the strata $\operatorname{ran}\mathsf{P}^{\mathrm{OPE}}_{=q}$ *have real bases whose union contains the unit;*

(ii) there are positive numbers δ_q^{ct} *with*

$$\delta_q^{\mathrm{ct}} \to 0, \qquad 4\delta_{q+1}^{\mathrm{ct}} < \delta_q^{\mathrm{ct}}, \qquad 2(q+1)\delta_q^{\mathrm{ct}} \le \delta_q^{\mathrm{pres}}\,, \tag{4.42}$$

where $\delta_q^{\mathrm{pres}} > 0$ *is the clearance of the prescription at degree* q*;*

(iii) for every state $\zeta \in \mathsf{X} \sqcup \{\mathrm{mix}\}$*, every arity* n *and every list* $A_i \in F_{d_i}$ *there is a distribution* $\mathfrak{D}_{\boldsymbol{A}}^{\zeta}$ *on* $\mathbb{R}^{4n}$*, built in Section 16 from these field projections, with*

$$\mathfrak{D}_{\boldsymbol{A}}^{\zeta}(f) = \int f(x) S_n^{\zeta}(\boldsymbol{A}; x)\, dx \qquad (f \in C_c^{\infty}(\mathrm{Conf}_n))\,. \tag{4.43}$$

For every compact K *it has a finite seminorm bound as in Definition 2.25; the order and constant may depend on* $\zeta, \boldsymbol{A}, K$ *and the fixed subtraction. No uniformity in the state is required by this statement.*

Resolve collisions into nested and disjoint subsets and localize their normal scales. For a colliding subset I, put $d_I = \sum_{i\in I} d_i$, $n_I = 4(|I|-1)$ and $L_I = \max(0, d_I + 1 - n_I)$. Retain field weights $w < L_I$ and subtract the normal Taylor polynomial of the test through order $L_I - 1 - w$ in each such term. For $0 < \varepsilon < 1$, on a shell of normal radius r_{sh} away from further partial collisions, the discarded part is bounded by $C r_{\mathrm{sh}}^{L_I - d_I - \varepsilon}$. A retained weight-$w$ coefficient has bound $C r_{\mathrm{sh}}^{w - d_I - \varepsilon}$,

while its test remainder contributes $r_{\mathrm{sh}}^{L_I-w}$. Including normal volume bounds the absolute shell integral, for some finite test order k_{ct}, by

$$C\|f\|_{C^{k_{\mathrm{ct}}}(\mathbb{R}^{4n})} r_{\mathrm{sh}}^{n_I+L_I-d_I-\varepsilon} \le C\|f\|_{C^{k_{\mathrm{ct}}}(\mathbb{R}^{4n})} r_{\mathrm{sh}}^{1-\varepsilon} \qquad (0 < r_{\mathrm{sh}} \le 1)\,. \tag{4.44}$$

Partial collisions within angular sectors are resolved by induction over finer partitions. Choose the finitely many exponent losses so that a positive power remains; the dyadic shell sums then converge absolutely. The common field projections make proper-subcluster subtractions agree. Real bases of the strata give (i). Choose $\delta_0^{\mathrm{ct}} \le \min(1, \delta_0^{\mathrm{pres}}/2)$ and recursively choose $\delta_{q+1}^{\mathrm{ct}} \le \min(\delta_q^{\mathrm{ct}}/8, \delta_{q+1}^{\mathrm{pres}}/[2(q+2)])$ to obtain (ii). The summed local test-seminorm bounds give a distribution. Away from all collision diagonals, the construction equals the ordinary correlation. Proved in Section 16.

Remark 4.21. The contact estimates use the same field projections as Proposition 4.18, independently of its separated and nested estimates. Uniqueness, positivity and Ward identities for contact extensions are not asserted.

5 Proof of Theorems A and B

The lattice family is chosen before the volume sequence. Full convergence at each fixed volume then leaves only a volume refinement to choose. All subsequent constructions use the resulting table and its quotient fields.

Proof of Theorems A and B.

Step 1: Choose the construction. Fix the group, its realization, probability Haar measure and trace inner product, and fix $\mathsf{b} > 0$ as in Theorem B. Nonabelianness gives $d_{\mathfrak{g}} > 0$. Since $G \subset U(n_G)$, each $\operatorname{ad} X$ is skew for (2.1), so

$$C_{\mathfrak{g}} = d_{\mathfrak{g}}^{-1} \sum_{\alpha} \|\operatorname{ad} e_\alpha^{\mathfrak{g}}\|_{\mathrm{HS}}^2 > 0\,. \tag{5.1}$$

The sum is positive because $[\mathfrak{g},\mathfrak{g}] \neq 0$; hence $\kappa_G > 0$. By Proposition 4.1 there are constants and a construction, chosen in the order of Section 6, including the additional lattice estimates required in Propositions 4.4, 4.5, 4.8, 4.15, 4.18 and 4.20; fix them.

Step 2: Choose the trajectories. Keep the level data, path actions, admissible classes and specified recursion of Proposition 4.1. Proposition 4.2 gives represented trajectories ending at $\bar\beta$ at level zero, their bare couplings $\beta_{N,p}$, and the tuning family within this recursion. Its uniform running law gives the slope in B1 of Theorem B.

Step 3: Fix the Wilson family. Use the local sections and normalized marked transfers of Proposition 4.1, and set

$$\mathcal{T} = \bigsqcup_{d\ge 0} V_d, \qquad I_{N,p}((d,\mathbf{c}),y;U) = O_{N,p}^{(d)}(\mathbf{c},y;U)\,. \tag{5.2}$$

The insertions are polynomials in the links on the fixed degree-d stencil, with fixed linear renormalization at each N,p. They are continuous and gauge invariant, proving Reg2–Reg4. At $m = N$, Proposition 4.2 gives Reg1 with $b_* = \kappa_G > 0$, since $s_N = \log(1/a_N)$. This proves regularity and the remaining lattice assertions B1 of Theorem B. The family, reference levels and token stencils are now fixed.

Step 4: Take the fixed-volume limits. Apply Propositions 4.4 and 4.5 to this family. For every p, n, v, obtain the kernel $f_{p,v}$, smooth on $\mathrm{NoHit}_{L_p,n}$, with all its fixed finite difference, interpolation and smear limits along the full sequence $N \to \infty$. This is B2 of Theorem B; its thresholds may depend on p.

Step 5: Choose one volume refinement. Let $\mathsf{p} : \mathbb{N} \to \mathbb{N}$ be strictly increasing. Proposition 4.6 supplies a strictly increasing q, $\mathsf{v} = \mathsf{p} \circ \mathsf{q}$, and one table S, with the field calculus and repeated-word estimate. By Proposition 4.6(a) the family converges to S along v in the sense of Definition 2.5. Its inner limits are the full limits $N \to \infty$ of Proposition 4.5 at every admissible volume, with the fixed-volume kernels identified by Proposition 4.6(b). This gives A-P1 in Theorems A and B, retaining v and S. Since $L_{\mathsf{v}(k)} \to \infty$, every fixed compact test support eventually lies in the open fundamental box.

Step 6: Normalize the stress tensor. Proposition 4.8 applies with its lattice comparison estimates for the fixed family. It supplies stable Maxwell frames, the conserved line, a nonzero current J and the raw shell maps. Proposition 4.10 supplies $c_\star$, the continuous $O(4)$ actions and the normalized tensor T, with both Ward identities and the joint comparison for all derivatives, arbitrary stress slots, transported derivatives and rotation generators. The scaling sequence of Proposition 4.10(e) is stable, so its ordinary frames are eventually exactly compatible with conjugation, H_4, units and inclusions. Together with Proposition 4.6, including $\dim_{\mathbb{C}} F_d = \dim_{\mathbb{C}} M_d^{\mathrm{Max}}$, these give the mixed assertions B3 of Theorem B. Refining comparison radii leaves the volume refinement fixed.

Step 7: Choose the phase. Proposition 4.11 gives the compression algebra, its compact metric spectrum X, the full-support probability measure ν and the phase tables. Each S^ξ satisfies E1–E7, has null spaces $\mathcal{N}_d(S)$ and thus descends to the same F_d, and has the same inclusions, derivatives, conjugation, $O(4)$ representations and stress laws. With the definition of Θ in Step 9, this proves B3 of Theorem B. Choose ξ_* from the full-measure set with one-dimensional OS ground space. This choice imposes no trace condition and is retained below.

Step 8: Reconstruct the theories. Apply Proposition 4.13 simultaneously to every $\xi \in \mathsf{X}$. Retain its theories and continuations, onto OS comparison maps, unique ground spaces, scalar bounded weak commutants, continued stress tensors, continuity and joint-test averages, and phase-uniform low-energy and heat bounds. These give B4 of Theorem B. For the selected hierarchy use $\mathfrak{W} = \mathfrak{W}^{\xi_*}$ and $\mathsf{F}_n = \mathsf{F}_n^{\xi_*}$. The family and its extraction remain fixed.

Step 9: Center the trace in the selected phase. With the tensor and phase just chosen, put

$$\tau = \sum_{a=1}^{4} T_{aa}, \qquad e_{\mathrm{tr}} = S^{\xi_*}(\tau), \qquad \Theta = \tau - e_{\mathrm{tr}} \mathbf{1}_4 \,. \tag{5.3}$$

Proposition 4.15 gives its nonzero scalar field, centered operator identity, full-radius asymptotic and unique positive spectral measure, including both high-energy test limits. Proposition 4.16 applies to this same trace and selected theory and gives all three non-freeness conclusions. These are B5 of Theorem B.

Step 10: Choose the products and subtractions. Proposition 4.18 gives one prescription—channel sets Ch_q, real fields $O_{q,\lambda}$, coefficients $c_{q,\lambda}$ and the associated field projections (Sec-

tion 16.1)—chosen before every spectator, state and truncation, with all its separated, integrated and nested estimates. Proposition 4.20, with the same projections, gives real stratum bases, scales δ_q^{ct} and contact distributions. Its proof uses its own lattice estimates, independently of the bounds of Proposition 4.18. These give B6 of Theorem B.

Step 11: Identify the hierarchy. Take $F_d = V_d/\mathcal{N}_d(S)$ and the injective response maps $\iota_d : F_d \to \mathcal{L}$ of Proposition 4.6, and put $\pi_{\mathrm{quo}}(d, \mathbf{c}) = (d, [\mathbf{c}])$. Changing a representative in slot i by $\mathbf{n} \in \mathcal{N}_{d_i}(S)$ changes $S_n(v; x)$ by a correlation with $\mathbf{n}$ in that slot. By multilinearity, permutation symmetry E1 and the definition of $\mathcal{N}_{d_i}(S)$, this correlation vanishes. The same argument applies in every phase by Step 7. Hence the quotient table satisfies

$$S_n^{\mathrm{mix}}(\pi_{\mathrm{quo}}(v_1), \dots, \pi_{\mathrm{quo}}(v_n); x) = S_n(v; x) \qquad (x \in \mathrm{Conf}_n)\,. \tag{5.4}$$

Use the phase data of Proposition 4.11, the selected theory and continuation of Proposition 4.13, the coefficients of Proposition 4.18 and the trace of Proposition 4.15. These supply H1–H8. On collision diagonals put $S_n^{\zeta} = 0$. Theorem A evaluates these functions on Conf_n: A-P1 through separated tests, R9–R12 and R14 by definition, and G_Θ at $r > 0$. In O2, the triangle inequality gives $|x_i - y_j| \geq \delta_{\mathrm{sp}} - M_{\mathrm{geo}} r > 0$ for $0 < r < \delta_{\mathrm{sp}}/(M_{\mathrm{geo}} + 1)$; decrease r_1 accordingly. Internal cluster separation and spectator clearance then put the whole configuration in Conf_n. The clause check below proves A-P1–A-P4 for this hierarchy. All assertions of Theorem B now hold. Setting $\mathsf{b} = 1$ and retaining A-P1–A-P4 proves Theorem A. □

The clauses of Theorem A. We check the data and each assertion in the order of their definitions.

Quantifiers and regularity. Steps 1–3, using Propositions 4.1 and 4.2, fix the strictly increasing positive integer sequences, probability Haar measure and regular family before p. Indeed, $D_N = \prod_{i<N} \ell_i$ is strictly increasing because $\ell_i \geq 3$, and L_p is strictly increasing by Proposition 4.1. Reg1 has the positive slope κ_G, uniformly in p; Reg2 is continuity; Reg3 is gauge invariance; Reg4 uses the fixed finite stencil. Proposition 4.6 then gives one increasing q for each increasing p.

H1–H8. Proposition 4.13 supplies H1 and H6, the selected relativistic datum and continuation. Proposition 4.6 supplies the finite-dimensional normed spaces and response maps H2; their quotient projections supply H3. Proposition 4.11 supplies H4 and, together with Proposition 4.6, the all-arity functions H5. Propositions 4.18 and 4.15 supply H7 and H8, respectively.

A-P1. For a separated test f, set $C_p^\infty(v; f) = \int f(x) f_{p,v}(x)\, dx$ at every admissible volume, and assign any value at inadmissible volumes. Proposition 4.5 gives the full inner limit there. Eventual admissibility follows from $L_{\mathsf{v}(k)} \to \infty$. Proposition 4.6(a) and (5.4) give the outer limit to $\int f(x) S_n^{\mathrm{mix}}(\pi_{\mathrm{quo}}(v_1), \dots, \pi_{\mathrm{quo}}(v_n); x)\, dx$, on the one refinement for every request.

A-P2, R1. Proposition 4.13(a)–(b) gives W0a–c: the translation spectral representation, forward-cone support and unique invariant vacuum line. It gives W1a–e: normalized vacuum, dense domain equal to the field-word span, continuous Schwartz smearing on each domain vector and the adjoint relation. It also gives W2a–d, the label representation law, domain and vacuum invariance and field covariance, and W3, bosonic spacelike locality.

R2–R6. Faithfulness in Proposition 4.13(a) gives R2. The injective response maps of Proposition 4.6(d) give R3. By E1 in Proposition 4.6(c), $S_1(\mathbf{u}_0(x_0)) = S_0 = 1$, so $\mathbf{1}_0 \neq 0$ and $\iota_0 \mathbf{1}_0 \neq 0$ by R3; R2 then gives R4. The directed-union property of Proposition 4.6(d) gives R5, and choosing a representative in each quotient gives the surjectivity R6.

R7–R12. Proposition 4.11 gives normalization and full support of ν (R7–R8), pointwise phase continuity (R11), and the phase-average identity (R12). Its Euclidean properties, with

Proposition 4.6(c) for the mixture, give permutation symmetry (R9) and smoothness on every Conf_n (R10).

R13–R15. Proposition 4.13(a) gives holomorphy in ordered complex times at fixed real spatial positions, agreement at ordered Euclidean times, and every product-test boundary limit for each strictly ordered height list. R15 retains the stated zero value for a nonintegrable positive-height integral; no extra integrability assertion is needed.

A-P3, O1–O2. Proposition 4.18(a) gives each coefficient bound with its own threshold (O1), and one coefficient system for every spectator, state and truncation (O2), with coefficient 1 in both bounds. The remainder threshold is uniform in positions and state.

A-P4, T1–T4. Proposition 4.15 gives selected-vacuum centering (T1) and the full positive-radius limit (T3). Its amplitude is $A_G = 3d_{\mathfrak{g}}/\pi^4 > 0$, since $d_{\mathfrak{g}} > 0$ (T2). Proposition 4.16(a) excludes equality with every finite Wick profile on every interval $(0, \delta)$ (T4).

6 The constants and the order of their choice

Proposition 4.1 asserts the existence of one choice of constants and one construction with all the stated properties. The table below describes that choice; its numerical restrictions are properties of this construction. Each row may use only preceding rows, and stage-indexed choices are made for every fixed $j \geq 4$ before the entry levels are chosen.

No.	Choice	Dependence, restrictions and use
1	$G \subset U(n_G)$, dh, metric, $d_{\mathfrak{g}}, C_{\mathfrak{g}}, b_0, \kappa_G$; $\mathsf{b} > 0$	The prescribed realization and penalty; $\kappa_G > 0$. These are fixed throughout.
2	Kernel cutoffs $\delta_{\mathrm{fr}}, \varepsilon_{\mathrm{pred}}$ and axial factor a_{ax}	G; positive sufficiently small cutoffs and $a_{\mathrm{ax}} \geq 16$ of the block-spin construction (Section 8), with the branch conditions at every patch. The penalty remains fixed. Proposition 4.1, B1 of Theorem B.
3	Homogeneous label bases, norms, local sections, stencils	G and degree; Taylor normalization, reality and H_4 covariance. These precede the ratios needed for their bounds. Propositions 4.4 and 4.5, B1.
4	$\mathfrak{q}_j, \varpi_j, \sigma_j$, clipping constants, $R_j^{\mathrm{rel}}, w_j^{\mathrm{path}}$, $A_j^{\mathrm{path}}, C_j, \omega_j^{\mathrm{RG}}$	Rows 1–3 and j; $\mathfrak{q}_j \geq 6$, $\varpi_j \geq 2$, both tending to infinity, $\sigma_j = 100$, $C_j \geq 1$, $R_j^{\mathrm{rel}} w_j^{\mathrm{path}} \geq 12$; positive clipping constants, relaxation lengths, widths and path bounds. Choose $\omega_j^{\mathrm{RG}} \geq 0$ at least the bound on the one-loop remainder of the coupling step (Section 9). Fix all constants entering ratio bounds, including the lattice estimates of Propositions 4.8, 4.15, 4.18 and 4.20.
5	Odd nondecreasing stage ratios ℓ_j^{st}	Rows 1–4; exceed every ratio threshold and satisfy $\omega_j^{\mathrm{RG}} / \log \ell_j^{\mathrm{st}} \leq \min(\kappa_G/4, j^{-2})$ and the logarithmic bound in Definition 2.7. Propositions 4.1 and 4.2.
6	Antitone exponents θ_j; coupling entry bounds β_j^{ent}	Rows 1–5; choose $0 < \theta_j \leq j^{-1}$. Take the finite maximum of the remaining coupling and smallness thresholds as β_j^{ent}, a lower bound on $\widehat{\beta}$ at levels using stage j (Section 11.1). Further entry bounds may be raised. Propositions 4.1, 4.2, 4.4, 4.5, 4.8, 4.15, 4.18 and 4.20.

7	$\bar\beta$, entry levels, minimum periods	Rows 1–6; choose $\bar\beta > 1$ at least the first-stage entry bound, with $\beta^{-1/2}\exp((\log\log\beta)^2) \le 1$ for every $\beta \ge \bar\beta$. Then choose entry levels and minimum periods for the cover and patch entrance conditions of Definition 2.7; these choices may be increased. Proposition 4.2.
8	$J_m, \ell_m, D_m, a_m, L_p$, $m_{\mathrm{ref}}(d)$	Rows 1–7; gentle growth, patch and cover fits, and volume divisibility; hence $0 < \theta_{J_m} \le J_m^{-1}$ by row 6. Require $J_{m_{\mathrm{ref}}(d)} \ge d+3$ and sufficient derivative orders thereafter. Propositions 4.2, 4.4 and 4.5.
9	$\widehat\beta_m, M_m^{\mathrm{cut}}, t_m, B_m$, $\epsilon_m^{\mathrm{reg}}, \epsilon_m^{\mathrm{for}}, \Lambda_m^{\mathrm{loop}}, \mathfrak{B}_m$	The formulas of Definition 2.7 determine these from rows 1–8. Since $s_m \ge 0$, row 7 gives $\widehat\beta_m \ge \bar\beta > 1$ and $t_m \le 1$. Propositions 4.1, 4.2 and 4.4.
10	Path actions, classes, recursion, trajectories, $\beta_{N,p}$, w_{tun}	Rows 1–9 and Propositions 4.1 and 4.2; the common level-zero endpoint and the specified recursion apply for every finite N, p. Used in Propositions 4.4 and 4.5, B1.
11	Marked transfers, terminal maps, renormalized insertions	Rows 1–10; normalize at the fixed $m_{\mathrm{ref}}(d)$. The entire Wilson family is fixed before a volume sequence is prescribed. Used in every continuum conclusion.
12	p, then $\mathsf{q}, S, F_d, \partial_a, \mathcal{L}$	Prescribe arbitrary strictly increasing p after row 11. Proposition 4.6 chooses one refinement and table, using the full inner limit of Proposition 4.5.
13	Comparison radii, frames, J, charges, $c_\star, R_d, T$	Row 12 and the fixed family; Proposition 4.8, then Proposition 4.10. Comparison radii may be refined while the volume refinement remains fixed.
14	Slab operators, X, ν, S^ξ, then ξ_*	Row 13 and Proposition 4.11; select ξ_* only for a one-dimensional OS ground space.
15	All-phase theories and continuations; Θ, μ_Θ	Rows 12–14; Proposition 4.13, then Proposition 4.15. Centering uses the same selected phase. Also used in Proposition 4.16.
16	One OPE prescription and coefficients; contacts, real bases, positive scales δ_q^{ct}	Rows 12–14 and the fixed family; Proposition 4.18, then Proposition 4.20 with unchanged projections. Require $4\delta_{q+1}^{\mathrm{ct}} < \delta_q^{\mathrm{ct}}$ and $2(q+1)\delta_q^{\mathrm{ct}} \le \delta_q^{\mathrm{pres}}$.
17	Thresholds N_0, k_0, r_0; estimate constants; local distribution seminorms	Exactly the fixed data preceding each threshold in its statement. There is no dependence on a later position, volume or state whenever uniformity is asserted. These are chosen for individual estimates.

The order is acyclic because the constants entering a ratio bound are fixed before that ratio, whereas the remaining smallness requirements are met by increasing entry levels. At each fixed stage only finitely many additional bounds are combined, and increasing entry levels preserves earlier inequalities. The reference coupling in row 7 exists because $-\frac{1}{2}\log\beta + (\log\log\beta)^2 \to -\infty$ as $\beta \to \infty$. Proposition 4.1 makes these choices jointly with the represented densities, marked transfers and additional lattice estimates. Rows 1–11 precede the prescribed volume sequence in row 12; the choices in rows 12–16 are supplied by the existence statements of Propositions 4.6,

4.8, 4.10, 4.11, 4.13, 4.15, 4.18 and 4.20. Row 9 is determined by formulas, and row 17 consists of thresholds for fixed requests.

Part II
Proofs

7 Fields, labels and norms

This section is under construction.

8 The renormalization step

This section is under construction.

9 Effective actions and the marginal flow

This section is under construction; its subsection on the marginal flow is below.

9.1 The marginal flow

The curvature-square term has the same scaling as the action in four dimensions. We extract its coefficient after each integration, compute its one-loop change, and absorb the remaining scalar correction into the coupling by an identity of densities. Throughout we use the single construction of Proposition 4.1.

Fix N, p and a step $m+1 \to m$, and write $\ell = \ell_m$ and $\Lambda = \Lambda_{n_{p,m}}$. Separate the path action as $P^{\mathrm{path}}_{N,m}(\beta, U) = \beta \mathcal{K}^{\mathrm{kin}}_{N,m}(U) + \mathcal{L}^{(2)}_{N,m}(U)$. The kinetic term is the H_4-average of the clipped plaquette terms, masked row squares $\frac{1}{2}\chi^{\mathrm{hlth}}_{m,x}|\mathcal{L}^{\mathrm{path}}_{\mathfrak{p}}|^2$, and cubic and quartic atoms; $\mathcal{L}^{(2)}_{N,m}$ is the quadratic-atom sum. Near the identity masks and clipping agree with their analytic expressions. The quadratic Taylor kernel $\widehat{K}^{\mathrm{kin}}_{N,m}$ has Maxwell normalization

$$[t^2]\widehat{K}^{\mathrm{kin}}_{N,m;(2,\nu),(2,\nu)}(te_1) = \tfrac{1}{2}\,. \tag{9.1}$$

Here ν is any colour in the fixed orthonormal basis, the kernel represents $D^2/2!$, and $[t^2]$ denotes Taylor coefficient. Momenta use unit lattice coordinates.

Lemma 9.1 (Extraction of the Maxwell term)**.** *For an analytic, real two-leg kernel $\widehat{Q}$ with the gauge, colour, exchange and signed-permutation symmetries of the coarse effective action, set*

$$\alpha^{(1)}_{F^2}(\widehat{Q}) = 2\,\mathrm{Re}[t^2]\widehat{Q}_{(2,\nu),(2,\nu)}(te_1)\,. \tag{9.2}$$

Then $\widehat{Q} - \alpha^{(1)}_{F^2}(\widehat{Q})\widehat{K}^{\mathrm{kin}}_{N,m}$ vanishes to fourth order at zero momentum. The level-$(m+1)$ quadratic atoms have zero coefficient in (9.2), also after composition with the stationary branch.

Proof. Gauge invariance gives the two Ward contractions with $e^{i\mathbf{p}^a} - 1$. Their lowest terms eliminate the constant term and make the linear coefficient totally antisymmetric in its three indices; reflection in one of these axes eliminates it. Simplicity makes the invariant quadratic colour

form a multiple of the trace metric. Reflections, permutations and the Ward contractions leave only $\frac{1}{2}\alpha^{(1)}_{F^2}(\widehat{Q})(|\mathbf{p}|^2\delta_{ab}-\mathbf{p}^a\mathbf{p}^b)\delta_{\nu\nu'}$. Equation (9.1) fixes the multiplier. After subtraction, the lower coefficients vanish, so reflection in all axes eliminates the cubic term. The quadratic-atom kernel is $\widehat{s}(-\mathbf{p})^{\mathsf{T}}\widehat{C}(\mathbf{p})\widehat{s}(\mathbf{p})$, where $\widehat{s}$ is the linearized curvature. Each curvature contributes one momentum factor; the vanishing zeroth and first moments give $\widehat{C}(\mathbf{p})=O(|\mathbf{p}|^2)$. Composition with the branch preserves this fourth-order vanishing by the alias calculation below, with an additional factor ℓ^{-2} in its bound. □

We first identify the determinant. The joint variables are fine links U and a frame at each block root. The joint action $\mathcal{S}_V$ is the β-coefficient of $P^{\mathrm{path}}_{N,m+1}$, without quadratic atoms, plus $\mathsf{b}\sum_e \mathrm{pr}(V_eQ_e^{\mathrm{pred}}(U)^{-1})$. It is invariant under fine gauge transformations acting on both variables. Near the identity, frames and predictors are the tag-independent minimizers of Definition 2.8. Its restriction to the axial slice is $S(\mathcal{A},\mathcal{Z})$, with coarse increments $\mathcal{A}$ and free fine-link variables $\mathcal{Z}$. The branch satisfies $\partial_{\mathcal{Z}}S(\mathcal{A},\mathcal{Z}_*(\mathcal{A}))=0$, $\mathcal{Z}_*(0)=0$. Section 8 gives $\partial^2_{\mathcal{Z}}S(0,0)\geq\eta_{\mathrm{Hess}}\mathrm{Id}$ uniformly in the torus. The analytic implicit function theorem on these finite-dimensional link spaces gives

$$D\mathcal{Z}_*(0)=-\big(\partial^2_{\mathcal{Z}}S(0,0)\big)^{-1}\partial_{\mathcal{A}}\partial_{\mathcal{Z}}S(0,0)\,. \tag{9.3}$$

This determines the free interpolant A_0. Fourth-order matching in Section 9 identifies the tree action with the coarse leading action through order four. Gaussian integration adds $\frac{1}{2}\log\det H_{\mathrm{ax}}-\log j_G$, where H_{ax} is the axial Hessian along the branch and j_G its product Haar chart density. Denote its infinite-volume quadratic kernel by $\widehat{Q}_{\mathrm{det}}$ (existence follows at the end of the next proof), and put $\alpha=\alpha^{(1)}_{N,m}\coloneqq\alpha^{(1)}_{F^2}(\widehat{Q}_{\mathrm{det}})$.

Let R_{g} insert joint infinitesimal gauge directions and R_{ax} insert the free axial links. Transport the root frame along its axial tree to every fine site; X_{fr} is minus the left-logarithmic derivative of this transported frame. Its covariance gives $X_{\mathrm{fr}}R_{\mathrm{g}}=\mathrm{Id}$. Write pr_{f} for projection onto fine links, so $\mathrm{pr}_{\mathrm{f}}R_{\mathrm{g}}=d_U$.

Lemma 9.2 (Determinant identification)**.** *Put $\mu_{\mathrm{reg}}=\ell^{-1}$ and let H_{jt} be the left-increment joint Hessian along the stationary branch. Define*

$$F_{\mathrm{gf}}=d_U^*\,\mathrm{pr}_{\mathrm{f}}+\mathfrak{m}_{\mu_{\mathrm{reg}}}X_{\mathrm{fr}},\qquad \mathsf{K}^{\mathrm{comp}}_{\mu_{\mathrm{reg}}}=H_{\mathrm{jt}}+F_{\mathrm{gf}}^{\mathsf{T}}F_{\mathrm{gf}},\qquad \mathsf{G}^{\mathrm{gh}}_{\mu_{\mathrm{reg}}}=F_{\mathrm{gf}}R_{\mathrm{g}}.$$

Near the trivial coarse field,

$$\tfrac{1}{2}\log\det H_{\mathrm{ax}}-\log j_G=\tfrac{1}{2}\log\det\mathsf{K}^{\mathrm{comp}}_{\mu_{\mathrm{reg}}}-\log\det\mathsf{G}^{\mathrm{gh}}_{\mu_{\mathrm{reg}}}\,. \tag{9.4}$$

Let $\widehat{Y}_{\mu_{\mathrm{reg}}}$ be the infinite-volume quadratic Taylor kernel of

$$\Gamma^{(1)}_{\mu_{\mathrm{reg}}}(U)=\tfrac{1}{2}\log\det(\mathcal{H}(U)+d_Ud_U^*+\mathcal{M}_{\mu_{\mathrm{reg}}}(U))-\log\det(d_U^*d_U+\mathfrak{m}_{\mu_{\mathrm{reg}}}(U))\,. \tag{9.5}$$

Then a constant C_{cmp}, independent of ℓ,N,p at fixed stage, satisfies

$$|\alpha-\alpha^{(1)}_{F^2}(\widehat{Y}_{\mu_{\mathrm{reg}}})|\leq C_{\mathrm{cmp}}\,. \tag{9.6}$$

Here $\mathcal{H}$ is the fine leading-action Hessian, $d_U\phi(y,a)=\phi(y)-\mathrm{Ad}(U(y,a))\phi(y+e_a)$, $\mathfrak{m}_{\mu_{\mathrm{reg}}}=\mu^{20}_{\mathrm{reg}}(d_U^*d_U+\mu^2_{\mathrm{reg}})^{-9}$, and $\mathcal{M}_{\mu_{\mathrm{reg}}}$ applies this regulator to each link component at its initial site.

The action (9.5) uses background-field Feynman gauge: vector gauge fixing $d_Ud_U^*$ and ghost $d_U^*d_U$, with infrared regulation at the block scale; see [1, Secs. 2.3 and 3.1] for the continuum convention. It is invariant under gauge transformations, colour rotations and unsigned axis permutations. We use only its coefficient (9.2); the full reflection hypothesis of Lemma 9.1 is imposed on $\widehat{Q}_{\mathrm{det}}$.

Proof. Step 1: The axial determinant. Order sites by height in the axial forest. Tree links equal 1_G; a gauge direction ξ changes a root frame by $-\xi(x)$ and a tree link $y \to y'$ by $\pm(\xi(y) - \xi(y'))$. Thus R_{g} is triangular on these coordinates with diagonal ± 1, while R_{ax} is the identity on free links. Hence $(R_{\mathrm{g}}, R_{\mathrm{ax}})$ is a basis with squared determinant one. Gauge invariance and axial stationarity make $\mathcal{S}_V$ critical in this basis. Differentiating gauge invariance at this critical point gives $H_{\mathrm{jt}} R_{\mathrm{g}} = 0$. In this basis the completed form is

$$\begin{pmatrix} \mathsf{G}^{\mathrm{gh}}_{\mu_{\mathrm{reg}}} & F_{\mathrm{gf}} R_{\mathrm{ax}} \\ 0 & \mathrm{Id} \end{pmatrix}^{\mathsf{T}} \begin{pmatrix} \mathrm{Id} & 0 \\ 0 & R_{\mathrm{ax}}^{\mathsf{T}} H_{\mathrm{jt}} R_{\mathrm{ax}} \end{pmatrix} \begin{pmatrix} \mathsf{G}^{\mathrm{gh}}_{\mu_{\mathrm{reg}}} & F_{\mathrm{gf}} R_{\mathrm{ax}} \\ 0 & \mathrm{Id} \end{pmatrix} .$$

If T_{chart} is the derivative of the axial chart into left increments, stationarity and Haar change of variables give $H_{\mathrm{ax}} = T_{\mathrm{chart}}^{\mathsf{T}} R_{\mathrm{ax}}^{\mathsf{T}} H_{\mathrm{jt}} R_{\mathrm{ax}} T_{\mathrm{chart}}$ and $j_G = |\det T_{\mathrm{chart}}|$. Taking determinants and logarithms proves (9.4). Positivity holds at zero and persists nearby. Any completing map with invertible $F_{\mathrm{gf}} R_{\mathrm{g}}$ gives this identity. The stated choice gives exactly $\mathsf{G}^{\mathrm{gh}}_{\mu_{\mathrm{reg}}} = d_U^* d_U + \mathfrak{m}_{\mu_{\mathrm{reg}}}$, and supplies the vector gauge-fixing term $d_U d_U^*$.

Step 2: Linearization and the predictor Hessian. The completed log-determinant difference, as a function of coarse and joint increments, is colour invariant. Since $\mathfrak{g}$ has no invariant vector, its first derivative at zero vanishes. The Hessian chain rule therefore allows us to replace the stationary branch by its linearization $\Phi(\mathcal{A})$, with coarse input $\mathcal{A}$, fine field $A_0 \mathcal{A}$ and its frame.

At $\Phi(\mathcal{A})$ split $\mathsf{K}^{\mathrm{comp}}_{\mu_{\mathrm{reg}}} = \mathsf{K}^{\mathrm{core}} + C_{\mathrm{ct}}$. The core is $\mathrm{pr}_{\mathrm{f}}^{\mathsf{T}} \mathcal{H} \mathrm{pr}_{\mathrm{f}} + (DQ^{\mathrm{pred}})^{\mathsf{T}} \mathsf{B}_{\mathrm{pen}} DQ^{\mathrm{pred}} + F_{\mathrm{gf}}^{\mathsf{T}} F_{\mathrm{gf}}$, where $\mathsf{B}_{\mathrm{pen}}$ is the penalty Hessian in predictor coordinates. The contact term C_{ct} is its first derivative contracted with $D^2 Q^{\mathrm{pred}}$, all in left-logarithmic coordinates. It vanishes at zero because the penalty is critical at $V_e = Q_e^{\mathrm{pred}}$. Stationarity identifies its linear coefficient with a column x_b of the coarse kinetic symbol. Expanding

$$D^2 \log\det A[v,w] = \mathrm{tr}\big(A^{-1} D^2 A[v,w] - A^{-1} DA[v] A^{-1} DA[w]\big) \tag{9.7}$$

for this split gives the contact contribution, evaluated on momentum te_1 and polarization $(2,\nu)$,

$$\frac{1}{4} \sum_b \big(x_b(t) \mathsf{y}_b(t) + x_b(-t) \mathsf{y}_b(-t)\big) - \frac{1}{8} \sum_{b,c} x_b(-t) x_c(t) \mathsf{f}_{bc}(t) .$$

Here y_b is the integrated mixed link–frame inverse contraction, and f_{bc} the integrated frame contraction of the squared inverse. The Ward identity gives $\mathsf{y}_b(t) = O(t)$, while $x_b(t) = O(t^2)$ and $\mathsf{f}_{bc}(t) = O(1)$. Thus this contribution is $O(t^3)$ and has zero coefficient (9.2).

Step 3: A comparison uniform in the block ratio. On the joint space compare the core to $\mathsf{K}^{\mathrm{aux}} = (\mathcal{H} + d_U d_U^* + \mathcal{M}_{\mu_{\mathrm{reg}}}) \oplus \mathrm{Id}_{\mathrm{frame}}$ at $U = \exp(A_0 \mathcal{A})$. Its determinant is the vector determinant in (9.5) composed with A_0. The ghost operators coincide at this same U, so cancel exactly. The common fine Hessian also cancels. The vector difference consists of the penalty term, the cross term $\mathrm{pr}_{\mathrm{f}}^{\mathsf{T}} d_U \mathfrak{m}_{\mu_{\mathrm{reg}}} X_{\mathrm{fr}}$ and its transpose, $X_{\mathrm{fr}}^{\mathsf{T}} \mathfrak{m}^2_{\mu_{\mathrm{reg}}} X_{\mathrm{fr}}$, and $-\mathcal{M}_{\mu_{\mathrm{reg}}} \oplus (-\mathrm{Id}_{\mathrm{frame}})$.

The penalty term and $\mathrm{Id}_{\mathrm{frame}}$ have rank at most $C d_{\mathfrak{g}}$ per coarse site. Every remaining term contains a regulator, whose Fourier decay is summable: in coordinates scaled by one block, its trace bound is controlled by

$$\sum_{j \in \mathcal{J}_\ell} (1 + |\mathbf{p} + 2\pi j|^2)^{-9} \le C \sum_{j \in \mathbb{Z}^4} (1 + |j|^2)^{-9} < \infty ,$$

where $\mathcal{J}_\ell$ is the centered set of ℓ^4 aliases and $\mathbf{p} \in [-\pi, \pi]^4$. Indeed $\mathfrak{m}_{\mu_{\mathrm{reg}}} = \mu_{\mathrm{reg}}^2 (1 + \mu_{\mathrm{reg}}^{-2} d^* d)^{-9}$ at zero. The factor μ_{reg}^2 is canceled by rescaling the quadratic form. The interpolation estimates

of Section 8, $\|\nabla^k A_0\|_{\mathrm{loc}} \le C\ell^{-1-k}$ for $0 \le k \le 2$, give the same trace bound for the first two background derivatives. Here ∇ is a fine-lattice difference, and $\|\cdot\|_{\mathrm{loc}}$ is the maximum weighted row and column sum, with weight $e^{\zeta|x-y|_1/\ell}$ for a fixed $\zeta > 0$. Finite rank, the displayed summability, and these interpolation bounds give a trace-norm bound Z_* per coarse site, independent of ℓ.

The inverse and first two background-derivative bounds in a strip $\mathfrak{S}_\eta = \{|\operatorname{Im} \mathbf{p}^a| < \eta\}$ are F_* and X_*. Subtract (9.7) for the two operators using $A^{-1} - B^{-1} = A^{-1}(B - A)B^{-1}$. Every product contains one trace-bounded difference, giving $F_* Z_* (1 + 3F_* X_* + 2(F_* X_*)^2)$ per coarse site. Cauchy's estimate gives a bound $C\eta^{-2}$ for (9.2). These strip bounds are fixed before ℓ; η_{Hess} is used only for the branch and positivity, and does not enter C_{cmp}.

Step 4: Aliasing and volume. The coarse kernel of $\Gamma^{(1)}_{\mu_{\mathrm{reg}}} \circ A_0$ sums the ℓ^4 fine Fourier aliases. At a nonzero alias the interpolant, modulo a gauge direction, is $O(t^2)$. Ward contractions remove the gauge parts, leaving $O(t^4)$. The zero alias agrees through order two with $\ell^4 \ell^{-2} \widehat{Y}_{\mu_{\mathrm{reg}}}(te_1/\ell)$; momentum rescaling contributes another ℓ^{-2}. Thus its multiplier is one. This proves (9.6). Analyticity gives, for fixed ℓ, a normalized torus-sum error $C_\ell e^{-c_\ell n_{p,m}}$. The Fourier integrals define $\widehat{Q}_{\mathrm{det}}$; testing on growing momentum meshes transfers the gauge, colour and signed-permutation symmetries of the coarse effective action to this kernel. □

Lemma 9.3 (The one-loop slope)**.** *There is C_{loop}, depending on the stage bounds and the fixed group and penalty, but not on ℓ, N, p, such that*

$$|\alpha^{(1)}_{N,m} + \kappa_G \log \ell| \le C_{\mathrm{loop}}\,, \qquad \kappa_G = \frac{11 C_{\mathfrak{g}}}{24\pi^2} > 0\,. \tag{9.8}$$

Proof. By simplicity, the invariant form $-\operatorname{tr}_{\mathfrak{g}}(\operatorname{ad} X)^2$ is a multiple of $|X|^2$; averaging over the orthonormal basis identifies the multiple as $C_{\mathfrak{g}}$. Thus $\operatorname{tr}_{\mathfrak{g}}(\operatorname{ad}(e^{\mathfrak{g}}_\nu)^2) = -C_{\mathfrak{g}}$ for each ν.

Use background $e^{itx^1} e_2 \otimes e^{\mathfrak{g}}_\nu$ and loop four-momentum $\mathbf{q}$, with q_a its ath coordinate. The continuum vector operator is $-D_A^2$ with spin term in $\operatorname{ad} F_{ab}$; the ghost is $-D_A^2$. Their orbital vertex is $-2iq_2$; the vector spin vertex has entries $\pm 2it$ in the $(1,2)$ plane. Momenta $\mathbf{q}$ and $\mathbf{q} + te_1$ run through the two propagators. The vector and ghost bubble traces are respectively $-C_{\mathfrak{g}}(4(-2iq_2)^2 - 8t^2)$ and $-C_{\mathfrak{g}}(-2iq_2)^2$. With weights $1/2$ and -1, their contribution to (9.7) is $-4C_{\mathfrak{g}}(q_2^2 + t^2)/(|\mathbf{q}|^2|\mathbf{q} + te_1|^2)$. Its $[t^2]$ coefficient is

$$h_{\mathrm{bub}}(\mathbf{q}) = 4C_{\mathfrak{g}} \left(\frac{q_2^2}{|\mathbf{q}|^6} - \frac{4q_1^2 q_2^2}{|\mathbf{q}|^8} - \frac{1}{|\mathbf{q}|^4} \right). \tag{9.9}$$

The first two terms expand the shifted propagator; the last is the spin contribution.

Let $k^{\mathrm{lat}}_{\mathrm{bub}}(\mathbf{q})$ denote $\frac{1}{2}\partial_t^2$ of the unintegrated second variation of (9.5), including the second-vertex term. The quadratic Taylor factor $1/2!$ cancels the 2 in (9.2), so

$$\alpha^{(1)}_{F^2}(\widehat{Y}_{\mu_{\mathrm{reg}}}) = (2\pi)^{-4} \int_{[-\pi,\pi]^4} \operatorname{Re} k^{\mathrm{lat}}_{\mathrm{bub}}(\mathbf{q})\, d^4\mathbf{q}\,. \tag{9.10}$$

Uniformly for $0 < \mu_{\mathrm{reg}} \le 1$, the lattice estimates give

$$\begin{aligned} |k^{\mathrm{lat}}_{\mathrm{bub}}(\mathbf{q})| &\le C_{\mathrm{ir}}(|\mathbf{q}| + \mu_{\mathrm{reg}})^{-4}, \\ |k^{\mathrm{lat}}_{\mathrm{bub}}(\mathbf{q}) - h_{\mathrm{bub}}(\mathbf{q})| &\le C_{\mathrm{sh}}(|\mathbf{q}|^{-3} + \mu_{\mathrm{reg}}^2 |\mathbf{q}|^{-6}) \quad (|\mathbf{q}| \ge \mu_{\mathrm{reg}})\,. \end{aligned} \tag{9.11}$$

Here we use the fourth-order matching estimates of Section 9: the quadratic, cubic and quartic kernels of the leading action minus Wilson vanish to momentum orders three, two and one. Thus the

completed free symbol is $|\mathbf{q}|^2\mathrm{Id}+O(|\mathbf{q}|^3)+O(\mu_{\mathrm{reg}}^2)$. Its inverse is bounded by $C(|\mathbf{q}|+\mu_{\mathrm{reg}})^{-2}$: the quadratic-form lower bound of Section 8 applies for $|\mathbf{q}|\ge\mu_{\mathrm{reg}}$, and $\mathfrak{m}_{\mu_{\mathrm{reg}}}\ge 2^{-9}\mu_{\mathrm{reg}}^2$ for $|\mathbf{q}|\le\mu_{\mathrm{reg}}$. The resolvent identity then bounds the difference from $|\mathbf{q}|^{-2}\mathrm{Id}$ by $C(|\mathbf{q}|^{-1}+\mu_{\mathrm{reg}}^2|\mathbf{q}|^{-4})$. Replacing one propagator or vertex by its difference and taking two external derivatives gives the second estimate in (9.11). The second-vertex term is bounded by $C(|\mathbf{q}|^{-2}+\mu_{\mathrm{reg}}^2|\mathbf{q}|^{-6})$ on the unit ball. The regulated versions of the same products give the first estimate.

Spherical averages give $\langle q_2^2\rangle=|\mathbf{q}|^2/4$, $\langle q_1^2q_2^2\rangle=|\mathbf{q}|^4/24$, and hence $\langle h_{\mathrm{bub}}\rangle=-11C_{\mathfrak{g}}/(3|\mathbf{q}|^4)$. Consequently

$$(2\pi)^{-4}\int_{\mu_{\mathrm{reg}}<|\mathbf{q}|\le 1} h_{\mathrm{bub}}\,d^4\mathbf{q}=-\frac{11C_{\mathfrak{g}}}{3}\frac{2\pi^2}{(2\pi)^4}\int_{\mu_{\mathrm{reg}}}^1\frac{dr}{r}=-\kappa_G\log\mu_{\mathrm{reg}}^{-1}\,. \tag{9.12}$$

The orbital average is $C_{\mathfrak{g}}/3$ and the spin average $-4C_{\mathfrak{g}}$. The unnormalized error integrals on $|\mathbf{q}|\le\mu_{\mathrm{reg}}$, $|\mathbf{q}|>1$, and the intervening shell are bounded by $C_{\mathrm{ir}}\pi^2/2$, $C_{\mathrm{ir}}(2\pi)^4$, and $3\pi^2C_{\mathrm{sh}}$ respectively. Lemma 9.2 proves (9.8). Nonabelianness makes $\sum_\nu|[X,e_\nu^{\mathfrak{g}}]|^2$ positive for some X, so $\kappa_G>0$. □

We now extract the correction from the regular activities. The Gaussian integration and connected-activity estimates of Section 10 give the following density before this extraction:

$$\mathcal{B}_{\beta^{\mathrm{in}}}^{(m)}(\mathrm{dens}\,\rho^{\mathrm{in}})=e^{v_{\mathrm{vac}}^{\mathrm{pre}}|\Lambda|-\beta^{\mathrm{in}}\mathcal{K}_{N,m}^{\mathrm{kin}}-\mathcal{L}_{N,m}^{(2)}-E_\Sigma^{\mathrm{pre}}-\alpha\mathcal{S}_0}\,\mathcal{G}_{\mathrm{for}}^{\mathrm{pre}}\,. \tag{9.13}$$

Here $E_\Sigma^{\mathrm{pre}}=\sum_i\chi_{X_i}E_i^{\mathrm{pre}}$, with $\chi_X=\mathbf{1}_{X\text{ healthy}}$, and $\mathcal{G}_{\mathrm{for}}^{\mathrm{pre}}$ is the forced sum of Definition 2.9. The function $\mathcal{S}_0$ is the H_4-average of the unclipped Wilson and expanded row-square terms, masked on their path supports. The nonquadratic determinant remainder, the effect of level-$(m+1)$ quadratic atoms on the fluctuation operator, and the torus remainder $C_\ell e^{-c_\ell n_{p,m}}$ are included in E_i^{pre}. The last is absorbed in its activity bound by choosing $n_{p,m}\ge n_{J_m}^{\min}$. These activities may have nonzero values and affine Hessians at the identity; both are removed below.

On a nonwrapping support, affine fields with the same curl differ by a gauge gradient. The Hessian of E_i^{pre} annihilates these gradients, and its first derivative vanishes by colour invariance. Thus simplicity gives a real form $c_i(\mathfrak{p},\mathfrak{q})$ on the six constant curls. Set $c_i=0$ when the support enlarged by the radius-two neighbourhood of o_i wraps. Let $\mathcal{F}_{\mathfrak{p}}(U,o)$ be the signed average of the four oriented plaquette curvatures in the $\mathfrak{p}$-plane based at o, so its derivative on an affine field is that field's constant curl. Define the unmasked projection and the subtracted function by

$$\mathsf{Q}_i^{\mathrm{quad}}(U)=\tfrac12\sum_{\mathfrak{p},\mathfrak{q}}c_i(\mathfrak{p},\mathfrak{q})\langle\mathcal{F}_{\mathfrak{p}}(U,o_i),\mathcal{F}_{\mathfrak{q}}(U,o_i)\rangle_{\mathfrak{g}},\qquad E_i^0=E_i^{\mathrm{pre}}-E_i^{\mathrm{pre}}(1)-\mathsf{Q}_i^{\mathrm{quad}}\,. \tag{9.14}$$

We mask the projection at its anchor and E_i^0 on its enlarged support $\widetilde{X}_i$. Reflections kill off-diagonal curl pairs in the anchor sum; permutations equate the diagonal pairs. Translation invariance makes $\delta=\delta\beta_{N,p,m}\coloneqq\sum_{o_i=o}c_i((1,2),(1,2))$ independent of o. Therefore

$$\sum_i\chi_{\{o_i\}}\mathsf{Q}_i^{\mathrm{quad}}=\delta\sum_o\chi_{\{o\}}\tfrac12\sum_{\mathfrak{p}}|\mathcal{F}_{\mathfrak{p}}(U,o)|^2\,. \tag{9.15}$$

For the diagonal entry use $\mathcal{A}_2(y)=r_{n_{p,m}}(y-o)^1e_\nu^{\mathfrak{g}}$, all other components zero. The direction $t_m\mathcal{A}$ satisfies the activity norm bound since $\varpi_{J_m}\ge 2$. The diagonal Hessian estimate yields

$$t_m^2|\delta|\le H_{\mathrm{reg}}\coloneqq\sup_o\sum_{o_i=o}e_i^{\mathrm{pre}}\le c_{\mathrm{res}}\epsilon_m^{\mathrm{reg}}/4\,. \tag{9.16}$$

The majorants e_i^{pre} and constant c_{res} are supplied by Section 10. Its entry bound $c_{\mathrm{res}}\epsilon_m^{\mathrm{reg}} \le 4t_m^2$ gives $|\delta| \le 1$.

Lemma 9.4 (Exact coupling reset)**.** *For each member ρ_u^{in} of a continuous admissible input curve in Proposition 4.1(b), let ρ_u^{out} have path action $P_{N,m}^{\mathrm{path}}$, coupling $\beta_u^{\mathrm{in}} + \alpha_{N,m}^{(1)} + \delta\beta_{N,p,m}(u)$, vacuum scalar $v_{\mathrm{vac}}^{\mathrm{pre}}(u) - a_{\mathrm{vac}}(u)/|\Lambda|$, and the regular activities and forced sum constructed below. Then*

$$\operatorname{dens}\rho_u^{\mathrm{out}} = \mathcal{B}^{(m)}_{\beta_u^{\mathrm{in}}}(\operatorname{dens}\rho_u^{\mathrm{in}}), \qquad |\delta\beta_{N,p,m}(u)| \le 1\,. \tag{9.17}$$

Both $\delta\beta_{N,p,m}(u)$ and $a_{\mathrm{vac}}(u)$ are continuous. Consequently $|\beta_u^{\mathrm{in}} - \beta(\rho_u^{\mathrm{out}}) - \kappa_G \log \ell_m| \le \omega_{J_{m+1}}^{\mathrm{RG}}$.

Proof. Suppress u. Put $a_{\mathrm{vac}} = \sum_i E_i^{\mathrm{pre}}(1)$. Define the mask discrepancy $\mathsf{D}_i^{\mathrm{mask}}$ by

$$E_i^{\mathrm{pre}}(1) + \chi_{\{o_i\}}\mathsf{Q}_i^{\mathrm{quad}} + \chi_{\widetilde{X}_i}E_i^0 - \mathsf{D}_i^{\mathrm{mask}} = \chi_{X_i}E_i^{\mathrm{pre}}\,.$$

It is zero when $\widetilde{X}_i$ is healthy, by (9.14), and otherwise is supported on a set containing a flagged site.

To identify the other new regular terms, expand each row square into its pairs of slots. Index the resulting quadratic terms, including plaquettes and their coefficients, by d. Write q_d for the unclipped term and q_d^{anc} for the expression obtained by replacing each transported curvature by the corresponding $\mathcal{F}_{\mathfrak{p}}(U, o_d)$. For plaquettes use the same replacement of the Wilson term by half the curvature norm squared. The kinetic normalization of Definition 2.9 gives $\sum_d q_d^{\mathrm{anc}} = \sum_o \frac{1}{2}\sum_{\mathfrak{p}} |\mathcal{F}_{\mathfrak{p}}(U,o)|^2$, also with the common anchor masks. Let χ_d be health on the combined path support enlarged to include these anchor plaquettes. The output regular sum E_Σ^{out} consists of

$$\sum_i \chi_{\widetilde{X}_i}E_i^0, \qquad \delta\sum_d \chi_d(q_d^{\mathrm{anc}} - q_d), \qquad -(\alpha+\delta)\sum_{k=3,4}\sum_i \chi_i \mathfrak{a}_i^{(k)}\,,$$

with the H_4 averages understood. On nonwrapping supports the first functions have zero affine Hessian by construction. In the second, every slot has the same linearized constant curl as its anchor analogue, so their affine Hessians agree. The last functions vanish to third order. All vanish at the identity.

On healthy supports, adding these three sums to $(\alpha+\delta)\mathcal{K}_{N,m}^{\mathrm{kin}}$ cancels the higher atoms and leaves $\alpha\mathcal{S}_0 + \delta\sum_o \chi_{\{o\}}\frac{1}{2}\sum_{\mathfrak{p}}|\mathcal{F}_{\mathfrak{p}}|^2 + \sum_i \chi_{\widetilde{X}_i}E_i^0$. At other configurations retain the discrepancies: B_{for} is the sum of $\mathsf{D}_i^{\mathrm{mask}}$ and the clip and health discrepancies in this calculation. Each discrepancy vanishes unless its support contains a flagged site. Thus, at every configuration,

$$\begin{aligned}(\alpha+\delta)\mathcal{K}_{N,m}^{\mathrm{kin}} + E_\Sigma^{\mathrm{out}} - B_{\mathrm{for}} &= E_\Sigma^{\mathrm{pre}} - a_{\mathrm{vac}} + \alpha\mathcal{S}_0,\\ \mathcal{G}_{\mathrm{for}}^{\mathrm{out}} &= e^{B_{\mathrm{for}}}\mathcal{G}_{\mathrm{for}}^{\mathrm{pre}}\,.\end{aligned} \tag{9.18}$$

The forced-activity expansion of Section 10 groups the local factors in the second line into disjoint supports with their flag inventories. Absolute support bounds justify the rearrangements and give the output activity bounds. Substitution in (9.13) proves (9.17), including configurations where the forced sum vanishes.

The sums defining δ and a_{vac} converge absolutely and locally uniformly along the input curve, by the activity bounds and summable tails of Section 10. Their summands are continuous in the activity derivative norms, so both scalars are continuous. Finally (9.8) and $|\delta| \le 1$ give the stated drift: row 4 of Section 6 chooses $\omega_j^{\mathrm{RG}} \ge C_{\mathrm{loop}} + 1$ before ℓ, which is possible because C_{loop} is independent of ℓ. □

This is the marginal estimate in Proposition 4.1; the output need not lie in $\mathfrak{B}_m$. Along the trajectories of Proposition 4.2, summation from the fixed endpoint gives

$$\left|\beta_{N,p,m} - \bar{\beta} - \kappa_G s_m\right| \le \sum_{i<m} \omega^{\mathrm{RG}}_{J_{i+1}} = o(s_m)\,. \tag{9.19}$$

Indeed, $\omega^{\mathrm{RG}}_{J_{i+1}} / \log \ell_i \le J_{i+1}^{-2} \to 0$; split at a fixed level and divide by $s_m = \sum_{i<m} \log \ell_i \to \infty$. The bound is uniform in $N \ge m$ and p. Reaching the endpoint and the uniqueness and holomorphic dependence of its tuning require Proposition 4.2.

10 The nonlinear step

This section is under construction.

11 The multiscale construction

This section is under construction.

12 Infinite volume

Proposition 4.5 removes the ultraviolet cutoff at each fixed volume. We now let the volume tend to infinity along a refinement of any prescribed increasing sequence. The bounds in Proposition 4.4(b), including one additional difference, give compactness of the fixed-volume kernels and all their derivatives on separated configurations. Thus the extraction can be made directly after taking the full ultraviolet limits. It retains more positional regularity than extraction on Gen_n alone would give.

Compactness of the kernels does not identify their fields. A lattice difference is represented by a coefficient vector in the next degree, but that vector need not remain bounded. We first identify exactly the coefficient vectors annihilated by every separated correlation. Normalizing by the size of the projected coefficient then produces a fixed derivative representative. The finite correlation tests used in this identification must be retained on the same volume refinement as the kernels. Finally, a second cell size controls the exterior of a repeated reflected word without introducing a factor for each repetition. This gives the estimate needed subsequently for slab operators.

We use the notation of Part I, in particular $M^{\mathrm{lat}}_{N,p}$ and $C^{\mathrm{lat}}_{N,p}$ for lattice moments and their smeared values. Limits involving auxiliary coefficient vectors below are ordered: first along a common ultraviolet subsequence at each fixed volume, then along the retained volumes. This auxiliary subsequence does not replace the full inner limits of Proposition 4.5.

Lemma 12.1 (Compactness after the ultraviolet limit)**.** *For every strictly increasing* p*, there is a strictly increasing* q *such that, with* $\mathsf{v} = \mathsf{p} \circ \mathsf{q}$*, the kernels* $f_{\mathsf{v}(k),v}$ *converge in* C^t *on every compact subset of* Conf_n*, for every fixed token list* v *and every* $t \ge 0$*. Their limits define a smooth multilinear table* S*. For every separated test* f*,*

$$\lim_{k\to\infty} \lim_{N\to\infty} C^{\mathrm{lat}}_{N,\mathsf{v}(k)}(v; f) = \int f(x) S_n(v; x)\, dx\,. \tag{12.1}$$

The inner limit exists at every f*-admissible volume, along the full ultraviolet sequence.*

Proof. Fix a compact set $K \subset \mathrm{Conf}_n$ and a compact neighbourhood K' contained in Conf_n. For all sufficiently large p, these sets lie in $\mathrm{NoHit}_{L_p,n}$ and the physical torus separation on K' is at least $2r > 0$, independently of p. For large N, every lattice tuple within sup-distance a_N of K' has separation at least r. Proposition 4.4(b), with this r and a fixed difference list $\boldsymbol{l}$, bounds $\Delta_{\boldsymbol{l}} M^{\mathrm{lat}}_{N,p}$ on these tuples by C_{diff}, for $N \geq N_0$ and every p. At fixed p and $x \in K'$, the multilinear interpolant is a convex combination of such values and converges to $\partial_{\boldsymbol{l}} f_{p,v}(x)$ by the moving-point assertion of Proposition 4.5. Hence

$$\sup_{p \geq p_0} \sup_{x \in K'} |\partial_{\boldsymbol{l}} f_{p,v}(x)| \leq C_{\mathrm{diff}} \,. \tag{12.2}$$

No ultraviolet threshold uniform in p is needed for this passage. One further derivative bounds differences along short line segments in K', and hence gives equicontinuity on K.

Apply the Arzelà–Ascoli theorem to these uniformly bounded, equicontinuous complex-valued functions on each compact set. Use compact exhaustions of Conf_n and diagonalize over arities, degree lists, basis token lists and derivative lists; these indices form a countable set. The domains contain each compact set eventually. Extend from basis lists by multilinearity. Integration against a compactly supported test shows that the limit assigned to $\partial_{i,a} f_{p,v}$ is the distributional derivative of the limit assigned to $f_{p,v}$. The continuous limits of all orders therefore define one smooth kernel with the asserted C^t convergence. The unit and arity-zero identities pass through the same limits; set the kernels to zero at coincident configurations.

Every fixed compact test support is eventually inside the open fundamental box, since $L_{\mathfrak{v}(k)} \to \infty$. Proposition 4.5 gives its full inner limit $\int f f_{p,v}$ at every admissible p. Uniform convergence on the support gives (12.1) and absolute integrability of $f S_n(v;\cdot)$. □

We record the cell estimates used both for growth bounds and for repeated words. Reflection positivity here is the Osterwalder–Seiler inequality for the Wilson measure: on an even periodic torus, with $\beta \geq 0$, a bounded gauge-invariant function F depending only on links with both ends strictly on one side of a site or bond reflection ϑ satisfies $\langle \overline{F \circ \vartheta} F \rangle \geq 0$ [4]; see also [5, Theorem 2.1]. For the site case, conditioning on the links in the two fixed planes gives an integral of squared conditional moduli. Its chessboard consequence is the Fröhlich–Israel–Lieb–Simon estimate: for a reflection tiling $\mathcal{C}$ with an even number of cells in each direction and bounded gauge-invariant interior functions F_C,

$$\left| \left\langle \prod_{C \in \mathcal{C}} (F_C)_C \right\rangle \right| \leq \prod_{C \in \mathcal{C}} \|F_C\|_{\mathcal{C}} \,. \tag{12.3}$$

Here the copies and roots are those of Part I. We use the abstract chessboard theorem [2, Theorem 4.1], applied successively in the four directions. The hypotheses hold because G is compact, the insertions are continuous and gauge invariant, the chosen couplings are nonnegative, and L_p is a multiple of four. Neither result requires connectedness of G. A translated tiling is the image of $\mathcal{C}^{\mathrm{tile}}_m(y)$ under a lattice translation; the root of an insertion in it is the root of its pullback to the base cell.

Lemma 12.2 (Cell bounds)**.** *Suppose centered roots at level k of an insertion and all its images under signed permutations and conjugation are at most A. In any translated cell of side a_m, $m \leq k$, the root of that insertion is at most A if its anchor has distance at least $3a_k$ from the boundary and its stencil fits inside distance a_k of the anchor.*

For the table of Lemma 12.1, with $n \geq 1$ and $D = 1 + \max_i d_i$, there is $C \geq 0$, depending on the token list, such that

$$|S_n(v;x)| \leq C r^{-4nD} \qquad (0 < r \leq 1,\ \mathrm{sep}_\infty(x) \geq r) \,. \tag{12.4}$$

More generally, take $m \le k \le l \le b$. Suppose main points are separated from one another and from all exterior points by more than a_m, exterior points are mutually separated by more than a_l, and the main points have at most P_0 residue classes modulo $a_m\mathbb{Z}^4$. If

$$10P_0 a_k < a_m, \qquad 6n_{\mathrm{ext}} a_b < a_l\,, \tag{12.5}$$

and, for all images under H_4 and conjugation, the centered roots at level k of the main insertions are at most A and those at level b of the exterior insertions are at most B ($A, B \ge 0$), then, writing $S(\mathit{main}, \mathit{exterior})$ for S at the concatenated list,

$$|S(\mathit{main}, \mathit{exterior})| \le A^{n_{\mathrm{main}}} B^{n_{\mathrm{ext}}}\,. \tag{12.6}$$

The root hypotheses are required eventually along the auxiliary ordered limits.

Proof. At centers of level-k cells aligned with the level-m tiling, reflect the observable into every level-m cell and apply (12.3) on the finer tiling, putting 1 in unused cells. Covariance makes each occupied factor an image of the original insertion, so taking the level-m root gives A. For intervening anchors write $R(z)$ for the root after translation by z lattice steps in one axis. Translate the whole configuration of reflected copies by one lattice step in that axis. The Wilson measure is invariant, and a reflection followed by a unit translation equals the reflection of the translate by -1. The translated configuration therefore carries the translate by $z+1$ in even cells and the reflected translate by $z-1$ in odd cells. The margins keep both interior. Equation (12.3) gives $R(z)^{|\mathcal{C}|} \le R(z+1)^{|\mathcal{C}|/2} R(z-1)^{|\mathcal{C}|/2}$, hence

$$R(z)^2 \le R(z-1)R(z+1)\,.$$

Log-convexity bounds R by its endpoint values between consecutive aligned centers. Applying this in each axis proves the first assertion; the stated margins keep all factors interior.

For (12.4), Proposition 4.4(a), with $\varepsilon = 1$, gives $C_{\mathrm{root}} a_k^{-D}$ for the finitely many images of the fixed labels, at every $k \ge F$, eventually in N, uniformly in p. Gentleness of the level system implies

$$a_{u+1} \ge c a_u^2\,. \tag{12.7}$$

Indeed $\log \ell_u / \log a_u^{-1} \to 0$ implies $\ell_u \le a_u^{-1}$ eventually; reduce $c > 0$ to cover the finitely many remaining levels. Consequently the first $u \ge F$ with $a_u < \delta \le 1$ satisfies $a_u \ge c_F \delta^2$, including $u = F$. Choose the first $m \ge F$ with $a_m < r/2$, and the first $k \ge m$ with $a_k < a_m/[6(n+1)]$. Then $a_m \ge c_1 r^2$ and $a_k \ge c_2 r^4$. In each axis the forbidden translations placing any anchor within $3a_k$ of a cell boundary occupy less than $6n a_k < a_m$. Choose a translation outside them. Distinct anchors occupy distinct cells, so the first assertion and (12.3) give $(C_{\mathrm{root}} a_k^{-D})^n$. Pass to the two limits using Proposition 4.5 and Lemma 12.1 to obtain (12.4).

For (12.6), avoid boundary strips of width $5a_k$ for the P_0 main residues. The first inequality in (12.5) permits this. Next translate by less than one level-l cell to place every exterior anchor at distance $3a_b$ from the finer boundaries; the second inequality permits this. Since $a_l \le a_k$, the main margins remain at least $3a_k$. Main insertions occupy distinct coarse cells and no coarse cell containing an exterior insertion. Several exterior insertions may share a coarse cell. For their product E_C, write $\mathcal{C}_m^{\mathrm{tile}}$ for the chosen coarse tiling, reflect the entire product into every coarse cell and apply the finer chessboard estimate. Its factors occupy distinct fine cells, also after reflection across coarse boundaries, whence

$$\|E_C\|_{\mathcal{C}_m^{\mathrm{tile}}}^{|\mathcal{C}_m^{\mathrm{tile}}|} \le B^{n_C |\mathcal{C}_m^{\mathrm{tile}}|}, \qquad \|E_C\|_{\mathcal{C}_m^{\mathrm{tile}}} \le B^{n_C}\,.$$

Apply the coarse chessboard estimate to the main factors and these exterior products. This proves the lattice inequality. For each fixed finite configuration pass to the two limits using Proposition 4.5 and Lemma 12.1. □

We need the following estimates for the construction of Proposition 4.1. They are not consequences of the statements of Propositions 4.1–4.5; their proof is deferred to Section 11.

Definition 12.3 (Ordered tails and coefficient errors)**.** Fix a volume refinement v and a strictly increasing ultraviolet sequence (N_j). A property holds *along ordered tails* if there are k_0 and thresholds $N_0(k)$ for $k \geq k_0$ such that it holds when $p = \mathsf{v}(k)$ and $N = N_j \geq N_0(k)$. Convergence along ordered tails means eventual membership in each neighbourhood of the limit. A coefficient family $\mathbf{w}_{N,p} \in V_d$ is *allowed* if, for every $\varepsilon > 0$, along ordered tails,

$$\|\mathsf{Z}_{N,p,d}^{-1}\mathbf{w}_{N,p}\|_\infty \leq a_N^{-\varepsilon}\,. \tag{12.8}$$

Bounded measurable observables $Q_{N,p}(y)$ have a *vanishing cell bound* if there are $\eta_{N,p}^{\mathrm{cell}} \geq 0$ tending to zero along ordered tails such that, for every $q, m \in \mathbb{N}$, there is $k \geq m$ with the following property along ordered tails: for every translated bond tiling $\mathcal{C}$ of physical side a_k and every site y in its base cell at lattice distance at least $\lfloor a_k/[8(q+1)a_N]\rfloor$ from the boundary, $Q_{N,p}(y)$ is interior and $\|Q_{N,p}(y)\|_{\mathcal{C}} \leq \eta_{N,p}^{\mathrm{cell}}$.

Lemma 12.4 (Corrected coefficients)**.** *For every d there are idempotents $\mathsf{P}_{N,p,d}^{\det}$ on V_d, bounded uniformly in N, p, with the following properties for every v and (N_j). Here* det *denotes the finite correlation detector of Section 11, and $q_d = \dim_{\mathbb{C}} M_d^{\mathrm{Max}}$.*

(i) Inclusion. *For $d' \leq d$ and $\mathbf{c} \in V_{d'}$, there is an allowed family $\mathbf{w}$ in V_d such that $O_{N,p}^{(d)}(\mathbf{w}_{N,p}, \cdot) - O_{N,p}^{(d')}(\mathbf{c}, \cdot)$ has a vanishing cell bound.*

(ii) Difference. *For $\mathbf{c} \in V_d$ and an axis a, there is an allowed family $\mathbf{w}$ in V_{d+1} such that*

$$O_{N,p}^{(d+1)}(\mathbf{w}_{N,p}, \cdot) - a_N^{-1}\big[O_{N,p}^{(d)}(\mathbf{c}, \cdot + e_a) - O_{N,p}^{(d)}(\mathbf{c}, \cdot)\big]$$

has a vanishing cell bound.

(iii) Kernel estimate. *If $\mathbf{w}$ is allowed and $\mathbf{w}_{N,p} \in \ker \mathsf{P}_{N,p,d}^{\det}$ for all N, p, there are measurable observables $Z_{N,p}(y)$, bounded at each N, p, such that $O_{N,p}^{(d)}(\mathbf{w}_{N,p}, \cdot) - Z_{N,p}$ has a vanishing cell bound. For every spectator list and $\delta > 0$, the moments with $Z_{N,p}$ in the tested slot vanish exactly along ordered tails, at every lattice tuple of physical separation at least δ.*

(iv) Equation count. *For all N, p,*

$$\dim V_d \leq \dim \ker \mathsf{P}_{N,p,d}^{\det} + q_d\,. \tag{12.9}$$

(v) Finite rank comparison. *There are a basis $\mathbf{m}_1, \dots, \mathbf{m}_{q_d}$ of M_d^{Max} and finitely many free spectator lists $\boldsymbol{B}_i^{\mathrm{Max}}$, $i = 1, \dots, s_d$. For every $\nu \in \mathbb{N}$ there are $r_\nu^{\mathrm{cmp}} > 0$, lattice-compatible separated configurations $x^{(\nu,i)} = r_\nu^{\mathrm{cmp}} y^{(\nu,i)}$ in one centered cell of side a_{k_ν}, and coefficient families $\mathbf{c}_{N,p}^{(\nu,j)} \in V_d$ and $\boldsymbol{b}_{N,p}^{(\nu,i)}$. For fixed ν, all these coefficients are bounded in N, p; column j uses the lift $\mathbf{c}^{(\nu,j)}$ of $\mathbf{m}_j$, independently of the row i, and the spectator degrees are fixed. Write $I_{\nu,i,j,h;N,p}$ for the lattice insertion at $x_h^{(\nu,i)}$, using $\mathbf{c}_{N,p}^{(\nu,j)}$ in slot 0 and $\boldsymbol{b}_{N,p}^{(\nu,i)}$ in the other slots. Let $F_{\nu,i,j,\mathcal{J}}^{\mathrm{Max}}$ be the free moment of $(\mathbf{m}_j, \boldsymbol{B}_i^{\mathrm{Max}})$ on the slot subset $\mathcal{J}$ at $y^{(\nu,i)}$, and $F_{\nu,i,j,\varnothing}^{\mathrm{Max}} = 1$. Define $\mathfrak{R}_{\nu,i,j,\varnothing} = 1$ and, suppressing i, j, N, p,*

$$\mathfrak{R}_{\nu,\mathcal{I}} = \prod_{h\in\mathcal{I}} I_{\nu,h} - \sum_{\varnothing\neq\mathcal{J}\subseteq\mathcal{I}} F_{\nu,\mathcal{J}}^{\mathrm{Max}}\mathfrak{R}_{\nu,\mathcal{I}\setminus\mathcal{J}}\,. \tag{12.10}$$

There are $\eta_\nu^{\mathrm{cmp}} \geq 0$ tending to zero such that, eventually in N, for every p, every insertion and every nonempty polynomial $\mathfrak{R}_{\nu,i,j,\mathcal{I}}$ is interior to that cell, and the polynomial's root is at most η_ν^{cmp}. Moreover $y^{(\nu,i)} \to y^{(i)} \in \mathrm{Gen}$, and the matrix of free full moments at $y^{(i)}$ has rank q_d.

Lemma 12.5 (Null coefficients and rank)**.** *After one further volume refinement, retaining S, the corrected projections and the bounded coefficients in the finite correlation tests have iterated limits along a common auxiliary ultraviolet subsequence. If P_d is the limiting projection, then*

$$\mathcal{N}_d(S) = \ker \mathsf{P}_d, \qquad \dim_{\mathbb{C}} F_d = q_d\,. \tag{12.11}$$

An error with a vanishing cell bound has vanishing smeared correlations against every fixed separated test and spectator list.

Proof. A diagonal extraction first in N, for every p, and then in the retained volumes includes every degree and every ν. The uniform bounds in Lemma 12.4 give bounded inner and outer matrix limits. All kernel limits from Lemma 12.1 persist. Limits preserve idempotence. Estimates for varying coefficient families remain valid on the refinement: extend such a family by zero outside the retained index pairs when applying (12.8) and the kernel estimate.

For the error assertion, fix separation $\delta > 0$ and n slots. Apply the error's cell bound with $q = n$ and m such that $a_m < \delta$, obtaining $k \geq m$. Choose $k' \geq k$ beyond all spectator reference levels with $3a_{k'} \leq a_k/[8(n+1)]$, and bound their roots by Proposition 4.4(a). Since $3a_{k'}/a_N$ is an integer, it is at most the rounded lattice margin. The boundary-strip argument of Lemma 12.2 places the slots in distinct cells with that margin. The chessboard estimate gives

$$\left| \left\langle E_{N,p}(y_i) \prod_{h \neq i} I_{N,p}(v_h, y_h) \right\rangle_{N,p} \right| \leq C_{\delta,v}\, \eta_{N,p}^{\mathrm{cell}}\,. \tag{12.12}$$

Smearing multiplies this by at most $\|f\|_\infty$ times a fixed bound on the volume of the cells meeting $\operatorname{supp} f$. This bound is independent of p, N.

Bounded coefficient families satisfy (12.8): the coefficient transport in Proposition 4.1(c) bounds their terminal coordinates by a fixed norm factor times $\prod_{m \leq N} C_{J_m}$. The entrance inequalities imply $\sum_{m \leq N} \log C_{J_m} = o(s_N)$, by comparison with $\sum_{m<N} \log \ell_m = s_N$. Apply the kernel estimate and (12.12) to $\mathbf{c} - \mathsf{P}^{\mathrm{det}}_{N,p,d}\mathbf{c}$. The smeared correlation maps converge in operator norm, because V_d is finite dimensional. Their limits annihilate $(1 - \mathsf{P}_d)\mathbf{c}$. Since S is smooth, vanishing against every separated test is equivalent to vanishing at every separated point. Thus $\ker \mathsf{P}_d \subset \mathcal{N}_d(S)$.

For the reverse dimension bound, fix one row and column of Lemma 12.4(v) and suppress their indices. In (12.10), the term $\mathcal{J} = \mathcal{I}$ is $F^{\mathrm{Max}}_{\nu,\mathcal{I}}$. Isolating it leaves $\mathfrak{R}_{\nu,\mathcal{I}}$ and the terms with nonempty proper $\mathcal{J}$. Applying (12.3) to each of these nonempty polynomials in the stated cell gives

$$\left| \left\langle \prod_{h \in \mathcal{I}} I_{\nu,h} \right\rangle - F^{\mathrm{Max}}_{\nu,\mathcal{I}} \right| \leq \eta_\nu^{\mathrm{cmp}} \sum_{\mathcal{J} \subsetneq \mathcal{I}} |F^{\mathrm{Max}}_{\nu,\mathcal{J}}| \leq W_\nu \eta_\nu^{\mathrm{cmp}}\,. \tag{12.13}$$

Here W_ν is the maximum, over the finitely many rows and columns, of $\sum_{\mathcal{J} \subseteq \mathcal{I}} |F^{\mathrm{Max}}_{\nu,\mathcal{J}}|$ for the full slot set $\mathcal{I}$. The rescaled configurations $y^{(\nu,i)}$ converge to the generic configurations $y^{(i)}$. Smoothness of the free subset moments makes W_ν bounded, so $W_\nu \eta_\nu^{\mathrm{cmp}} \to 0$. Passing through the coefficient and kernel limits preserves (12.13). Row i of the limiting matrix is the functional $[\mathbf{c}] \mapsto S(\mathbf{c}(x_0^{(\nu,i)}), \boldsymbol{B}^{(\nu,i)}(x^{(\nu,i)}))$, with limiting spectator coefficients $\boldsymbol{B}^{(\nu,i)}$; column j is the class of the limiting lift of $\mathbf{m}_j$. Each row annihilates $\mathcal{N}_d(S)$ and thus defines a functional on F_d. The

matrices approach the free matrix of rank q_d. A nonzero q_d-minor remains nonzero for large ν, proving $\dim F_d \geq q_d$. Nearby idempotents have the same rank: if $\|\mathsf{P}-\mathsf{Q}\| < 1$, then $1-\mathsf{Q}$ is injective from $\ker \mathsf{P}$ into $\ker \mathsf{Q}$; interchange P, Q. Passing (12.9) to P_d gives $\dim F_d \leq \dim \operatorname{ran} \mathsf{P}_d \leq q_d$. Equality proves (12.11). □

Lemma 12.6 (Fixed representatives)**.** *There are linear coefficient maps $\kappa_{d,e} : V_d \to V_e$ for $d \leq e$ and $\tau_{d,a} : V_d \to V_{d+1}$ such that, against all separated spectator configurations,*

$$S(\kappa_{d,e}\mathbf{c}, \dots; x) = S(\mathbf{c}, \dots; x), \qquad S(\tau_{d,a}\mathbf{c}, \dots; x) = \partial_{x_0^a} S(\mathbf{c}, \dots; x)\,. \tag{12.14}$$

They induce compatible injective inclusions $\jmath_{d,e}$ and derivatives $\partial_a : F_d \to F_{d+1}$. Every positional derivative of S satisfies a polynomial separation bound.

Proof. For the difference representative, torus summation by parts gives the exact identity

$$a_N^{4n} \sum_y f(\hat{y}) \Delta_{i,a} M^{\mathrm{lat}}_{N,p}(v; y) = a_N^{4n} \sum_y \frac{f(\hat{y} - a_N e_{i,a}) - f(\hat{y})}{a_N} M^{\mathrm{lat}}_{N,p}(v; y)\,. \tag{12.15}$$

On a fixed compact neighbourhood of the test support, the backward quotient differs from $-\partial_{i,a} f$ by at most $a_N \|\partial^2_{i,a} f\|_\infty / 2$. Proposition 4.4(b) bounds the moment there; the weighted number of contributing sites is bounded by a fixed Euclidean volume. Thus the error tends to zero independently of the expanding fundamental box. Equations (12.1) and (12.12) show that the corrected degree-$(d+1)$ representative $\mathbf{w}_{N,p}$ has target smeared limit $-\int \partial_{i,a} f\, S$. The inclusion representative has target $\int fS$.

For a spectator list $\boldsymbol{B}$ and separated test f, put $L_{N,p}(\mathbf{c}) = C^{\mathrm{lat}}_{N,p}((e, \mathbf{c}), \boldsymbol{B}; f)$. Let $\mathsf{P} = \mathsf{P}^{\det}_{N,p,e}$ for either representing degree e. Applying Lemma 12.4(iii) to the normalized remainder gives

$$L_{N,p}\left(\frac{\mathbf{w}_{N,p} - \mathsf{P}\mathbf{w}_{N,p}}{1 + \|\mathsf{P}\mathbf{w}_{N,p}\|} \right) \longrightarrow 0\,. \tag{12.16}$$

Indeed $\mathbf{w}/(1 + \|\mathsf{P}\mathbf{w}\|)$ still satisfies (12.8), the normalized projected vector is bounded, and their difference is in $\ker \mathsf{P}$.

The pairs

$$\left(\frac{1}{1 + \|\mathsf{P}\mathbf{w}\|}, \frac{\mathsf{P}\mathbf{w}}{1 + \|\mathsf{P}\mathbf{w}\|} \right)$$

belong to a compact finite-dimensional set. Let $\mathfrak{F}$ be the filter on pairs (N, p) generated by the ordered tails

$$\{(N_j, \mathsf{v}(k)) : k \geq k_0,\ N_j \geq N_0(k)\}\,.$$

An iterated limit is a limit along $\mathfrak{F}$: first choose k_0 for the outer limit, then $N_0(k)$ for each inner limit. Fix an ultrafilter $\mathfrak{U} \supset \mathfrak{F}$ and a limit point $(\lambda, \mathbf{z})$ of the pairs along it. Every spectator limit is retained, $\mathsf{P}_e \mathbf{z} = \mathbf{z}$, and $\lambda + \|\mathbf{z}\| = 1$. Operator-norm convergence of $L_{N,p}$ and (12.16) give $L(\mathbf{z}) = \lambda \mathfrak{t}_L$, where $\mathfrak{t}_L$ is the target smeared limit identified above. If $\lambda = 0$, all such L annihilate $\mathbf{z}$; Lemma 12.5 gives $\mathbf{z} \in \ker \mathsf{P}_e$, contradicting $\|\mathbf{z}\| = 1$. Therefore $\mathbf{z}/\lambda$ is a fixed representative for every spectator at once. Choose it on a basis and extend linearly. The distributional identities become (12.14) by smoothness.

The identities preserve null coefficients and hence descend to F_d. Equality against all spectators makes the inclusions injective and gives $\jmath_{e,f}\jmath_{d,e} = \jmath_{d,f}$. Repeated differentiation replaces slots by fixed higher-degree tokens. Applying (12.4) to each resulting token list bounds every coordinate derivative. The operator norm of a positional Fréchet derivative is bounded by the finite sum of its coordinate coefficients. This proves E7, with constants depending on the labels and derivative order. For arity zero use $S_0 = 1$. □

Lemma 12.7 (Euclidean properties)**.** *The table S satisfies E1–E7. Its inclusion and derivative maps preserve conjugation; the inclusions preserve units and commute with H_4, and*

$$h\partial_a = \sum_b h_{ba}\partial_b h \qquad (h \in H_4)\,. \tag{12.17}$$

Proof. Permutation symmetry, reality, the unit identity and signed-permutation covariance hold for the lattice insertions once N exceeds their fixed reference levels. Approximate positions and a translation vector by the spacing-a_N grid. Proposition 4.5, followed by Lemma 12.1, passes these identities and lattice translation invariance to S. Smoothness supplies continuity at nongeneric separated positions. This proves E1–E3; E6 and E7 were proved above.

For E5, fix finitely many joint tests f_i with compact support on positive-side separated configurations in one axis. Their lattice-smeared observables F_i depend only on links strictly inside the positive half-torus for large p, N: compact support has positive distance from the reflecting plane, and the physical stencil radius tends to zero. Osterwalder–Seiler positivity gives

$$\sum_{i,j} \overline{z_i} z_j \langle \overline{F_i \circ \vartheta} F_j \rangle_{N,p} \geq 0\,.$$

Each matrix entry is a lattice smear against the reflected joint test $\overline{f_i}\otimes f_j$, including the corresponding reflected labels. Its support is compact and separated: each half is separated internally, and opposite halves have a strictly positive axis separation. Thus (12.1) passes every entry to its continuum integral. Smoothness bounds the kernel on this compact support, giving absolute integrability. The cone of positive semidefinite matrices is closed, proving E5. Normalized smooth bumps tending to any prescribed positive-side words give E4 by continuity. Other axes use the same argument.

Finally compare correlations of the two sides of each asserted field identity. Reality commutes with differentiation. For (12.17), differentiate $S(h\boldsymbol{A}; hx) = S(\boldsymbol{A}; x)$ in x_i^a and apply the chain rule; its coefficients are h_{ba}. The all-spectator null-space definition makes these correlation identities field identities. The same argument for inclusion representatives proves equivariance and preservation of the distinguished unit class. □

Lemma 12.8 (Repeated reflected words)**.** *For every degree bound $d_{\max}$ and coefficient norm bound $R_{\mathrm{lab}} \in \mathbb{R}$, there are $C_{\mathrm{tr}} > 0$ and $q_{\mathrm{tr}} \in \mathbb{N}$ with the following property. A nonempty length-n slab word w of width $h_{\mathrm{sl}} > 0$, with those bounds and buffer $0 < r \leq 1$, fits a width $0 < h_{\mathrm{sl}}^{\mathrm{fit}} = n_{\mathrm{lat}} a_m \leq h_{\mathrm{sl}}$, with $m, n_{\mathrm{lat}} \in \mathbb{N}$ chosen before the positive exterior word w_{ext}. For that width there is $B_{w_{\mathrm{ext}}}^{\mathrm{ext}} < \infty$ such that*

$$\big\|[(A_{w^\dagger} A_w)^{n_{\mathrm{rep}}} w_{\mathrm{ext}}]\big\|_{\mathrm{OS}}^2 \leq B_{w_{\mathrm{ext}}}^{\mathrm{ext}}\, [C_{\mathrm{tr}}^n (n/r)^{q_{\mathrm{tr}} n}]^{4n_{\mathrm{rep}}} \qquad (n_{\mathrm{rep}} \geq 0)\,. \tag{12.18}$$

Proof. Proposition 4.4(a), with $\varepsilon = 1$, and the finite number of degrees and signed images give F, J such that all allowed single-insertion roots at $k \geq F$ are at most $J a_k^{-D}$, where $D = d_{\max}+1$. Enlarge the coefficient ball to include every image, and take $J > 0$. The constants F, J, D precede n, r and the word.

Choose the first $m \geq F$ with $a_m < r/32$, and put $h_{\mathrm{sl}}^{\mathrm{fit}} = a_m \lfloor h_{\mathrm{sl}}/a_m \rfloor$. Since w is nonempty and has buffer r, this width is positive, contains w, and leaves buffer at least $r/2$. By (12.7), $a_m \geq c_1 r^2$. Choose a fixed small $\alpha > 0$ with $20\alpha < c_1$, and the first $k \geq F$ with $a_k < \alpha(r/n)^2$. Then $a_k < a_m$, so $k > m$. The first-crossing lower bound yields

$$20 n a_k < a_m, \qquad a_k \geq c_2 (r/n)^4, \qquad (J a_k^{-D})^n \leq (J c_2^{-D})^n (n/r)^{4Dn}\,. \tag{12.19}$$

These choices precede the exterior.

The OS norm squared in (12.18) is the correlation of $4n_{\mathrm{rep}}n$ main insertions and two copies of w_{ext}. Main insertions are mutually separated by at least $r/2$, and separated from the exterior by at least $r/2$. Since the fitted width is an integer multiple of a_m, their residues modulo $a_m\mathbb{Z}^4$ belong to the $2n$ patterns of w and its reflected word, independently of n_{rep}. For a nonempty exterior choose $l \geq k$ with $a_l < \min\{\mathrm{sep}_\infty(w_{\mathrm{ext}}), 2\min_i x_i^4(w_{\mathrm{ext}})\}$. Within either exterior copy, translations and reflection preserve separation; between the shifted copy $\tau_{2n_{\mathrm{rep}}h_{\mathrm{sl}}^{\mathrm{fit}}e_4}w_{\mathrm{ext}}$ and its reflection, the time separation is at least $2\min_i x_i^4(w_{\mathrm{ext}})$. Thus l is independent of n_{rep}. Choose $b \geq l$, beyond all exterior reference levels, with $12|w_{\mathrm{ext}}|a_b < a_l$. The centered bounds at b give a finite common root bound X for the exterior labels and their images. Lemma 12.2 gives

$$\left\|[(A_{w^\dagger}A_w)^{n_{\mathrm{rep}}}w_{\mathrm{ext}}]\right\|_{\mathrm{OS}}^2 \leq X^{2|w_{\mathrm{ext}}|}(Ja_k^{-D})^{4n_{\mathrm{rep}}n}\,.$$

The finer tiling bounds each exterior group only once, even if several exterior points share a coarse cell. Set $B_{w_{\mathrm{ext}}}^{\mathrm{ext}} = \max(1, X^{2|w_{\mathrm{ext}}|})$, $C_{\mathrm{tr}} = Jc_2^{-D}$ and $q_{\mathrm{tr}} = 4D$. Equation (12.19) proves the claim. For the empty exterior omit its factors and take $B_{w_{\mathrm{ext}}}^{\mathrm{ext}} = 1$; for the empty slab word the positive-translation contraction semigroup supplies the assertion. □

Proof of Proposition 4.6. Let q be the composite of the refinements of Lemmas 12.1 and 12.5, and $\mathsf{v} = \mathsf{p} \circ \mathsf{q}$; all limits above persist along subsequences. Write $\mathrm{Gen}_n^{(p)}$ for the periodic generic set of Proposition 4.6(b). For a separated test f, let $C_p^\infty(v; f)$ equal $\int f f_{p,v}$ at admissible volumes and zero otherwise. Lemma 12.1 proves (a), and its restrictions $K_{p,v} = f_{p,v}|_{\mathrm{Gen}_n^{(p)}}$ prove (b). Lemma 12.7 proves (c).

Lemmas 12.5 and 12.6 prove the dimension and calculus assertions in (d). For slot i, apply (12.14) after the simultaneous permutation exchanging slots 0 and i (E1). The response map has kernel exactly $\mathcal{N}_d(S)$, hence is injective on F_d. Any finite sum of responses lies in F_e for the largest of its degrees, by the compatible inclusions. Thus their images form the asserted directed union $\mathcal{L}$; quotient surjectivity supplies token representatives. The distinguished coefficient $\mathbf{u}_d = a_{m_{\mathrm{ref}}(d)}^d\mathbf{1}$ has eventual lattice insertion 1, so its class is $\mathbf{1}_d$. Lemma 12.8 proves (e), with the level k there equal to m in that lemma. □

13 The Maxwell comparison, Ward identities and the stress tensor

This section is under construction.

14 Phases and reconstruction

This section is under construction.

15 The trace law and non-freeness

This section is under construction.

16 The operator product expansion and contact extensions

This section is under construction.

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